



AKTU



B. Tech II-Year

Digital Electronics

Unit-3 in One Shot

Pdf Notes + AKTU PYQs + Imp. Questions



By Swati Ma'am



AKTU B.Tech II-Year

Subject : DIGITAL ELECTRONICS



Unit-3 : Syllabus

Sequential Logic And Its Applications: Storage elements: latches & flip flops, Characteristic Equations of Flip Flops, Flip Flop Conversion, Shift Registers, Ripple Counters, Synchronous Counters,

Other Counters: Johnson & Ring Counter

Latches & Flip flops

Sequential Circuits

These circuits have memory i.e output of these circuits at any instant of time , depends on both current inputs and past states (previous outputs).

These can be achieved by incorporating memory elements like **flip-flops and latches** into a feedback loop with combinational logic.

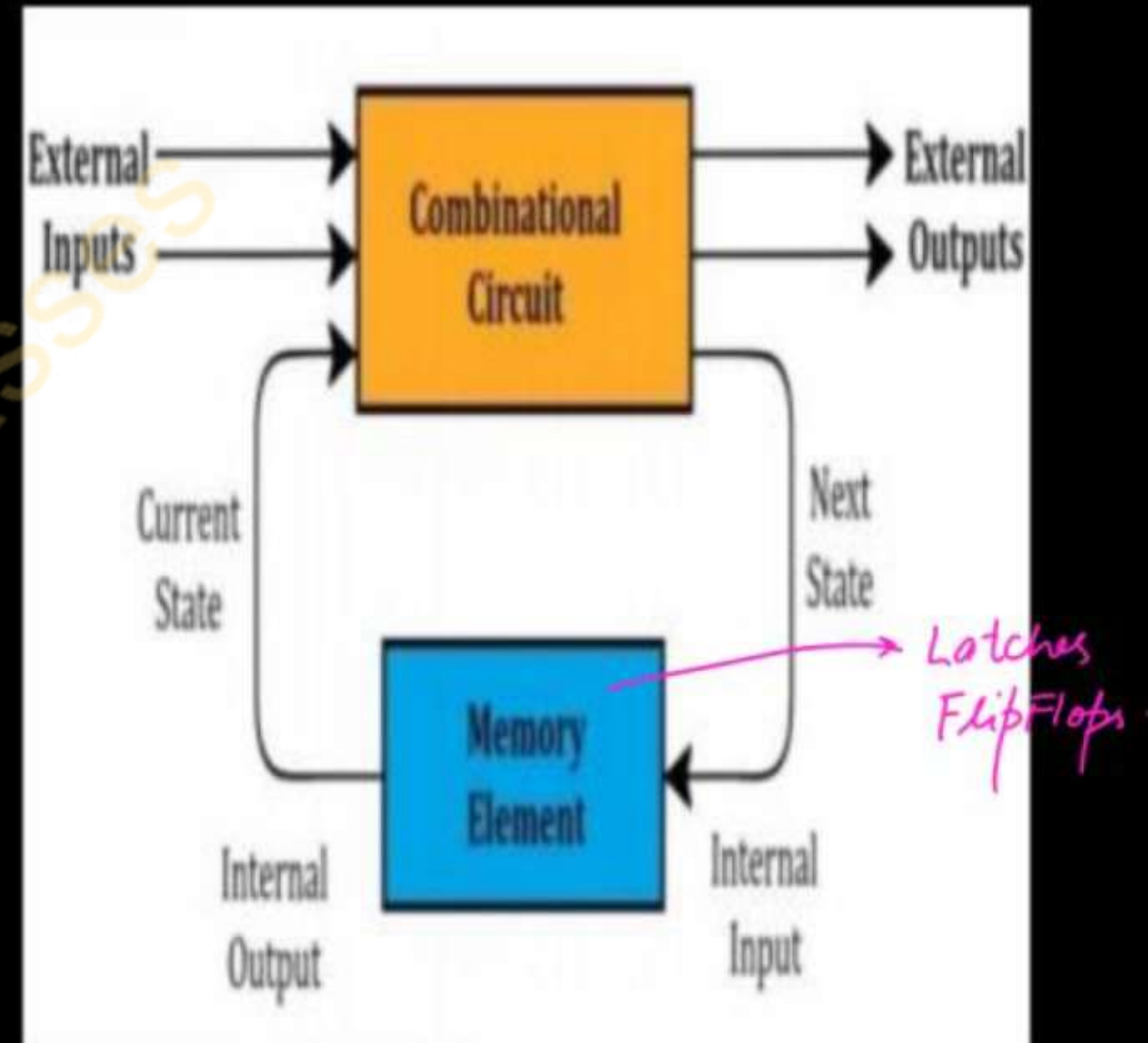
Examples –

Counters

Shift Registers

FSMs(Finite State Machine) etc.

Illustrate the term Sequential Logic



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Analysis of Sequential Circuits

Circuit Given



Write Input and Output Equations



Make State Table

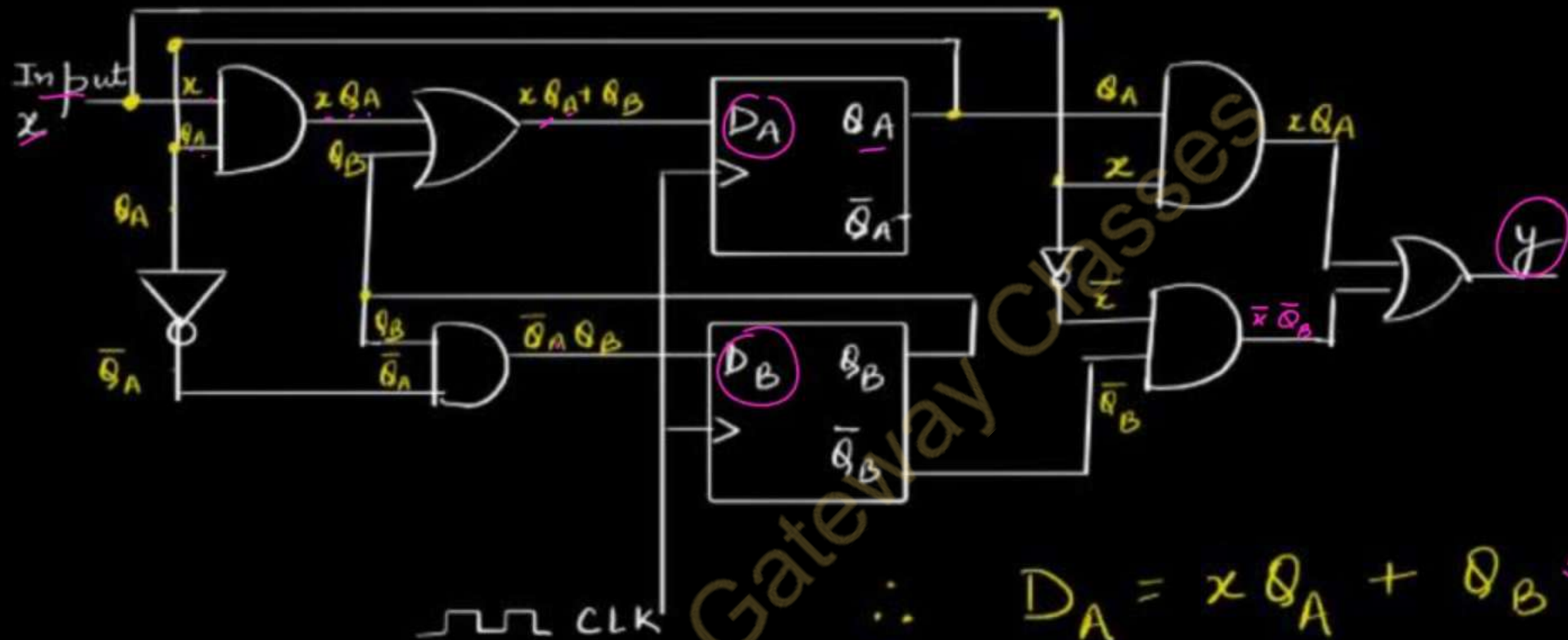
PS, Input, NS, output



Draw State Diagram

Gateway Classes

Analyze the following Sequential Circuit



$$\therefore D_A = xQ_A + Q_B \checkmark$$

$$D_B = \bar{Q}_A Q_B \checkmark$$

$$y = xQ_A + \bar{x}\bar{Q}_B \checkmark$$

$$D_A = xQ_A + Q_B$$

$$D_B = \bar{Q}_A Q_B$$

$$y = xQ_A + \bar{x}\bar{Q}_B$$

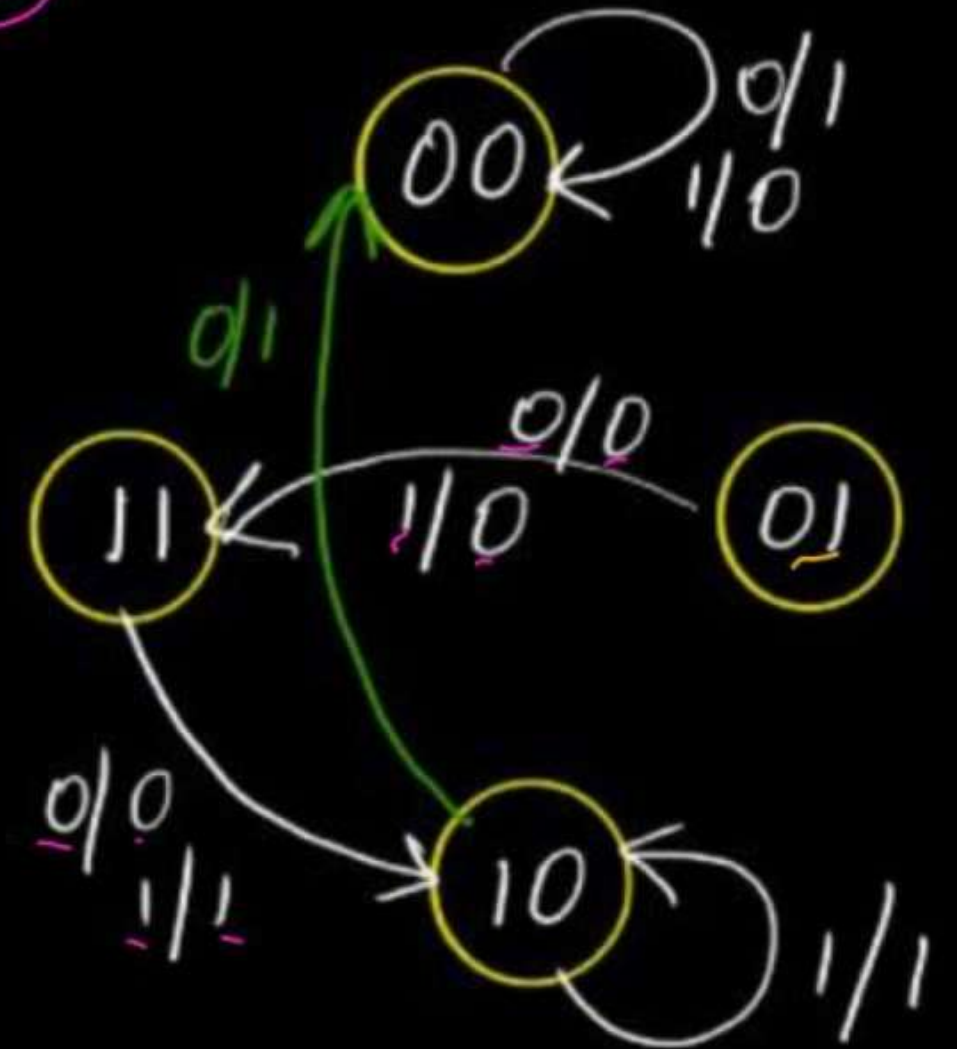
state Table

<u>Present State</u>		<u>Input</u>	<u>Next state</u>		<u>Output</u>
Q_A	Q_B		D_A	D_B	
A	B	x			y
0	0	0	0	0	1
0	0	1	0	0	0
0	1	0	1	1	0
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	1
1	1	0	1	0	0
1	1	1	1	0	1

State Diagram



PS $\xrightarrow{I/p/O/p}$ NS



Latches & Flip flops

Latches and flip-flops are the basic elements for storing information.

One latch or flip-flop can store one bit of information.

→ 0 or 1

Explain the term storage elements.

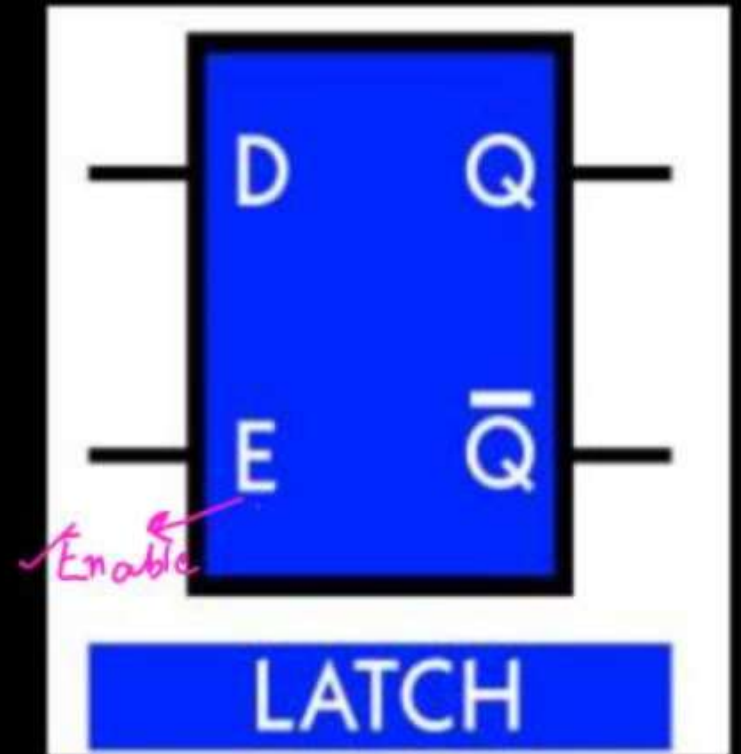
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Gateway Classes

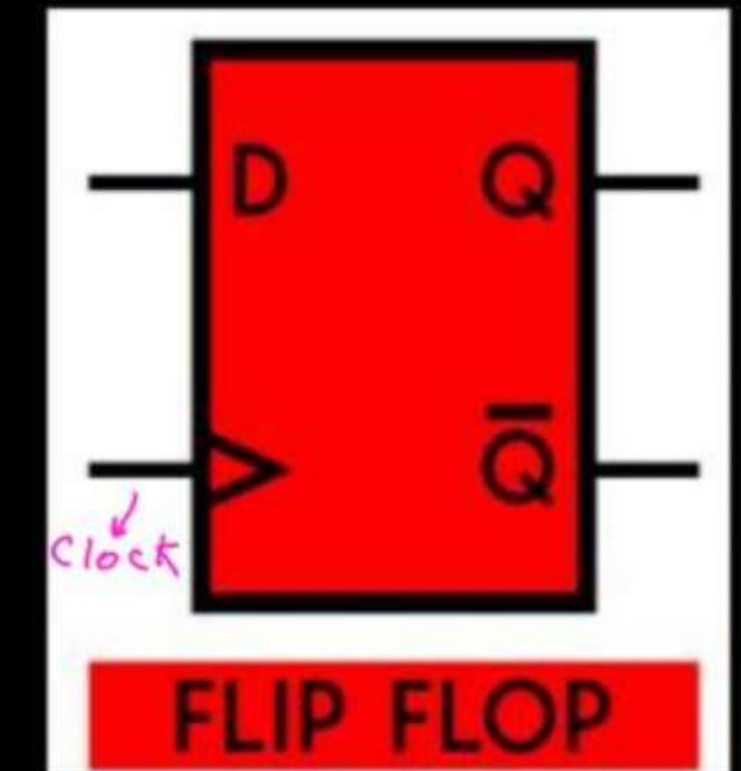
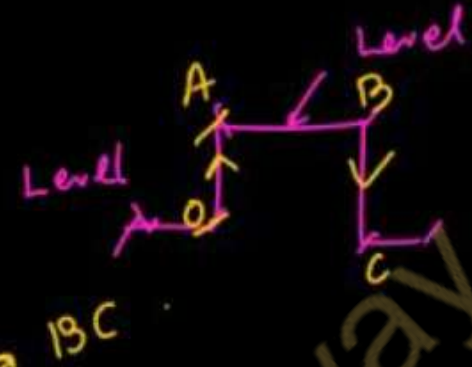
Latches & Flip flops

The main difference between latches and flip-flops is that :-

For latches, their outputs are constantly affected by their inputs as long as the enable signal is asserted. In other words, when they are enabled, their content changes immediately when their inputs change.



Flip-flops, on the other hand, have their content change only either at the rising or falling edge of the enable signal (clock signal). After the rising or falling edge of the clock, the flip-flop content remains constant even if the input changes.



Flip Flops

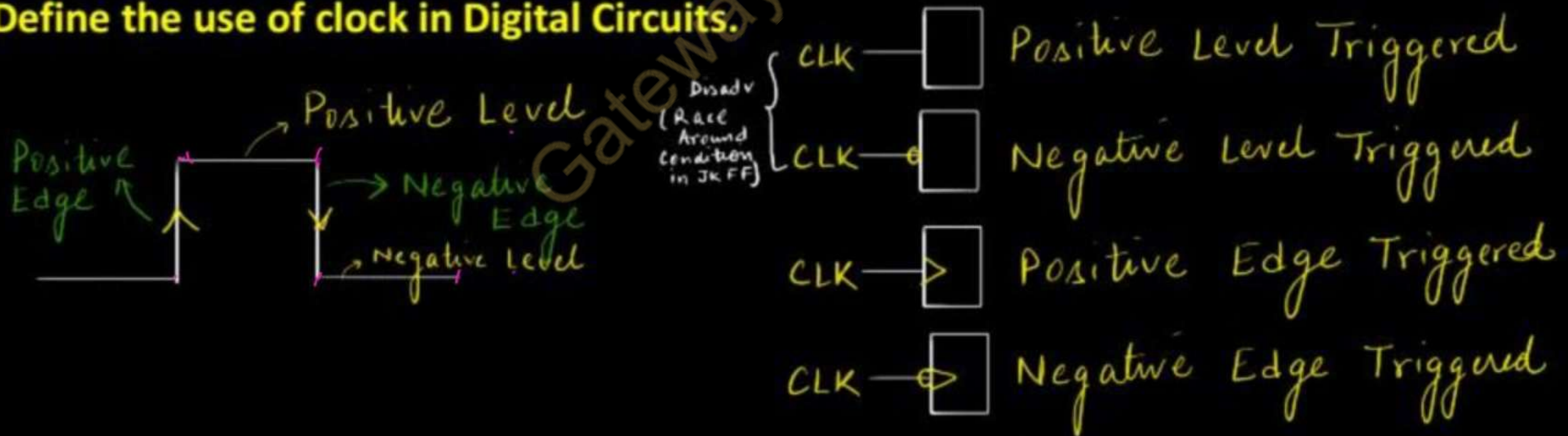
Flip-flop is a basic digital memory circuit, which stores one bit of information.

They are used as memory elements in sequential circuits.

Their content change only either at the rising or falling edge of the enable signal (**clock signal**).

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Define the use of clock in Digital Circuits.



Types of Flip flops

SR Flip Flop (Set-Reset Flipflop)

JK Flip Flop

D Flip Flop (Delay or Data Flipflop)

T Flip Flop (Toggle Flipflop)

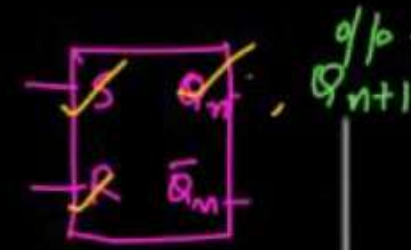
Gateway Classes

SR Flip Flop

It is a Flip Flop with two inputs, one is S and the other is R.

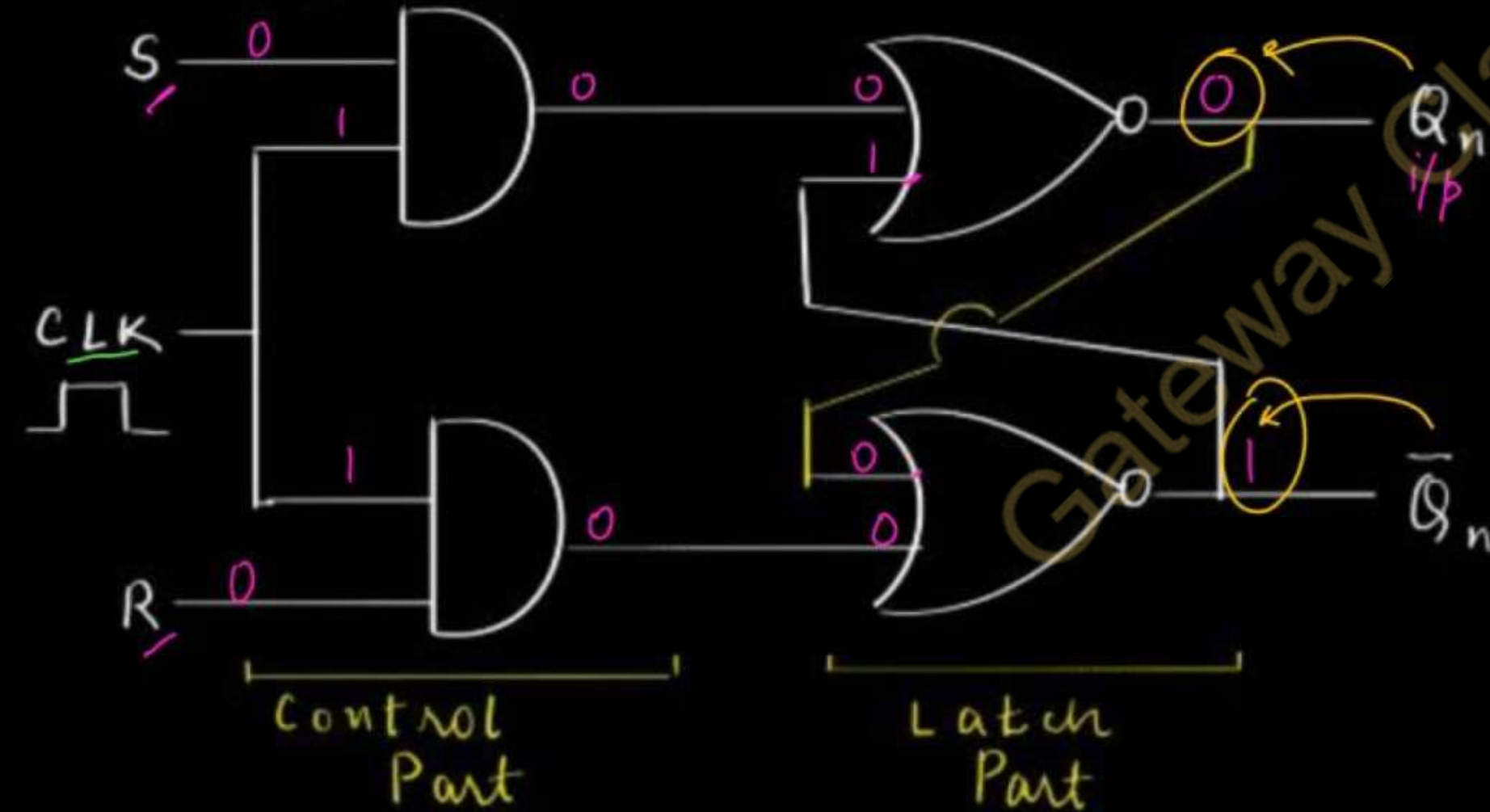
S here stands for **Set** and R here stands for **Reset**.

Set basically indicates set the flip flop which means output 1 and reset indicates resetting the flip flop which means output 0.



Q_n :- Input
 Q_{n+1} :- Output
 S & R :- Control Pins

Q_{n+1}
 o/p



Truth Table

3 inputs

Output

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

Q_n (later)

Reset
 (pin is high)

Set
 (pin is high)

Invalid

Q_n and $\bar{Q}_n$ are not coming complementary to each other

Truth Table

S	R	Q _n	Q _{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

K-map (for Q_{n+1})
for Characteristic Equation

SR	00	01	11	10
Q _n 0			X	1
Q _n 1	1		X	1

$$Q_{n+1} = S + \bar{R}Q_n$$



Elaborate the Characteristic equations of S-R Flip-Flop.

Functional Table
(Compressed Truth Table,
Q_n hide)

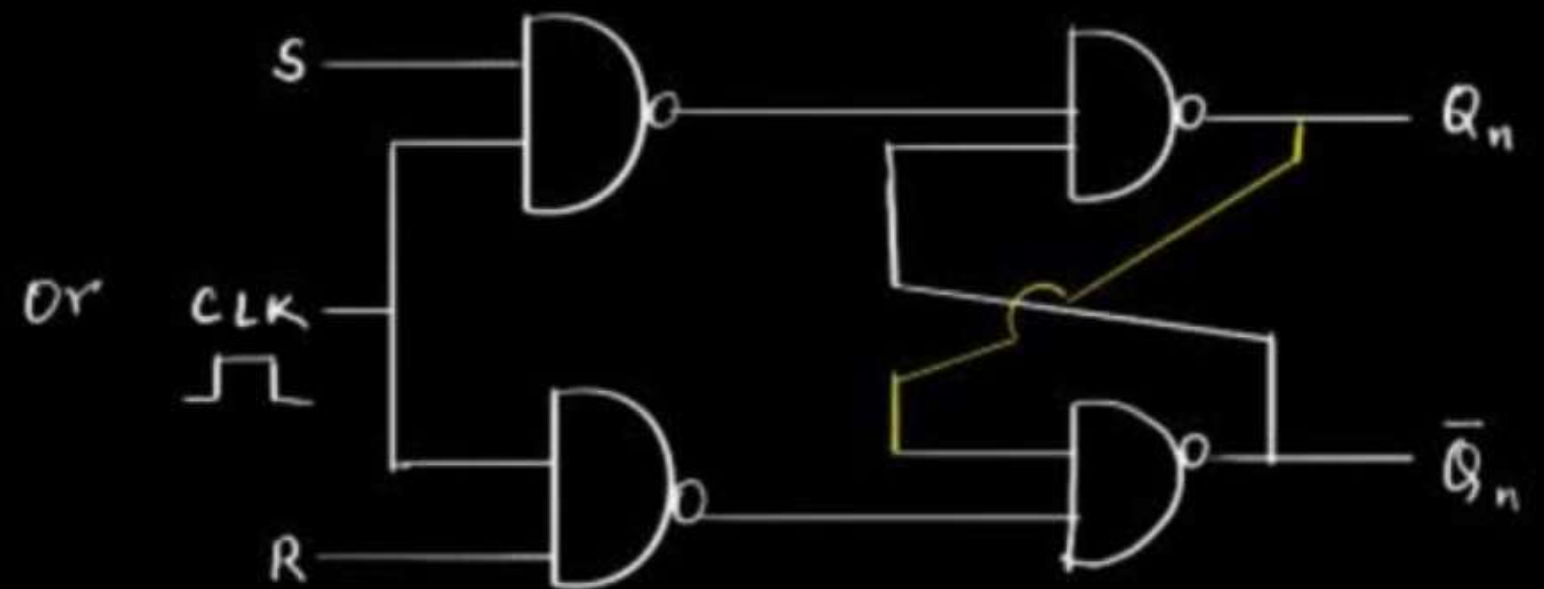
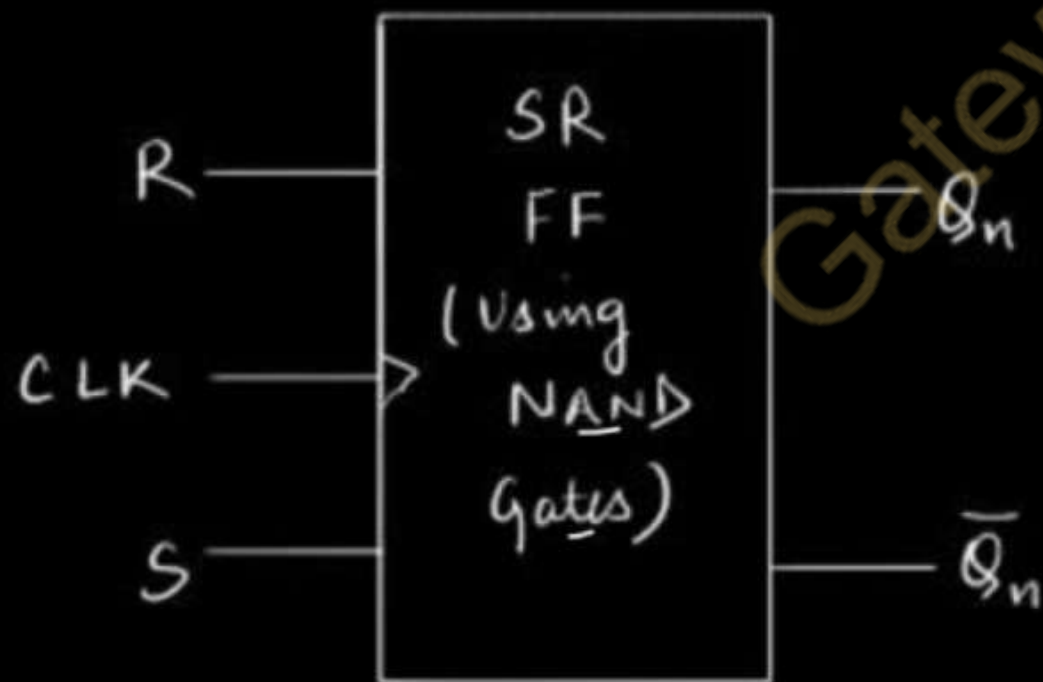
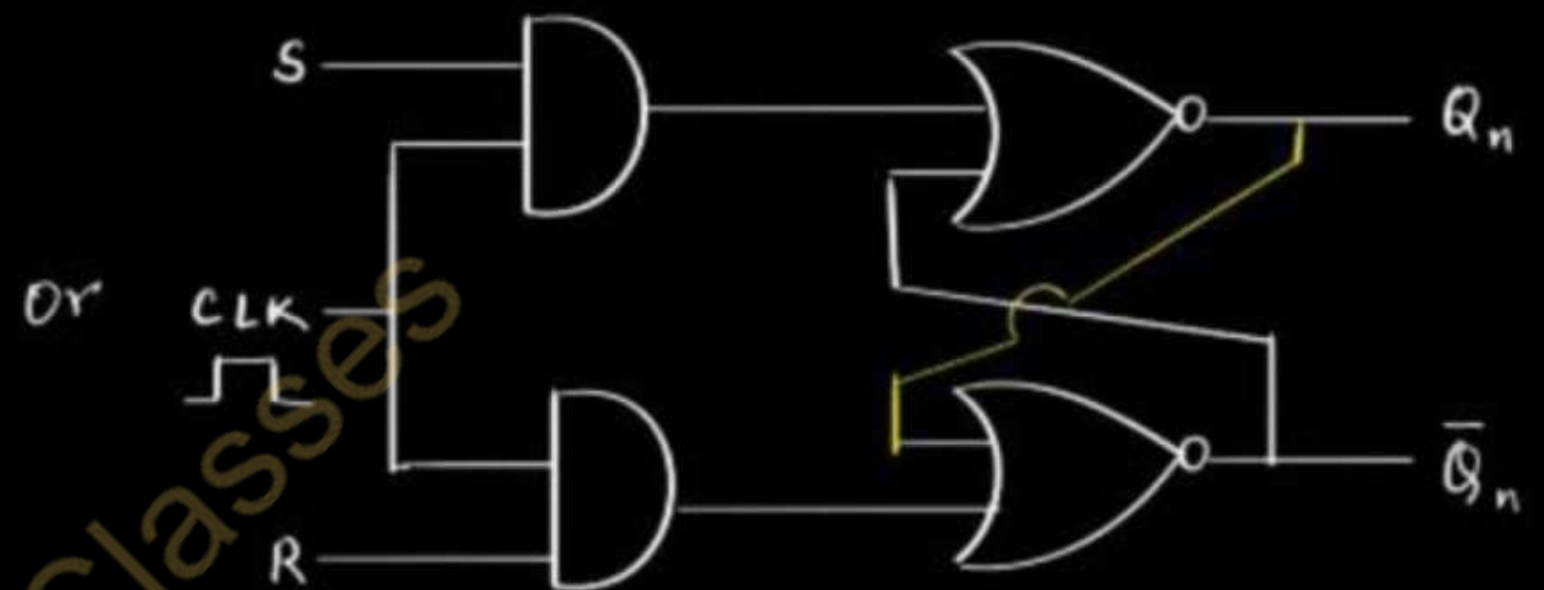
S	R	Q _{n+1}
0	0	Q _n
0	1	0
1	0	1
1	1	X

Excitation Table
(Q_n Q_{n+1} S R)
Used in FF conversion

Q _n	Q _{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

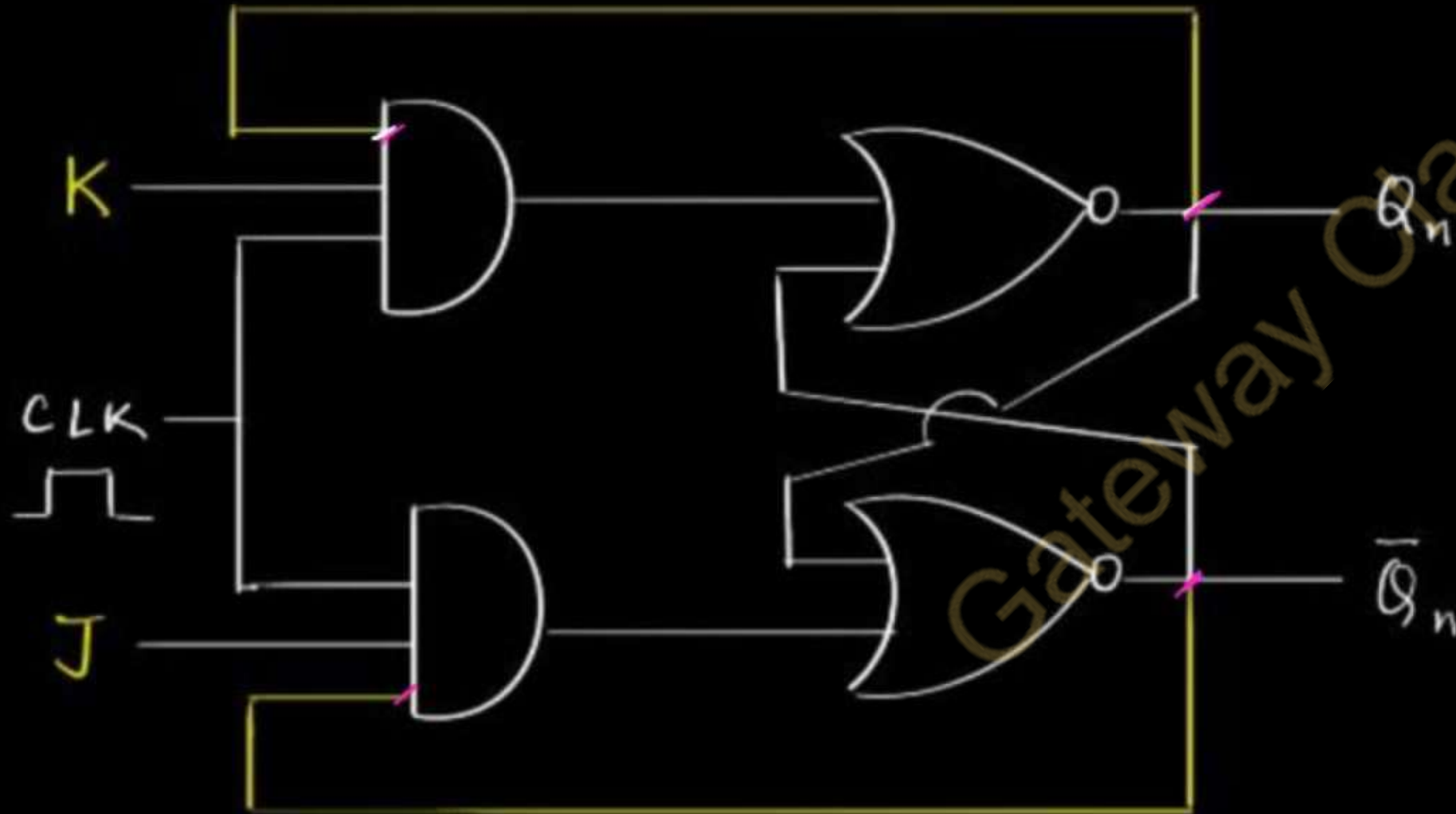
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Basic Diagram



JK Flip Flop

SR Flipflop had invalid states (at $S=1$ & $R=1$), which was overcome by JK Flipflop.



Truth Table

J	K	Qn	Qn+1
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

SR
JK
Master-slave
JK.

Truth Table

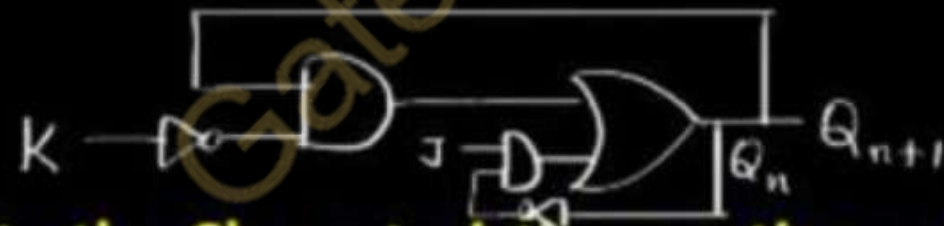
J	K	Q _n	Q _{n+1}
<u>0</u>	<u>0</u>	<u>0</u>	<u>0</u>
0	0	1	1
<u>0</u>	<u>1</u>	<u>0</u>	<u>0</u>
0	1	1	0
1	0	<u>0</u>	<u>1</u>
1	0	1	<u>1</u>
1	1	<u>0</u>	<u>1</u>
1	1	1	<u>0</u>

K-map ^{for Q_{n+1}}
for Characteristic Equation

J \ K	00	01	11	10
0			1	1
1	1			1

Annotations: $J\bar{Q}_n$ (pointing to cell 11), $\bar{K}Q_n$ (pointing to cell 10)

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$



Functional Table
(Compressed Truth Table,
Q_n hide)

J	K	Q _{n+1}
0	0	Q _n
0	1	0
1	0	1
1	1	Q _n '

Annotation: $\bar{Q}_n$ (pointing to Q_n' in the last row)

Excitation Table
(Q_n Q_{n+1} J K)

Used in FF conversion

Q _n	Q _{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Annotation: 0, 1 (pointing to the first column)

Elaborate the Characteristic equations of J-K Flip-Flop.

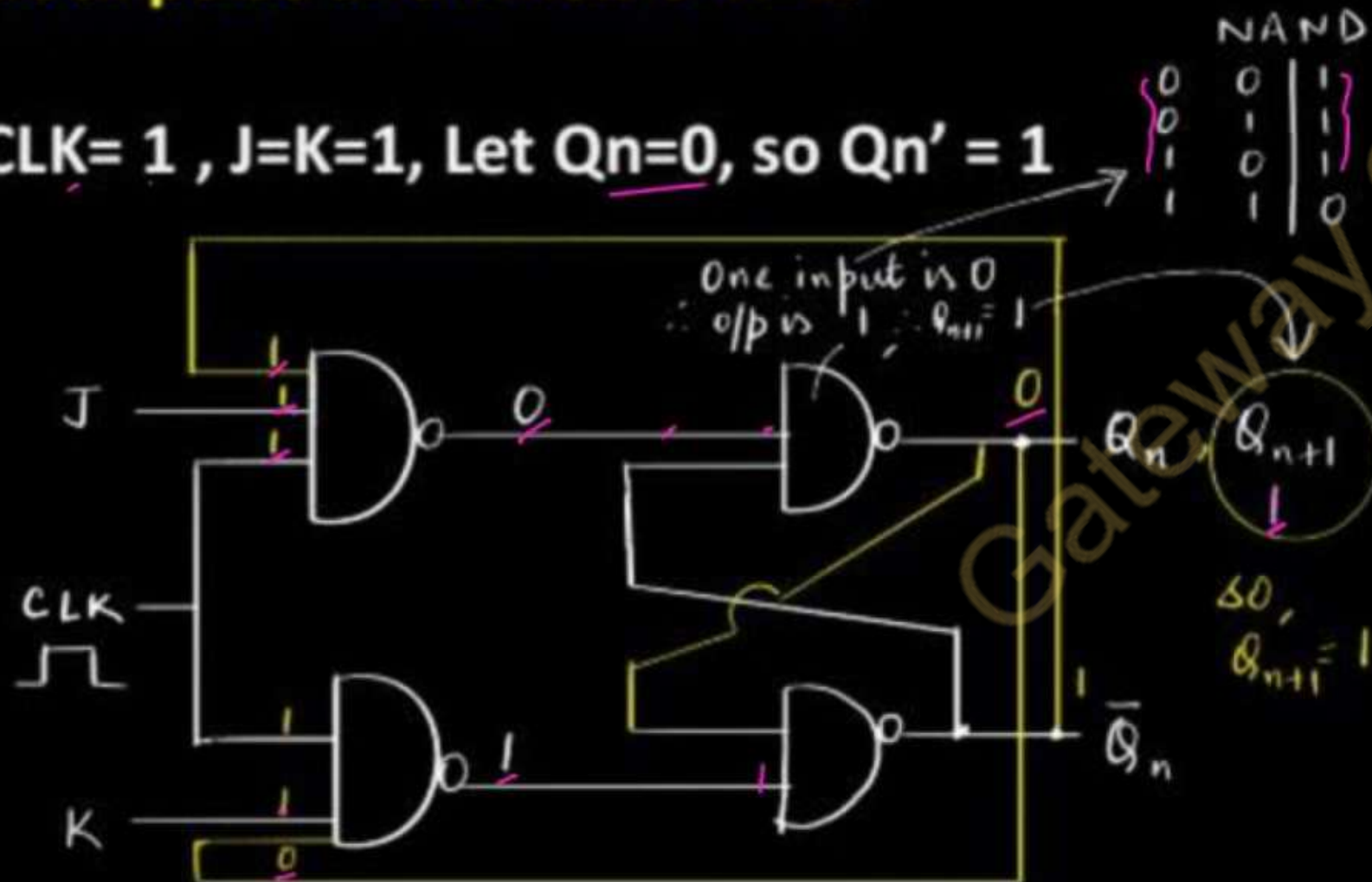
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RACE AROUND CONDITION

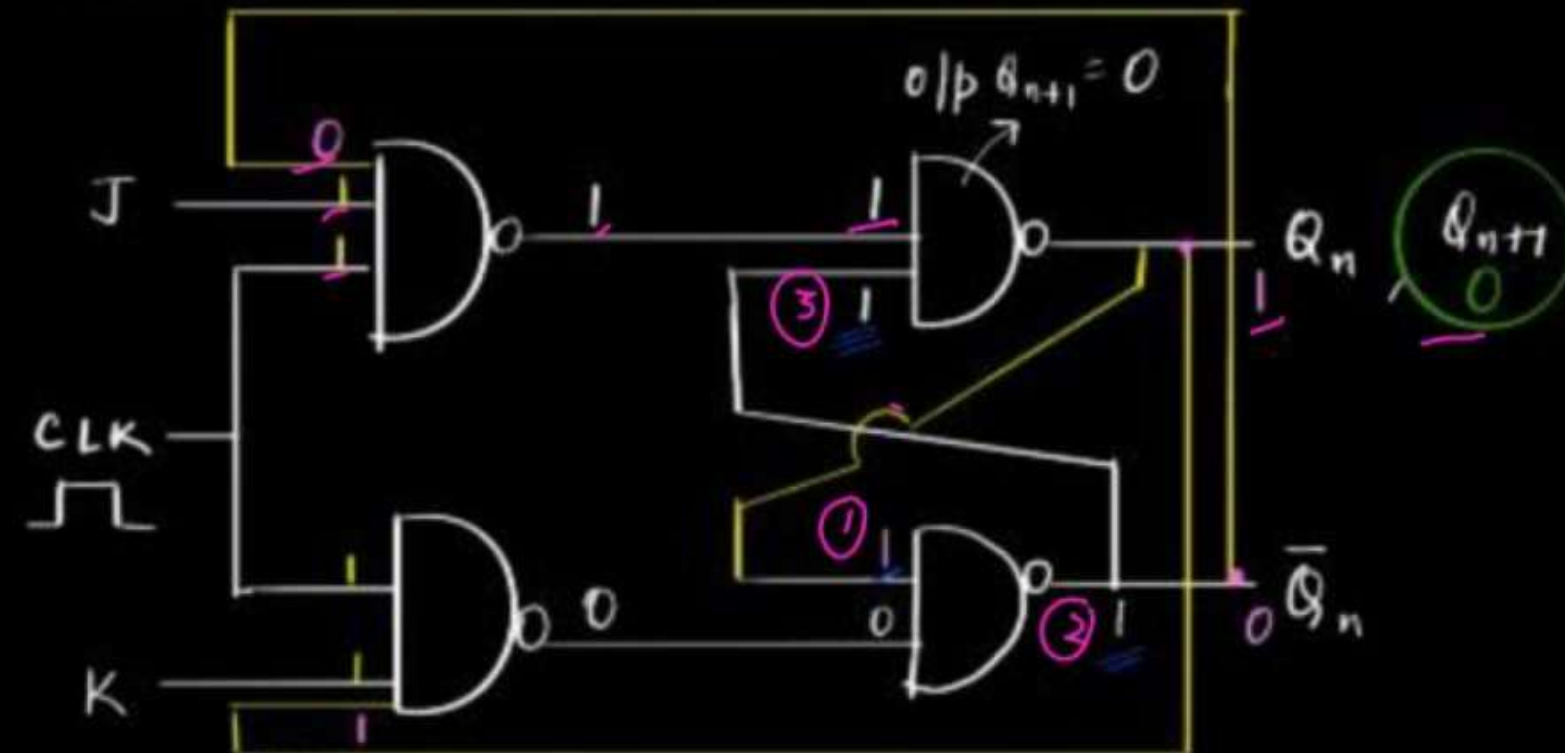
It is change of output without any change in input in a JK Flipflop when $J=1, K=1$.

The output oscillates (is not certain), sometimes it is 1, sometimes it is 0, for same clock pulse and when $J=K=1$.

CLK= 1 , J=K=1, Let $Q_n=0$, so $Q_n' = 1$



Now, CLK is still 1 so its duration is much more, so this JK FF still works. ($Q_{n+1} = 1$ is feedback so now $Q_n = 1$, $\bar{Q}_n = 0$)



A race occurs between 0 and 1
 Q_{n+1} $\bar{Q}_{n+1}$

Q_{n+1} fluctuates (oscillates), for some time it is 0, after some time it is 1
for same input $J=K=1$ and for same clock.

This is called **Race around condition**.

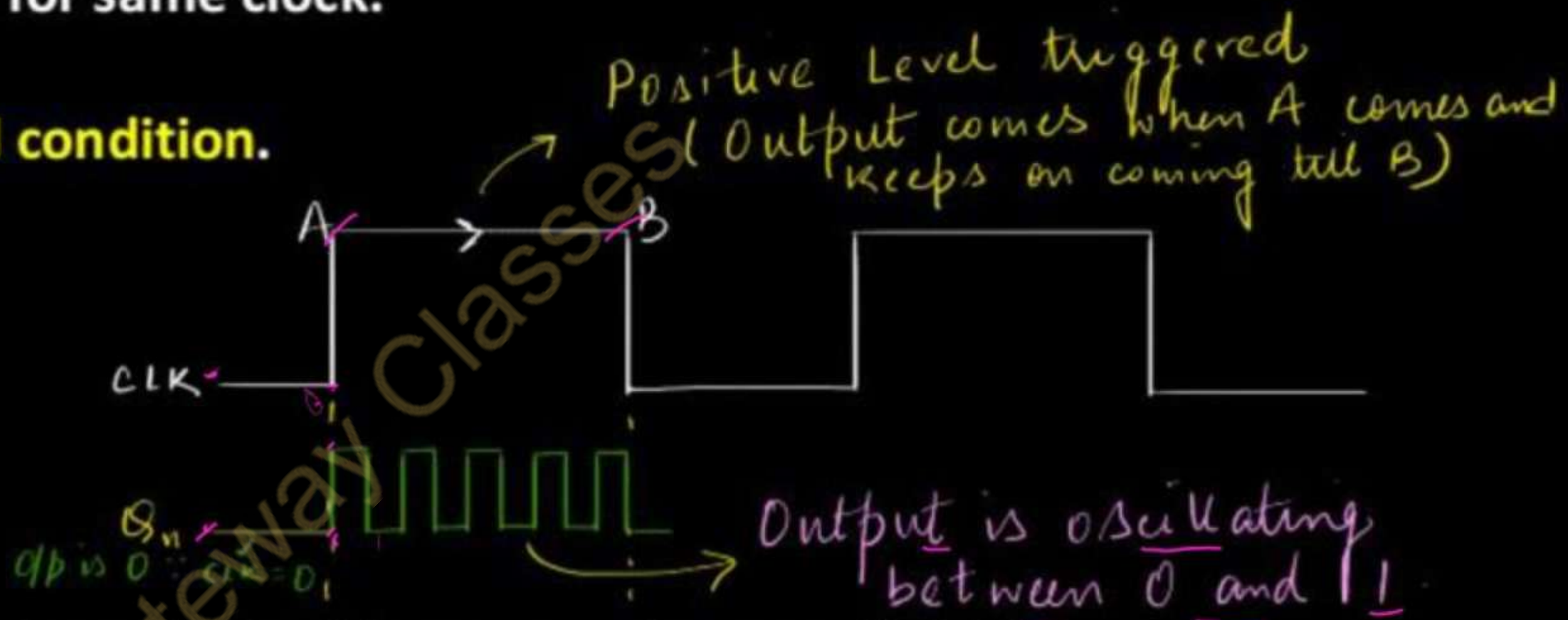
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It is due to

Long duration of Clock

And

Feedback



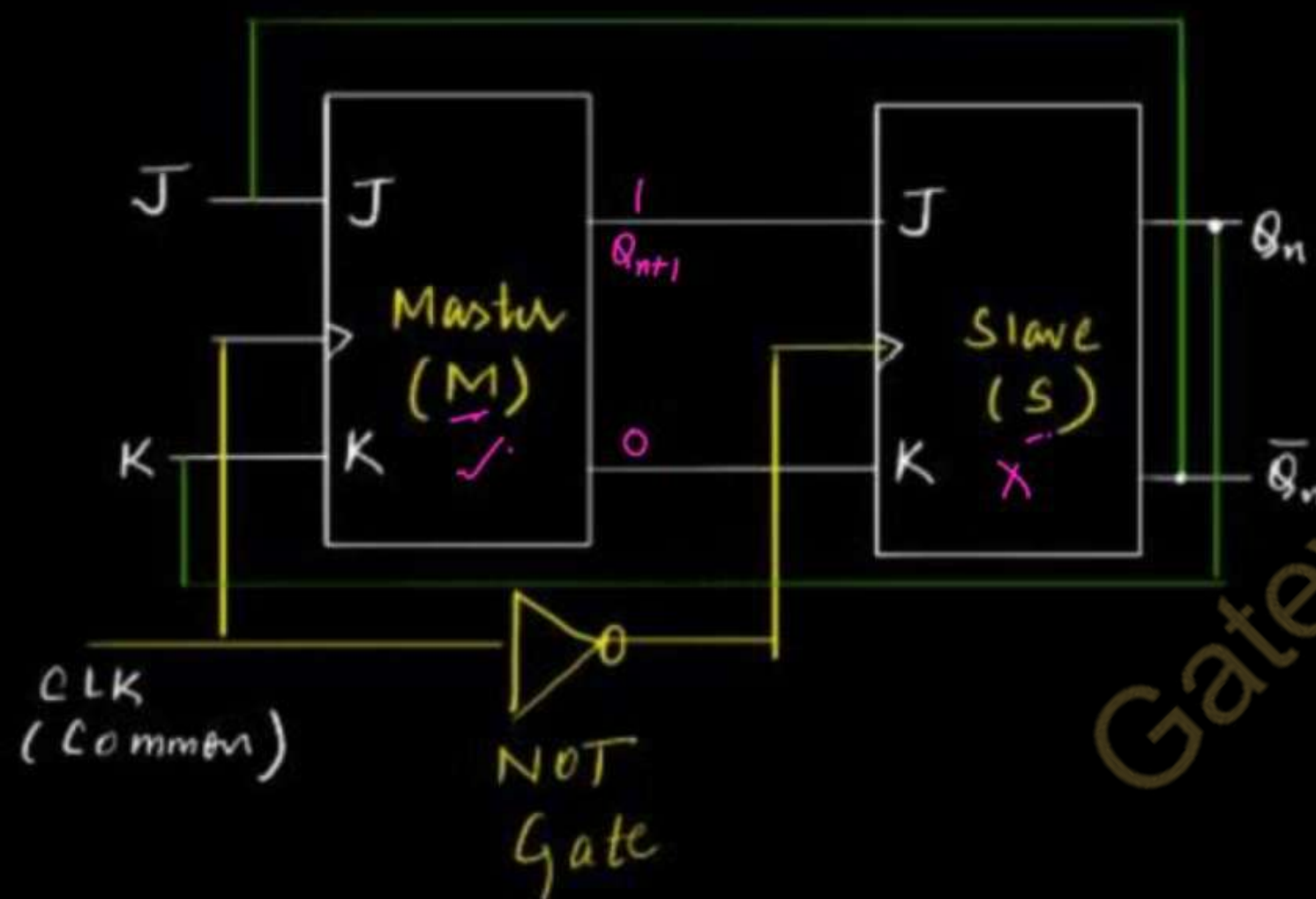
It should be avoided since input is fixed but output is unpredictable.

Therefore **Master-Slave JK Flipflop** is used.

↓
1 Master and 1 Slave
(Output comes only once between A and B)

Master Slave JK Flip Flop

is used to avoid Race-around condition.



Step 1

CLK = 1 , M active , S is disabled (off or hold) stores previous value

J=K=1 ,

Let $Q_n = 0$,
(previous state)

so $Q_n' = 1$

Truth Table

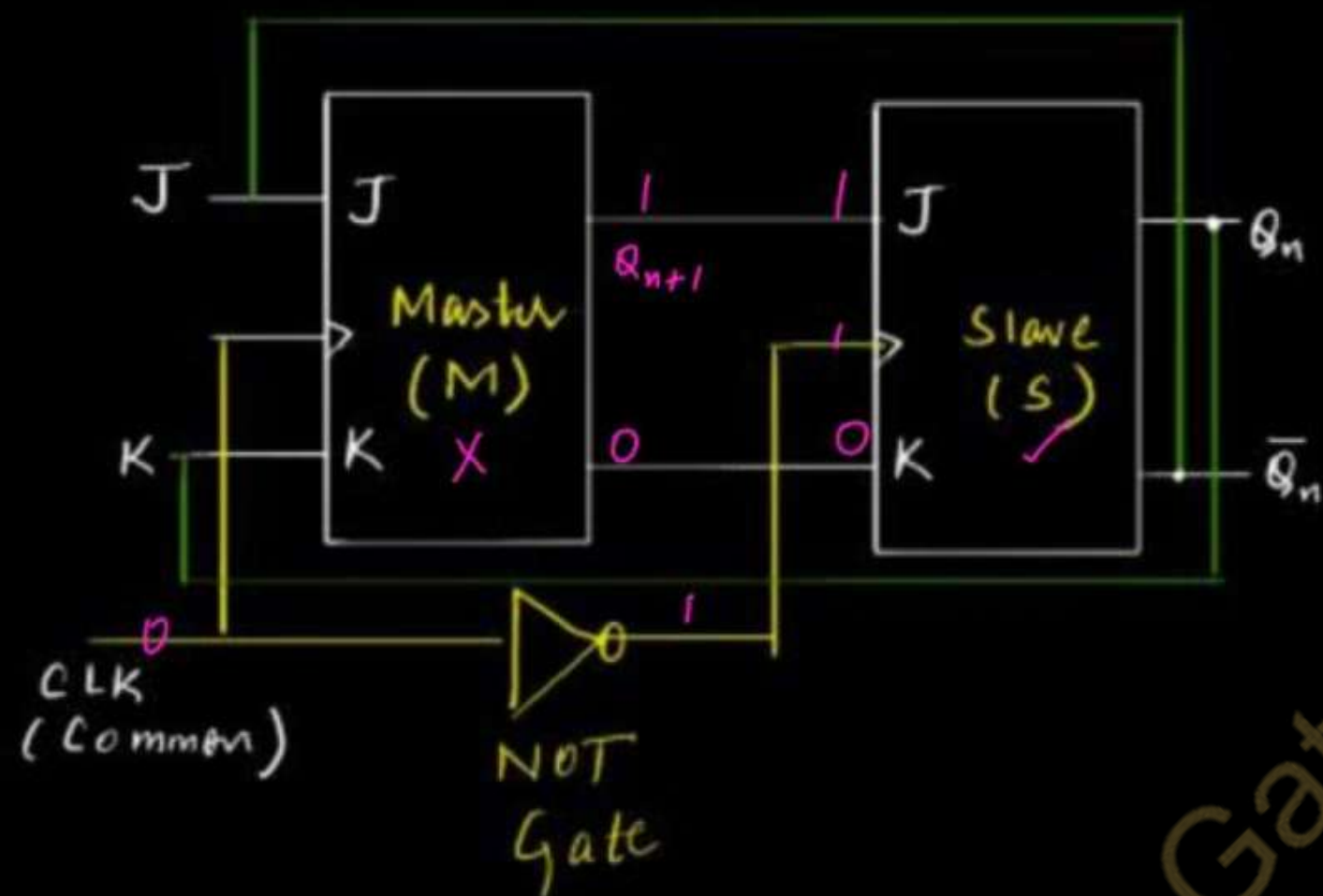
(For M)

J	K	Q_n	Q_{n+1}
1	1	0	1
1	1	1	0

→ now this is J
for S -
J=1 for S
K=0 for S (complementary)

Output of M = 1, But S is OFF ,so no feedback.

Master Slave JK Flip Flop



Step 2

CLK = 0 , M is disabled (off or hold) , S is active.

For S , J=1 , K=0 (from Step 1)

Truth Table

(For S)

J	K	Qn	Qn+1
<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>
<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>

Feedback works ,but M is off , so of no use.

Final Output is 1 .

So, now the output does not oscillates , it is fixed ,for inputs (J=K=1).

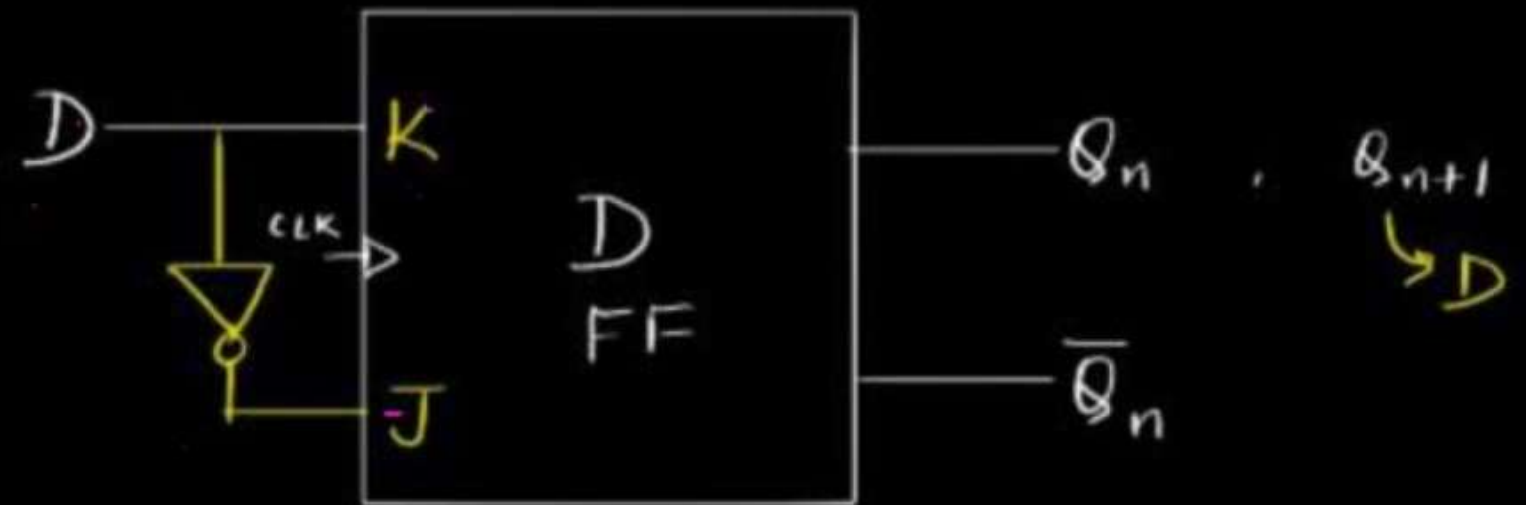
D Flip Flop

A D Flip-Flop (**Delay** or **Data Flip-Flop**) stores a single bit of data, transferring the input (D) to the output (Q) only on a clock edge, effectively delaying the data.

It has a data input (D), a clock input, and two outputs (Q and its inverse, Q-bar).



or



(D FF from JK FF)

Truth Table for JK FF

J	K	Q _n	Q _{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Truth Table for D FF

D	Q _n	Q _{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

K-map
For
Q_{n+1}

D \ Q _n	0	1
0		1
1		1

$$Q_{n+1} = D$$

This means
D FF is acting as
a Buffer to store data



Functional Table
(Compressed Truth Table,
Q_n hide)

D	Q _{n+1}
0	0
1	1

Excitation Table
(Q_n Q_{n+1} D)

Used in FF conversion

Q _n	Q _{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

→ We will consider these
4 cases for D FF : only in these 4 cases
J and K are complementary to each other.
(J=0, K=1 and J=1, K=0)

T Flip Flop

T flip flop is known as **Toggle Flip Flop** because it is able to toggle its output depending upon on the input.

Toggle basically indicates that the bit will be flipped i.e., either from 1 to 0 or from 0 to 1.



Truth Table for $\underline{T}$ FF

$\underline{T}$	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

K-map

for Q_{n+1}

T	0	1
0		1
1	1	

$$Q_{n+1} = T\bar{Q}_n + \bar{T}Q_n$$

$$= T \oplus Q_n$$

Functional Table
(Compressed Truth Table,
 Q_n hide)

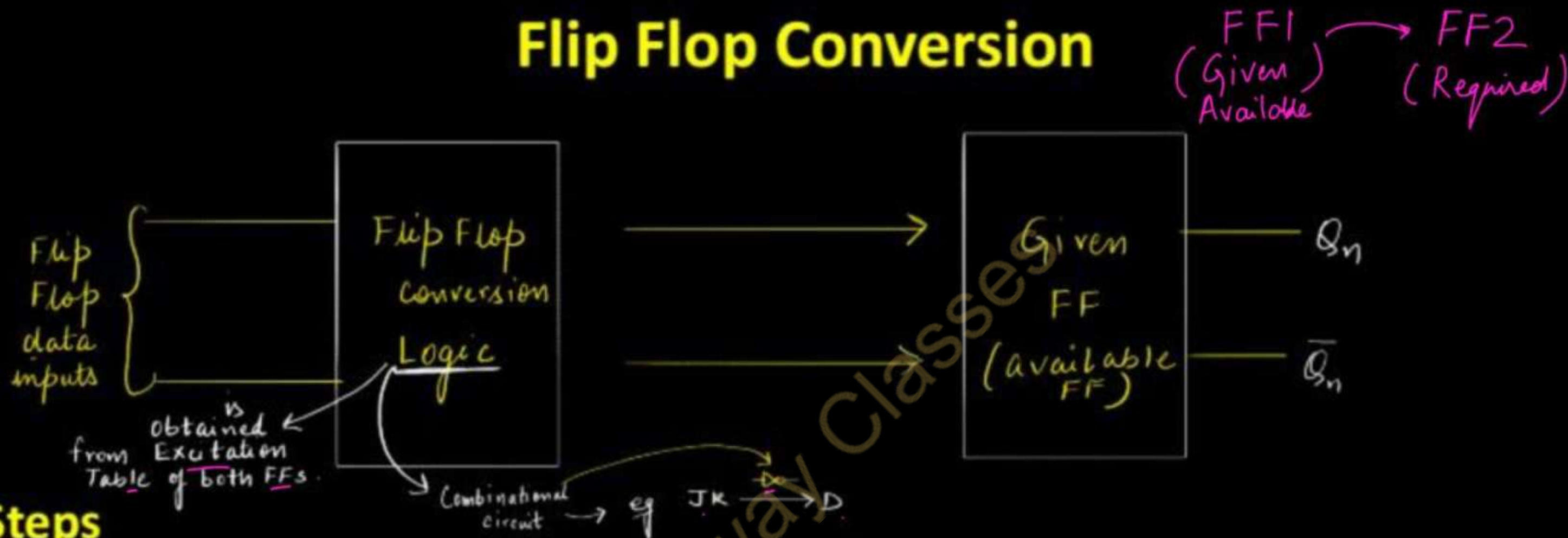
$\underline{T}$	Q_{n+1}
0	Q_n
1	Q_n'

Excitation Table
($Q_n Q_{n+1} T$)
Used in FF conversion

Q_n	Q_{n+1}	$\underline{T}$
0	0	0
0	1	1
1	0	1
1	1	0



Flip Flop Conversion



Steps

1. Characteristic (Truth Table) for Required FF.
- ✓ 2. Excitation Table for Available FF.
3. Extend Table of Step 1 by adding from Step 2.
4. K-Map for available FF (Leave Column of Q_{n+1} .)
5. Logic Diagram

SR TO JK FLIPFLOP

Step 1 Step 2 Step 3

J	K	Q _n	Q _{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Q _n	Q _{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

J	K	Q _n	Q _{n+1}	S	R
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	X	0
1	1	0	1	1	0
1	1	1	0	0	1

Steps

- 1.Characteristic (Truth Table) for Required FF.
- 2.Excitation Table for Available FF.
- 3.Extend Table of Step 1 by adding from Step 2.
- 4.K-Map for available FF (Leave Column of Q_{n+1}).
- 5.Logic Diagram

J	K	Q _n	Q _{n+1}	S	R
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	X	0
1	1	0	1	1	0
1	1	1	0	0	1

Step 4

K Map for S

K Map for R

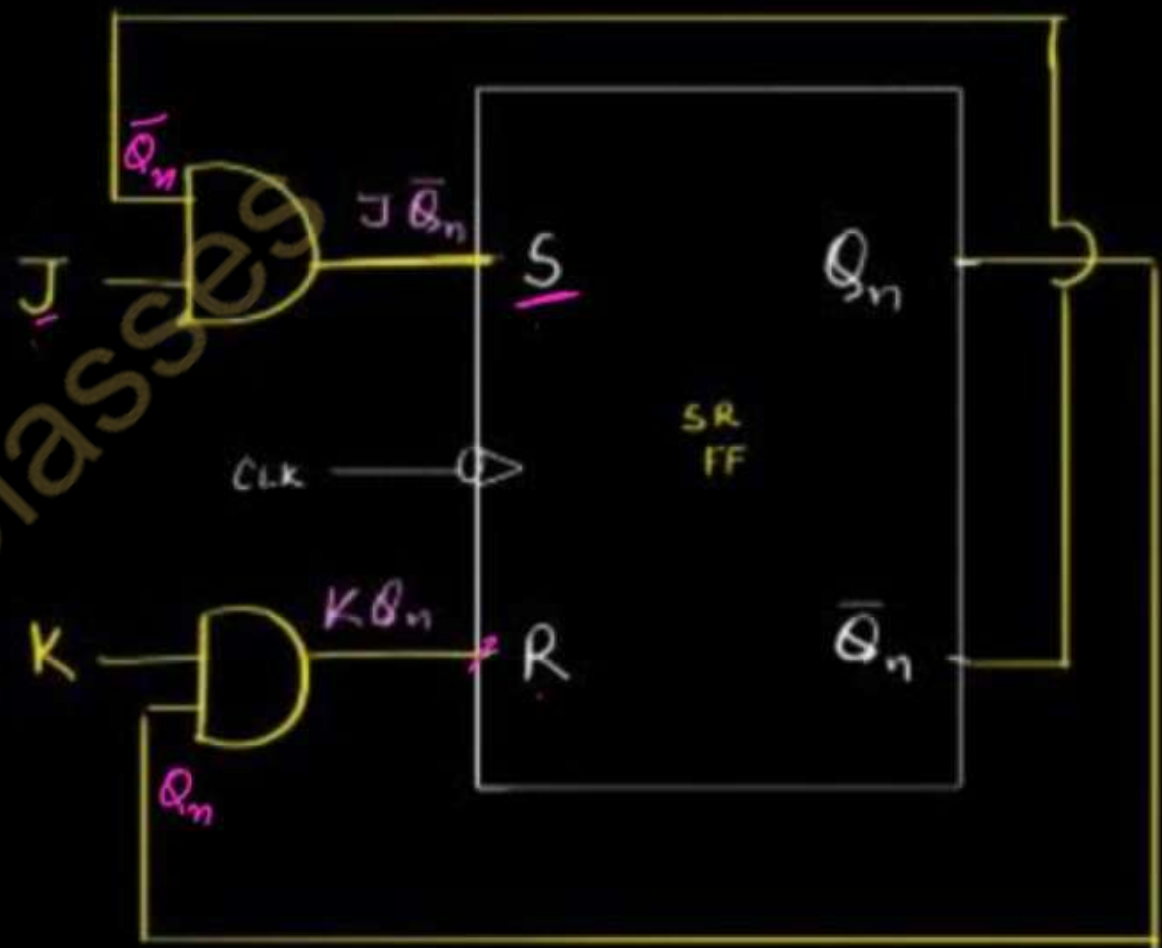
JK	00	01	11	10
Q _n 0			1	1
Q _n 1	X			X

$$S = J\bar{Q}_n$$

JK	00	01	11	10
Q _n 0	X	X		
Q _n 1		1	1	

$$R = KQ_n$$

imp
Q_{n+1} is not taken



SR TO D FLIPFLOP

Step 1

TT of D

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

Step 2

ET of SR

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Steps

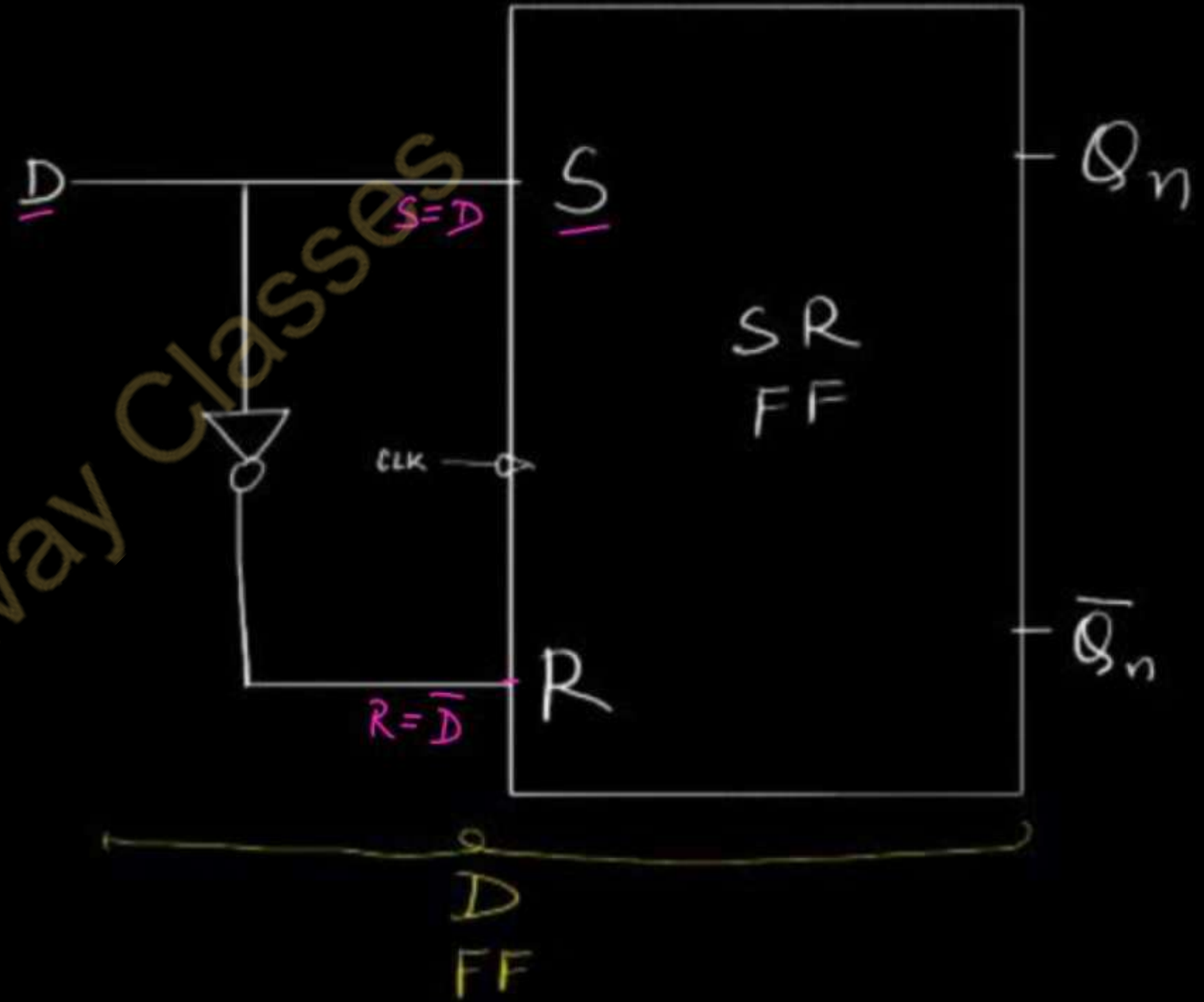
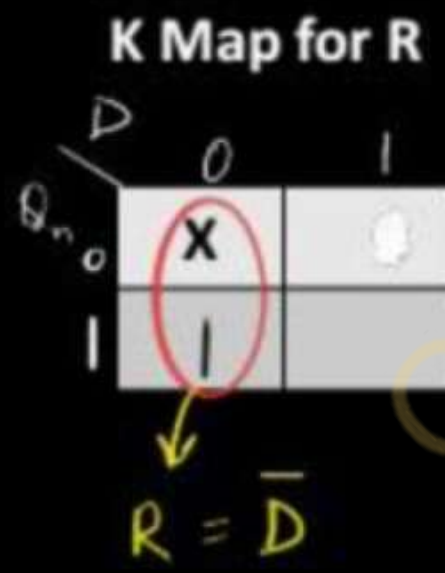
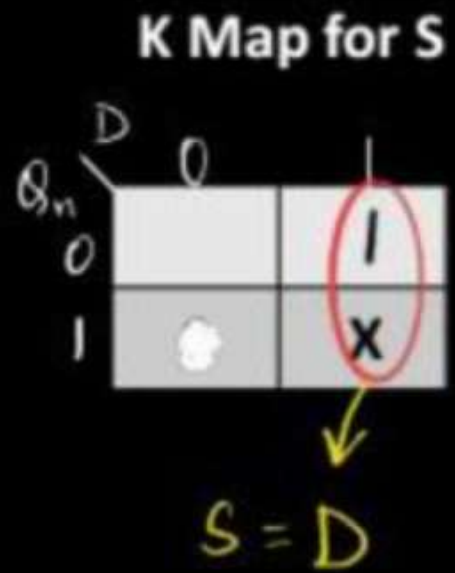
- 1.Characteristic (Truth Table) for Required FF.
- 2.Excitation Table for Available FF.
- 3.Extend Table of Step 1 by adding from Step 2.
- 4.K-Map for available FF (Leave Column of Q_{n+1}).
- 5.Logic Diagram

Step 3

D	Q_n	Q_{n+1}	S	R
0	0	0	0	X
0	1	0	0	1
1	0	1	1	0
1	1	1	X	0

D	Q _n	Q _{n+1} ^x	S	R
0	0	0	0	X
0	1	0	0	1
1	0	1	1	0
1	1	1	X	0

Step 4



SR TO T FLIPFLOP

Step 1

TT of T

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

Step 2 ET of SR.

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Steps

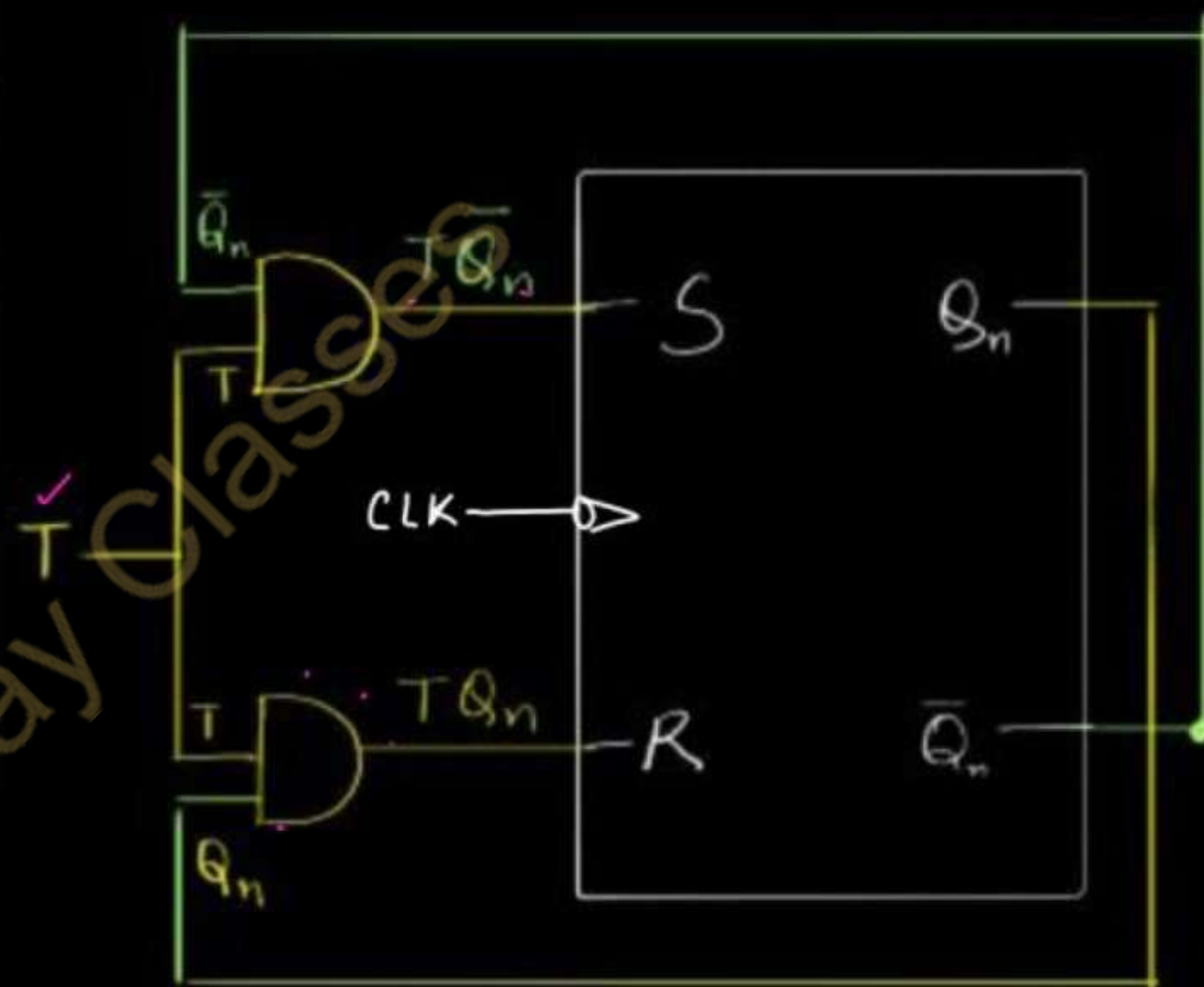
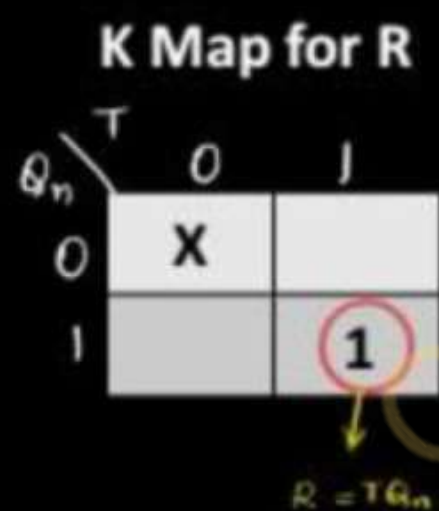
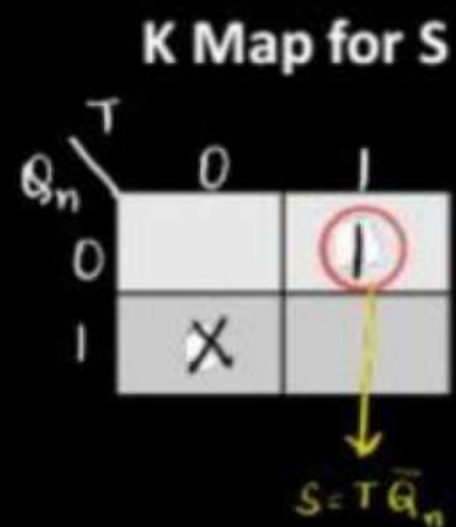
- 1.Characteristic (Truth Table) for Required FF.
- 2.Excitation Table for Available FF.
- 3.Extend Table of Step 1 by adding from Step 2.
- 4.K-Map for available FF (Leave Column of Q_{n+1}).
- 5.Logic Diagram

Step 3

T	Q_n	Q_{n+1}	S	R
0	0	0	0	X
0	1	1	X	0
1	0	1	1	0
1	1	0	0	1

T	Q _n	Q _{n+1} [*]	S	R
0	0	0	0	X
0	1	1	X	0
1	0	1	1	0
1	1	0	0	1

Step 4



JK TO SR FLIPFLOP

Step 1

TT of SR

S	R	Q _n	Q _{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

Step 2

ET of JK

Q _n	Q _{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Step 3

Steps

- 1.Characteristic (Truth Table) for Required FF.
- 2.Excitation Table for Available FF.
- 3.Extend Table of Step 1 by adding from Step 2.
- 4.K-Map for available FF (Leave Column of Q_{n+1}).
- 5.Logic Diagram

S	R	Q _n	Q _{n+1}	J	K
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	X	1
1	0	0	1	1	X
1	0	1	1	X	0
1	1	0	X	X	X
1	1	1	X	X	X

S	R	Q _n	Q _{n+1}	J	K
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	X	1
1	0	0	1	1	X
1	0	1	1	X	0
1	1	0	X	X	X
1	1	1	X	X	X

Step 4 K Map for J

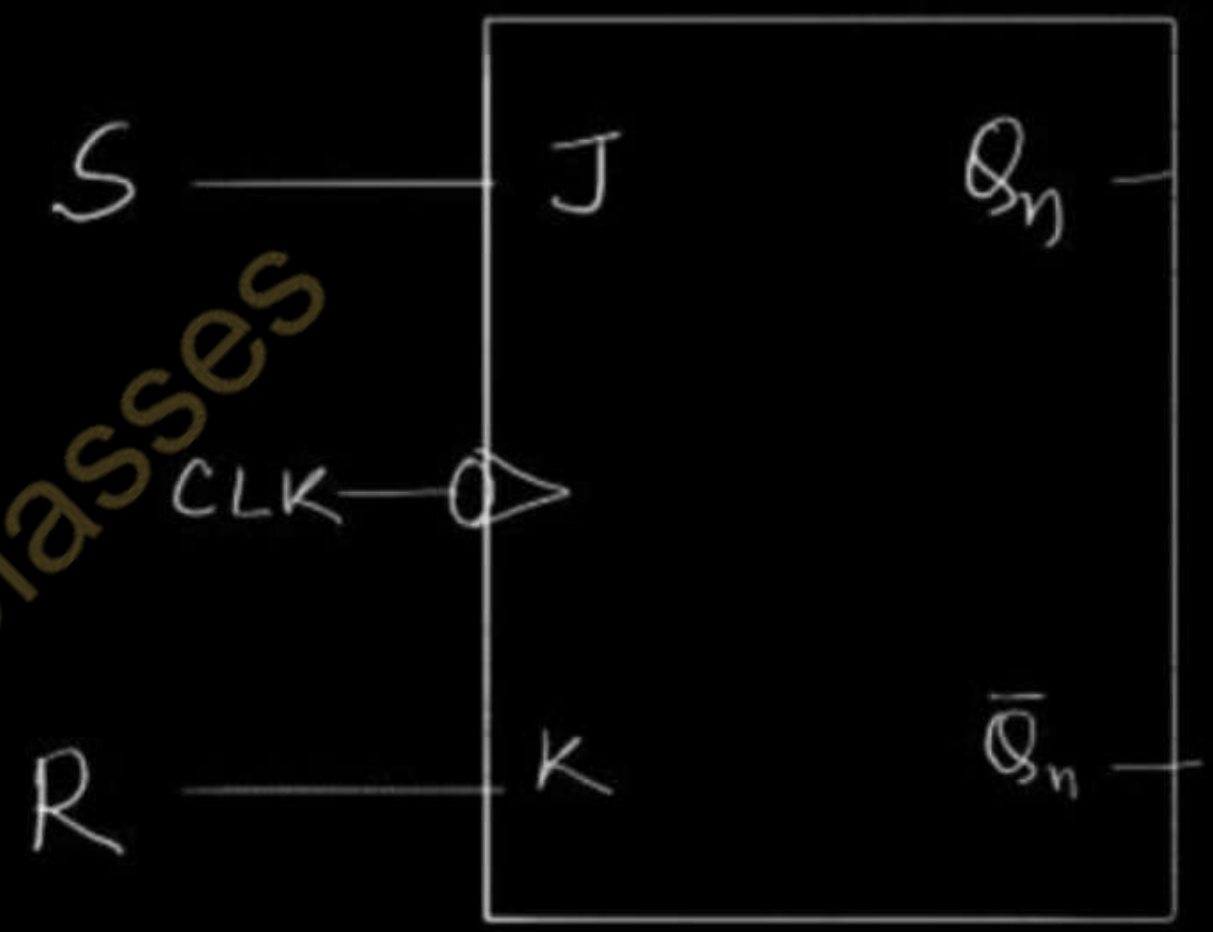
SR	00	01	11	10
Q _n 0			X	1
1	X	X	X	X

↓
S
J = S

K Map for K

SR	00	01	11	10
Q _n 0	X	X	X	X
1		1	X	

↓
R
K = R



JK TO D FLIPFLOP

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Steps

- 1.Characteristic (Truth Table) for Required FF.
- 2.Excitation Table for Available FF.
- 3.Extend Table of Step 1 by adding from Step 2.
- 4.K-Map for available FF (Leave Column of Q_{n+1}).
- 5.Logic Diagram

Step 1

TT of D

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

Step 2

ET of JK

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Step 3

D	Q_n	Q_{n+1}	J	K
0	0	0	0	X
0	1	0	X	1
1	0	1	1	X
1	1	1	X	0

D	Q_n	Q_{n+1}	J	K
0	0	0	0	X
0	1	0	X	1
1	0	1	1	X
1	1	1	X	0

Step 4

K Map for J

$Q_n \backslash D$	0	1
0		1
1	X	X

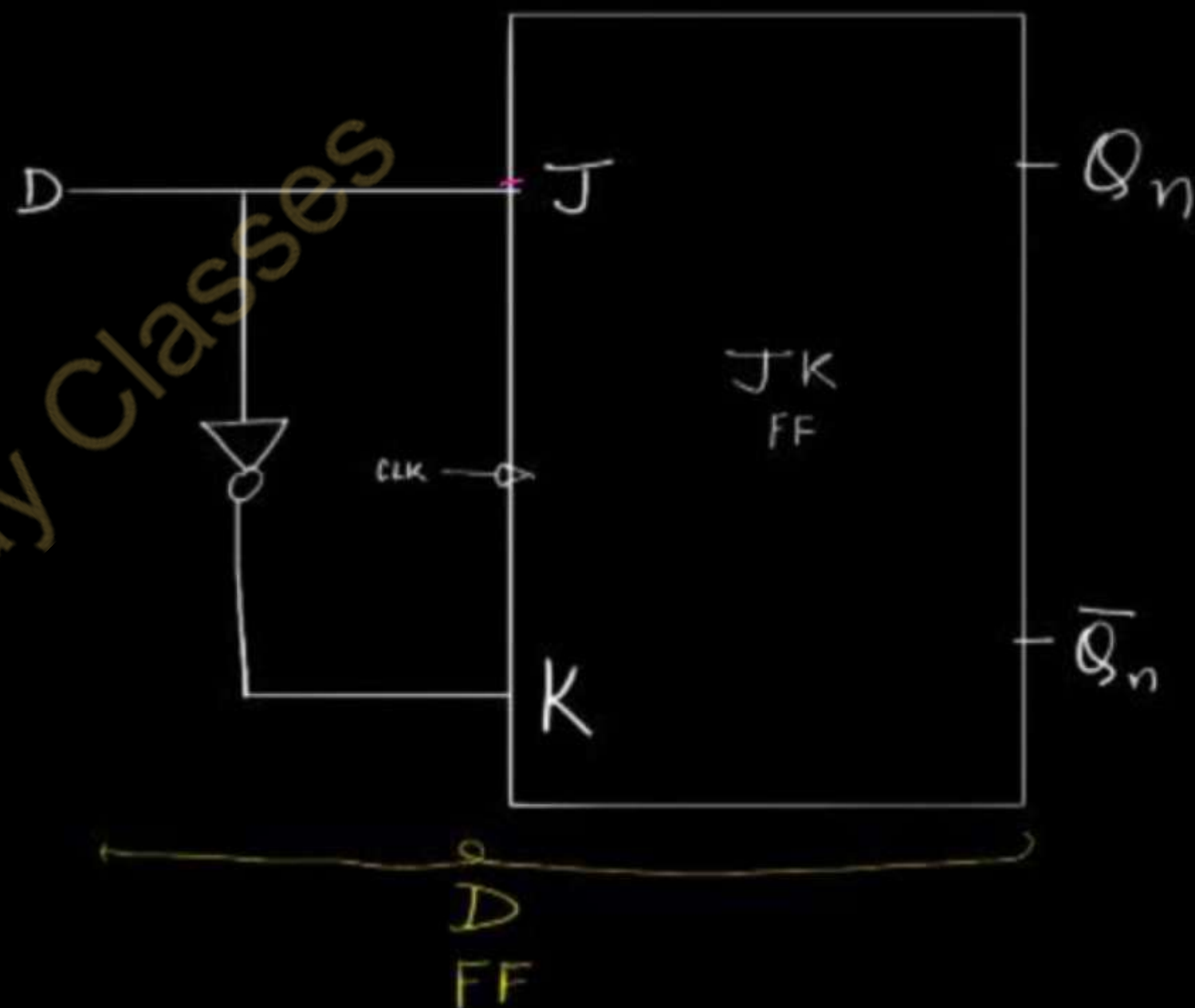
$J = D$

K Map for K

$Q_n \backslash D$	0	1
0	X	X
1	1	

$K = \bar{D}$

Step 5



JK TO T FLIPFLOP

Steps

- 1.Characteristic (Truth Table) for Required FF.
- 2.Excitation Table for Available FF.
- 3.Extend Table of Step 1 by adding from Step 2.
- 4.K-Map for available FF (Leave Column of Q_{n+1}).
- 5.Logic Diagram

Step 1

TT of T

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

Step 2 ET of JK

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Step 3

T	Q_n	Q_{n+1}	J	K
0	0	0	0	X
0	1	1	X	0
1	0	1	1	X
1	1	0	X	1

T	Q _n	Q _{n+1}	J	K
0	0	0	0	X
0	1	1	X	0
1	0	1	<u>1</u>	X
1	1	0	X	<u>1</u>

Step 4

K Map for J

	T	0	1
Q _n	0		1
	1	X	X

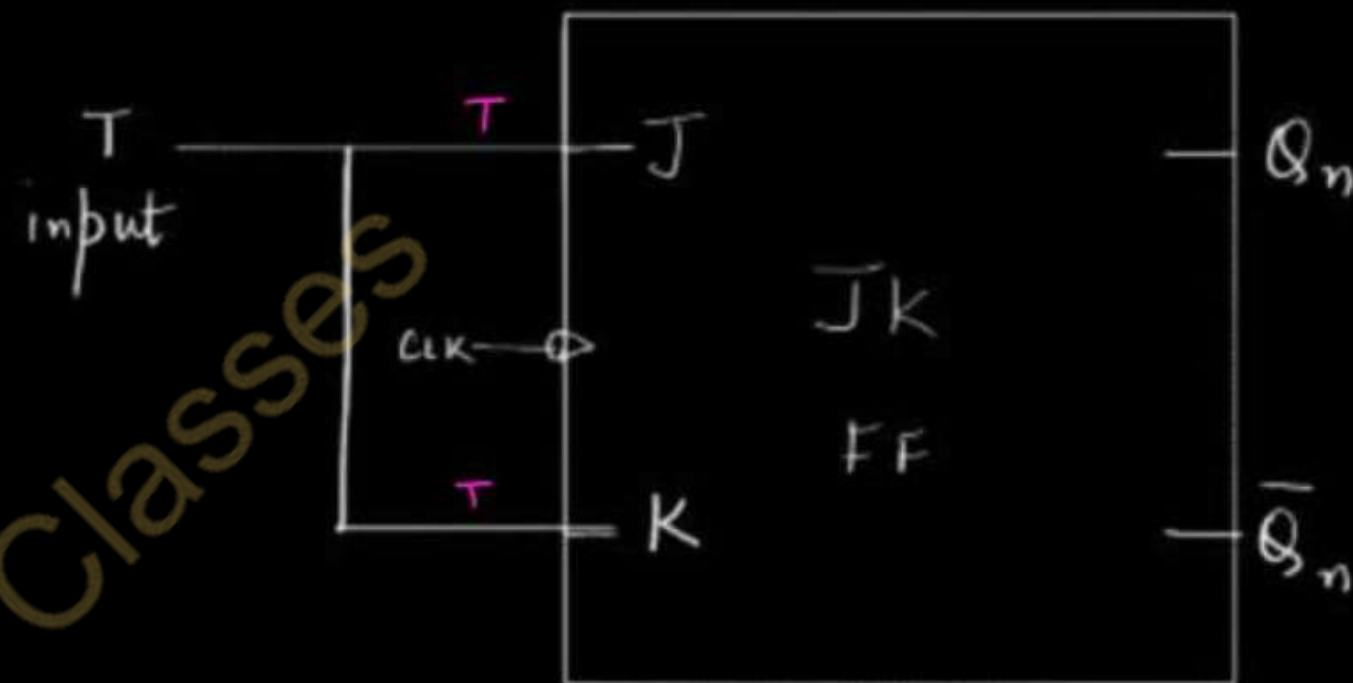
↓
J = T

K Map for K

	T	0	1
Q _n	0	X	X
	1		1

↓
K = T

Step 5



D TO SR FLIPFLOP

Step 1

S	R	Qn	Qn+1
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

Step 2

Qn	Qn+1	D
0	0	0
0	1	1
1	0	0
1	1	1

Steps

- 1.Characteristic (Truth Table) for Required FF.
- 2.Excitation Table for Available FF.
- 3.Extend Table of Step 1 by adding from Step 2.
- 4.K-Map for available FF (Leave Column of Qn+1.
- 5.Logic Diagram

Step 3

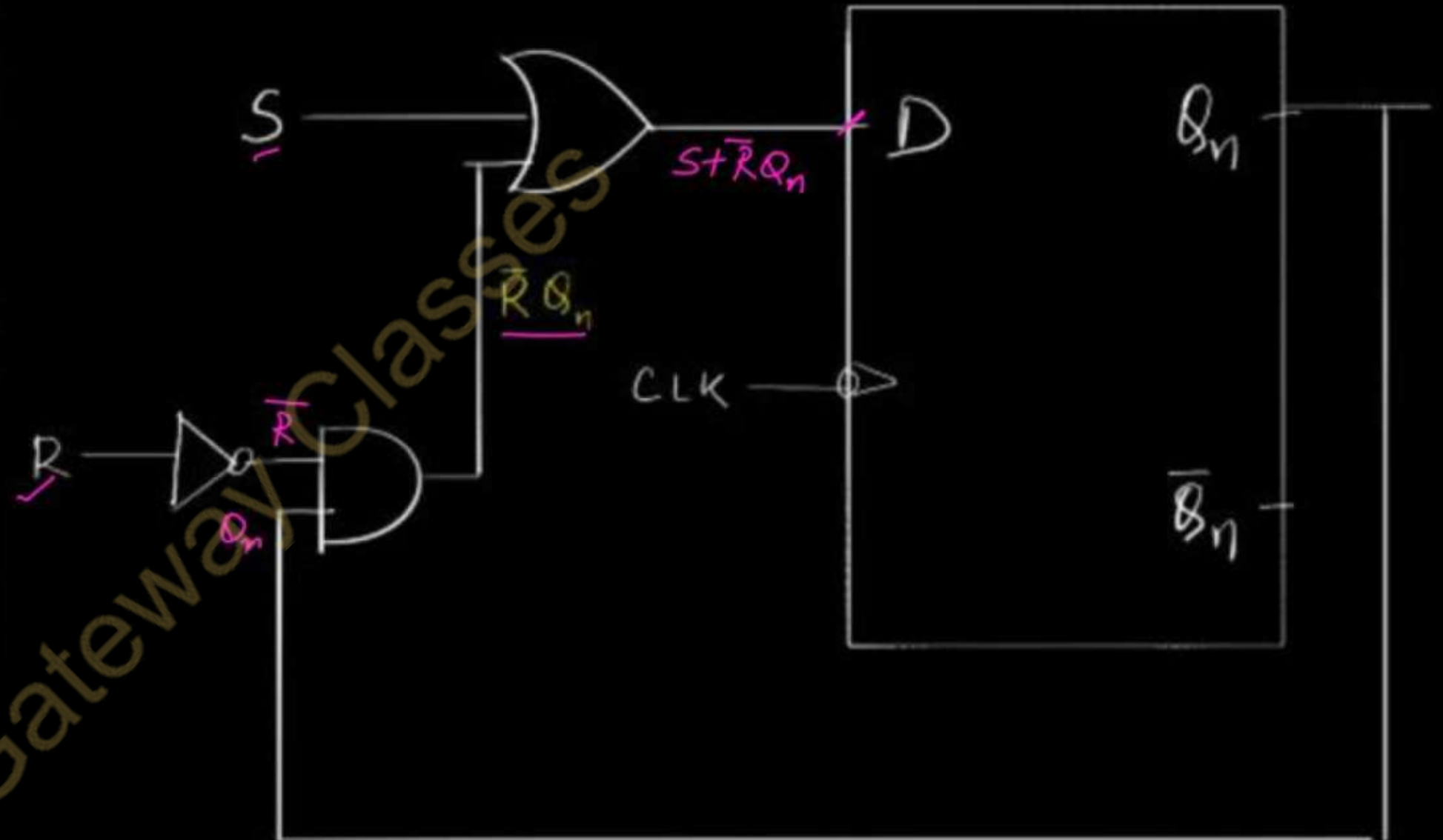
S	R	Qn	Qn+1	D
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	X	X
1	1	1	X	X

S	R	Q_n	Q_{n+1}	D
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	X	X
1	1	1	X	X

Step 4 K Map for D

$Q_n \backslash SR$	00	01	11	10
0			X	1
1	1		X	1

$\downarrow S$
 $\downarrow \bar{R}Q_n$

$$D = S + \bar{R}Q_n$$


D TO T FLIPFLOP

Step 1

T	Q _n	Q _{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

Step 2

Q _n	Q _{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Step 3

T	Q _n	Q _{n+1}	D
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

Steps

- 1.Characteristic (Truth Table) for Required FF.
- 2.Excitation Table for Available FF.
- 3.Extend Table of Step 1 by adding from Step 2.
- 4.K-Map for available FF (Leave Column of Q_{n+1}).
- 5.Logic Diagram

T	Q _n	Q _{n+1}	D
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

Step 4

K Map for D

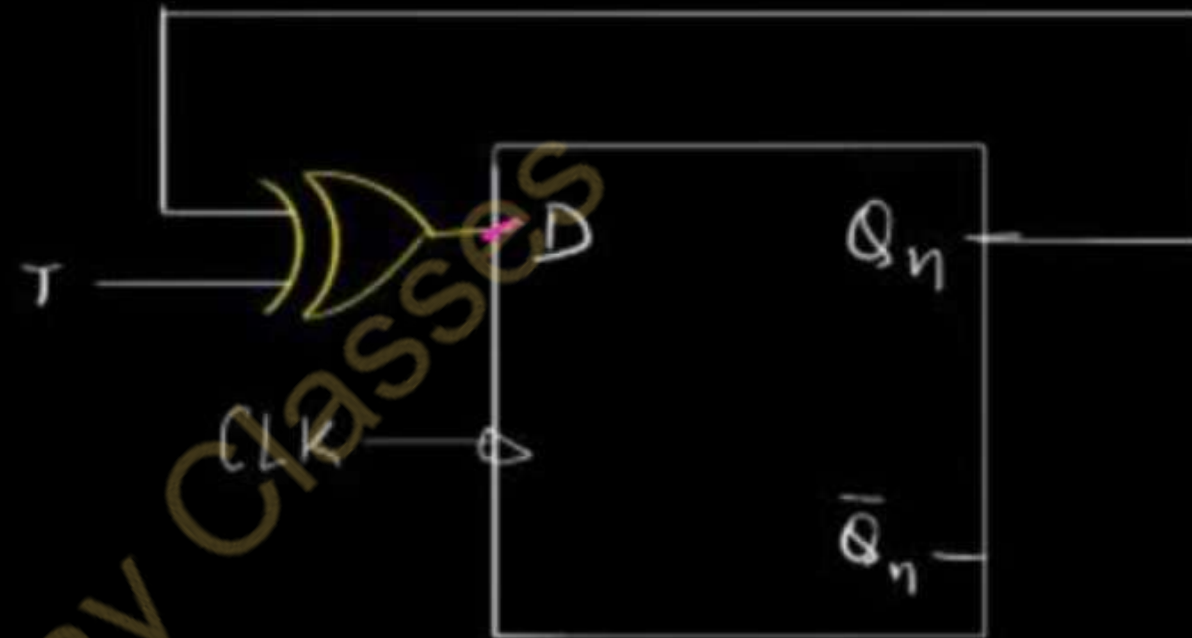
		T	
		0	1
Q _n	0		1
	1	1	

$\bar{T}Q_n$ $T\bar{Q}_n$

$$D = \bar{T}\bar{Q}_n + TQ_n$$

$$= T \oplus Q_n$$

Step 5



D TO JK FLIPFLOP

Step 1

TT of JK

J	K	Q _n	Q _{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Step 2

ET of D

Q _n	Q _{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Step 3

J	K	Q _n	Q _{n+1}	D
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

Steps

- 1.Characteristic (Truth Table) for Required FF.
- 2.Excitation Table for Available FF.
- 3.Extend Table of Step 1 by adding from Step 2.
- 4.K-Map for available FF (Leave Column of Q_{n+1}).
- 5.Logic Diagram

J	K	Q_n	Q_{n+1}	D
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

Step 4

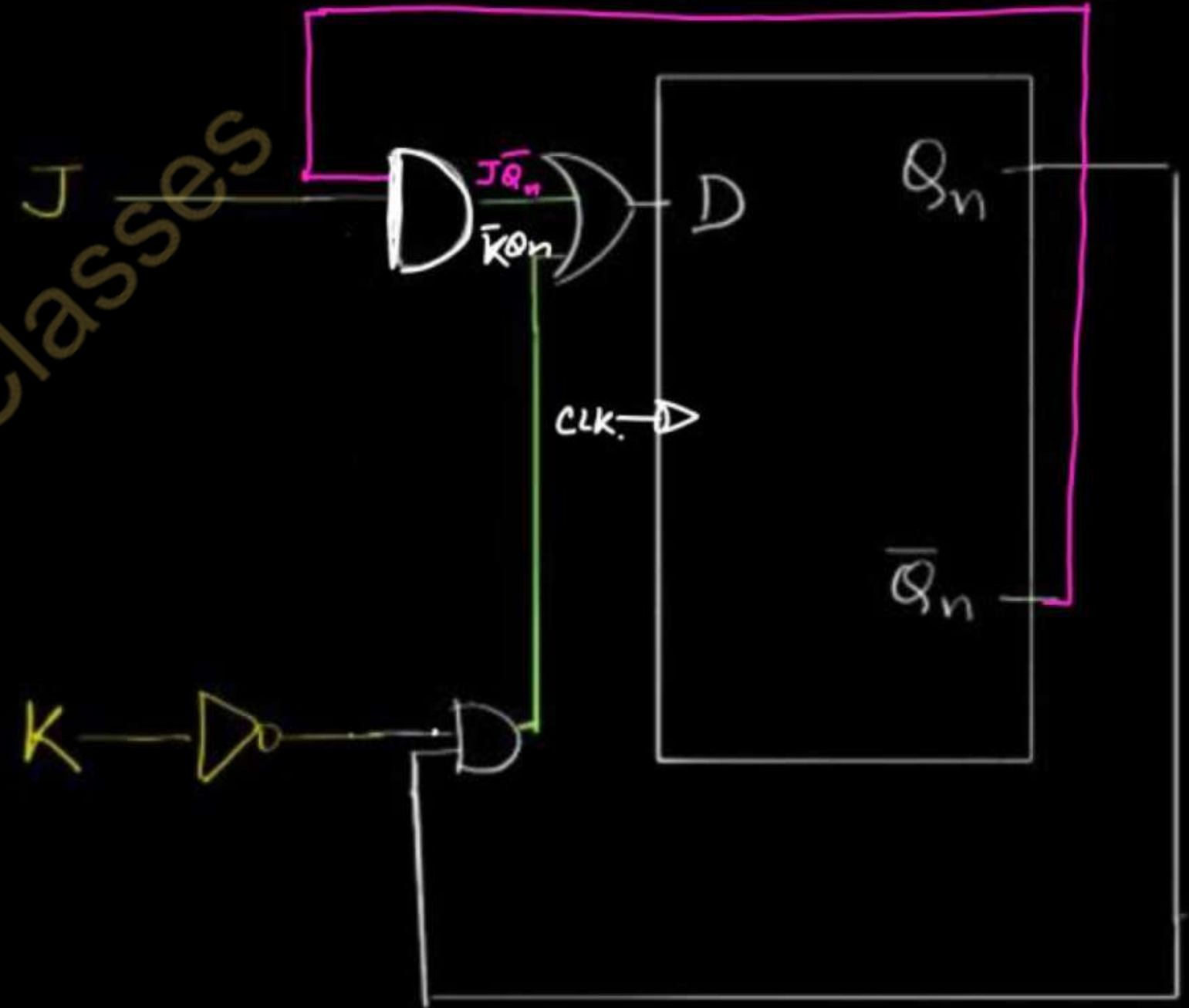
K Map for D

Q_n	JK			
	00	01	11	10
0			1	1
1	1			1

$$D = J\bar{Q}_n + \bar{K}Q_n$$

Step 5

$$D = J\bar{Q}_n + \bar{K}Q_n$$



T TO SR FLIPFLOP

Step 1

S	R	Q _n	Q _{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

Step 2

Q _n	Q _{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Steps

- 1.Characteristic (Truth Table) for Required FF.
- 2.Excitation Table for Available FF.
- 3.Extend Table of Step 1 by adding from Step 2.
- 4.K-Map for available FF (Leave Column of Q_{n+1}.
- 5.Logic Diagram

Step 3

S	R	Q _n	Q _{n+1}	T
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	0	X	X
1	1	1	X	X

T TO JK FLIPFLOP

Step 1

TT of JK

J	K	Q _n	Q _{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Step 2

ET of T

Q _n	Q _{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Step 3

J	K	Q _n	Q _{n+1}	T
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	0	1	1
1	1	1	0	1

Steps

- 1.Characteristic (Truth Table) for Required FF.
- 2.Excitation Table for Available FF.
- 3.Extend Table of Step 1 by adding from Step 2.
- 4.K-Map for available FF (Leave Column of Q_{n+1}).
- 5.Logic Diagram

J	K	Q_n	Q_{n+1}	T
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	0	1	1
1	1	1	0	1

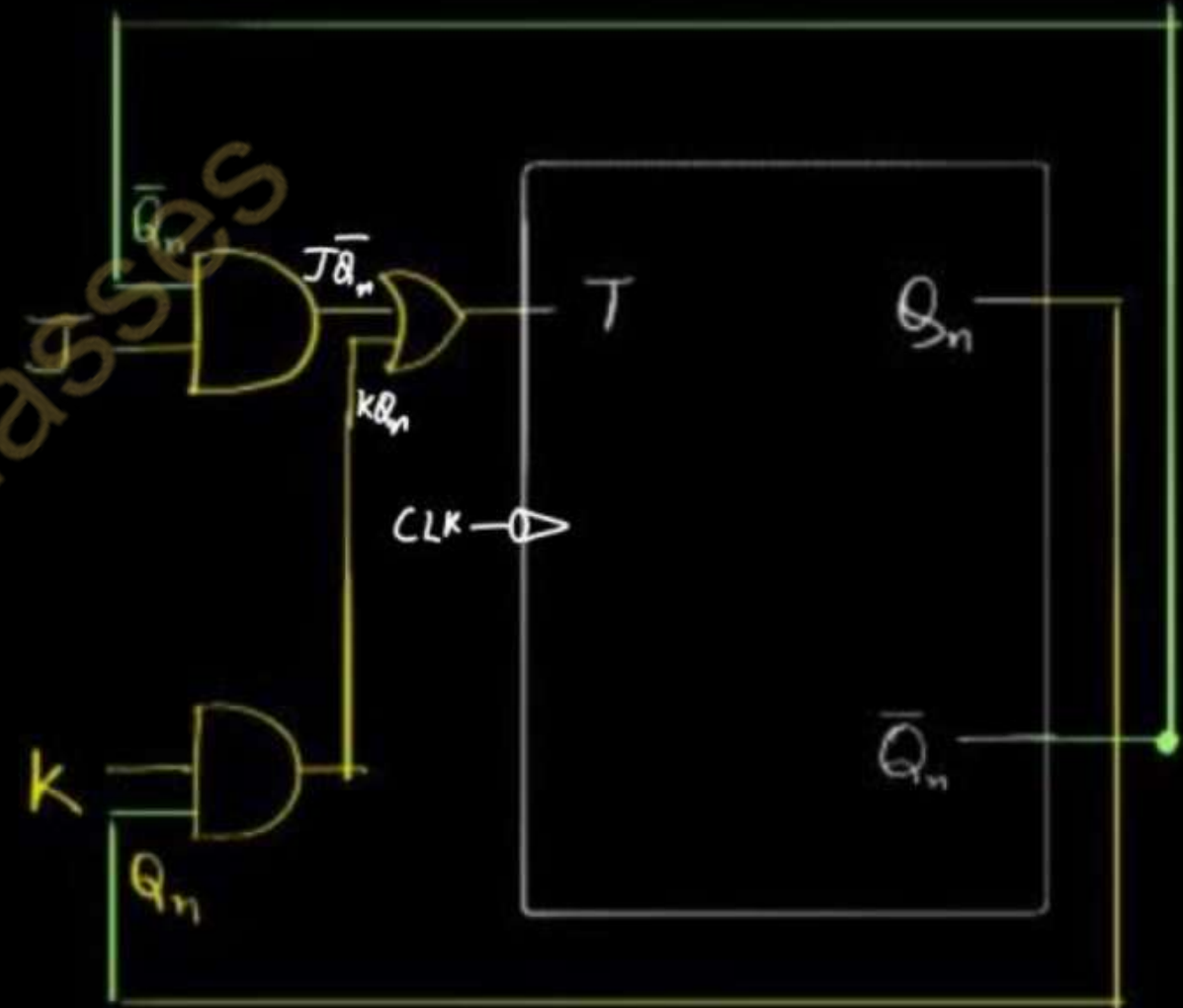
Step 4

K Map for T

$J \backslash K$	Q_n			
	00	01	11	10
0			1	1
1		1	1	

$$T = J\bar{Q}_n + KQ_n$$

Step 5



T TO D FLIPFLOP

Steps

- 1.Characteristic (Truth Table) for Required FF.
- 2.Excitation Table for Available FF.
- 3.Extend Table of Step 1 by adding from Step 2.
- 4.K-Map for available FF (Leave Column of Q_{n+1}).
- 5.Logic Diagram

Step 1

TT of D

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

Step 2

ET of T.

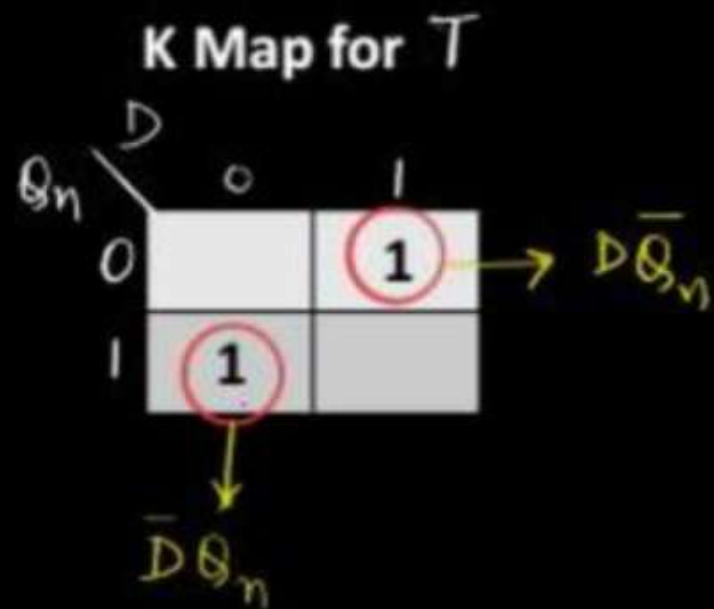
Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Step 3

D	Q_n	Q_{n+1}	T
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

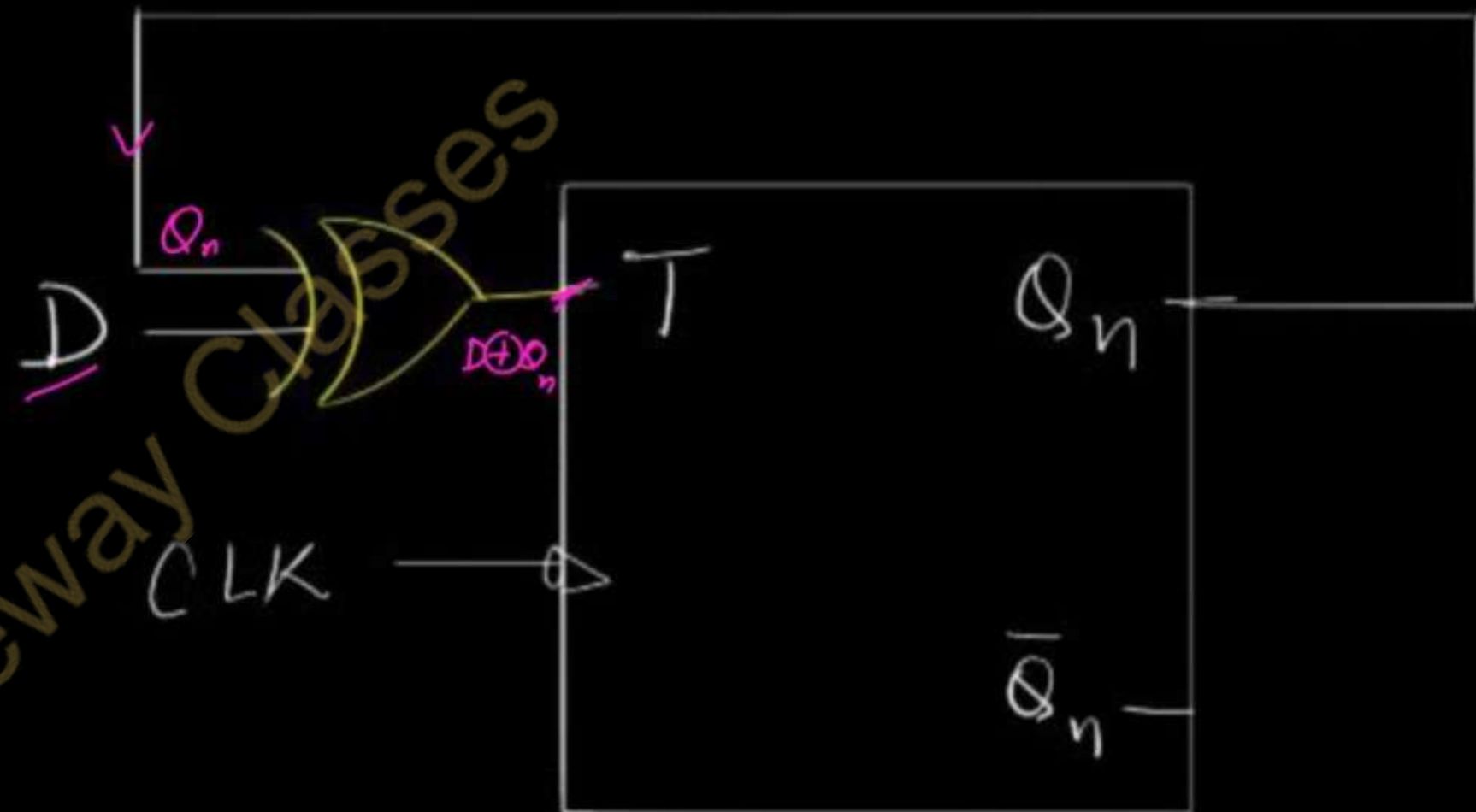
D	Q_n	Q_{n+1}	T
0	0	0	0
0	1	0	1 ✓
1	0	1	1 ✓
1	1	1	0

Step 4



$$\begin{aligned}
 T &= D \bar{Q}_n + \bar{D} Q_n \\
 &= D \oplus Q_n
 \end{aligned}$$

Step 5



SHIFT REGISTERS

Shift registers are fundamental digital circuits made of connected flip-flops that store and shift binary data, moving bits left, right, or both, with applications in serial-to-parallel conversion, data storage etc.

They are used in

Calculators

0001
4321

27

27

2

27

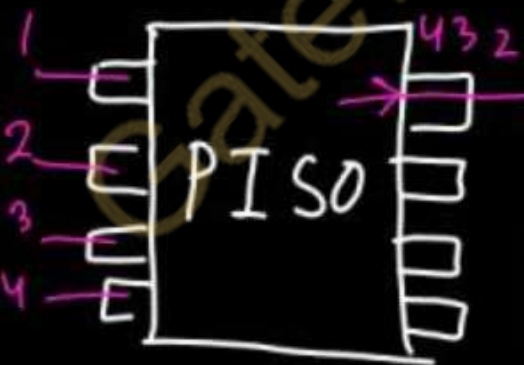
TYPES OF SHIFT REGISTERS

SISO (Serial-In Serial-Out)

SIPO (Serial-In Parallel-Out)

PISO (Parallel-In Serial-Out)

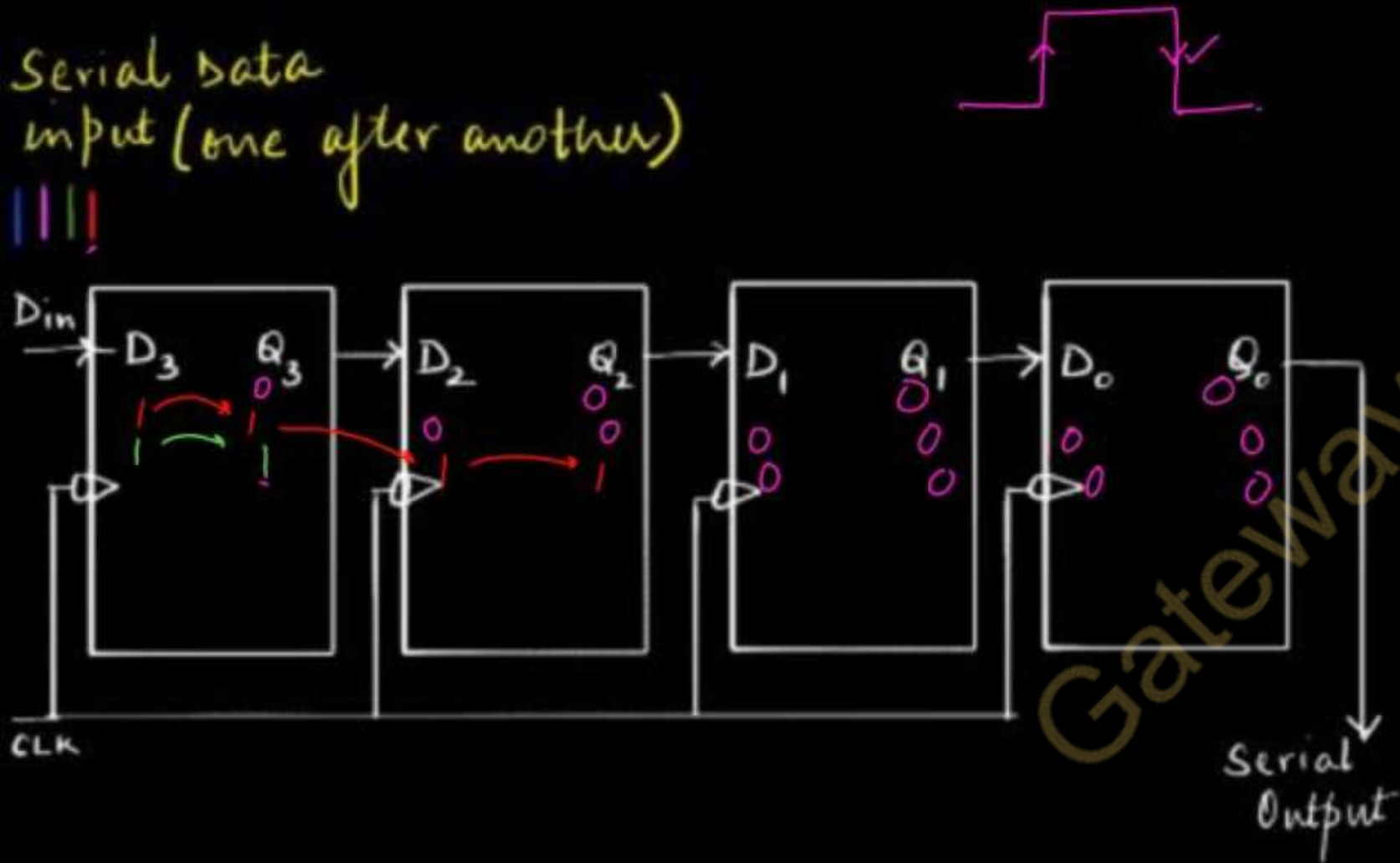
PIPO (Parallel-In Parallel-Out)



Data shifting can be unidirectional (left or right only) or bidirectional (both directions).

SISO (Serial-In Serial-Out)

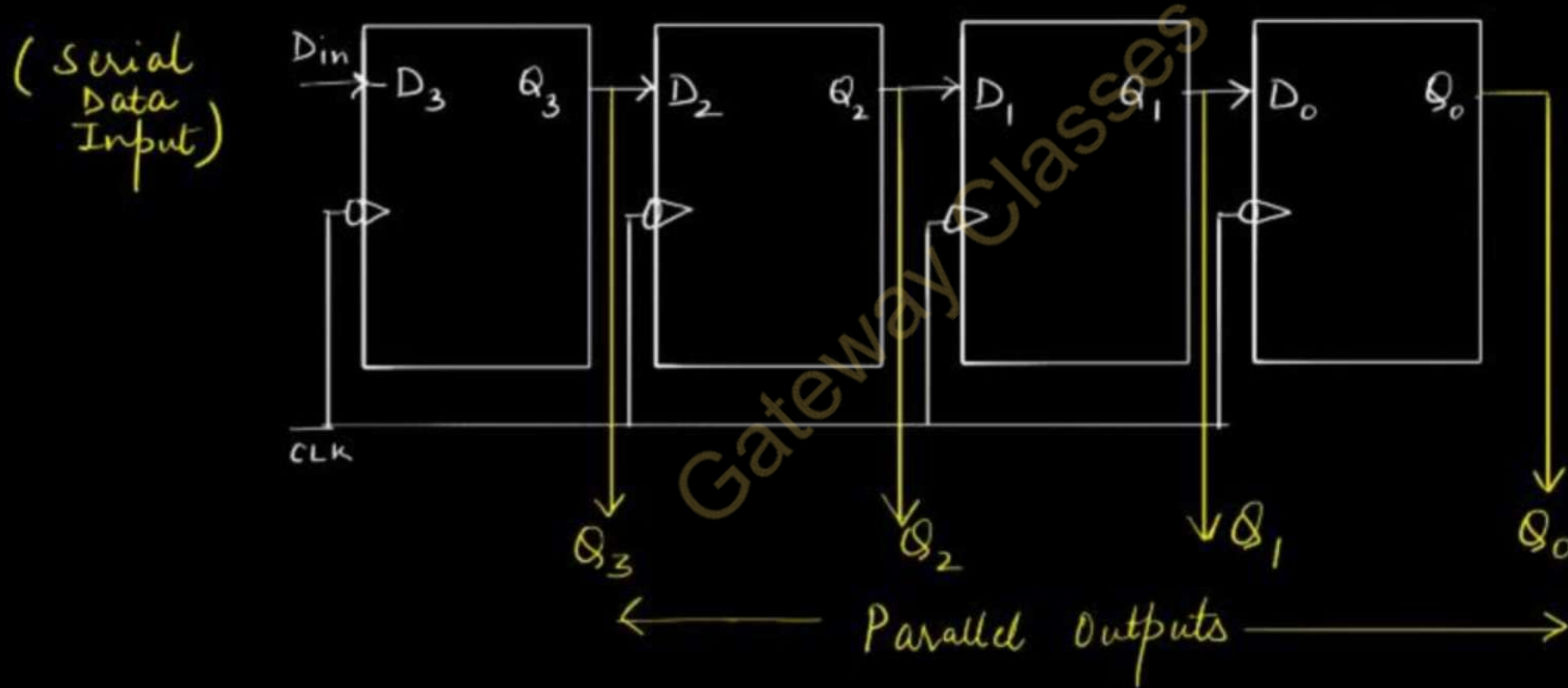
Data enters serially (one bit at a time) and exits serially.



CLK	Q3	Q2	Q1	Q0
Initially	0	0	0	0
1 st Falling Edge	1	0	0	0
2 nd Falling Edge	1	1	0	0
3 rd Falling Edge	1	1	1	0
4 th Falling Edge	1	1	1	1

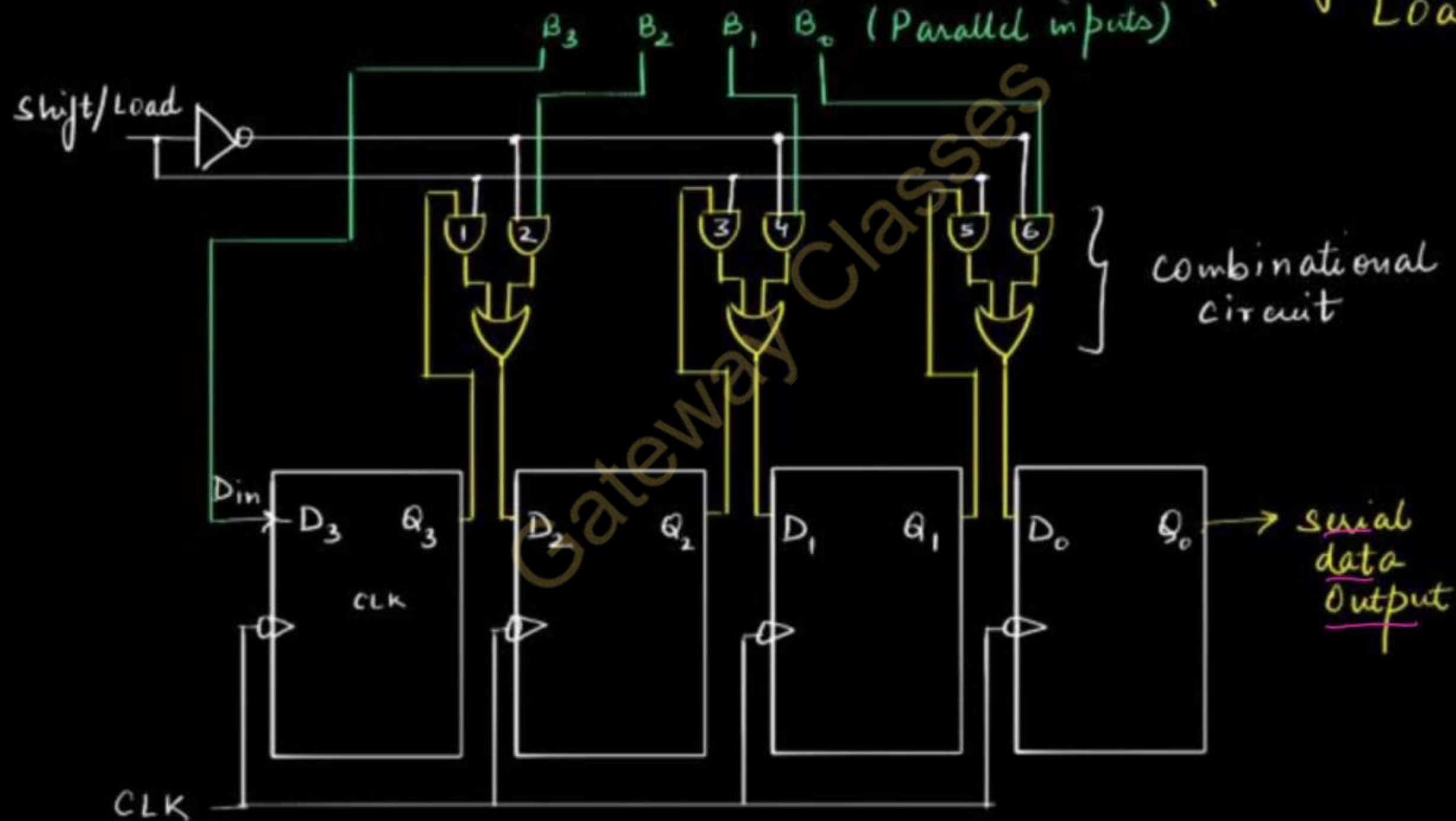
SIPO (Serial-In Parallel-Out)

Serial input, but all bits are available in parallel at the outputs.



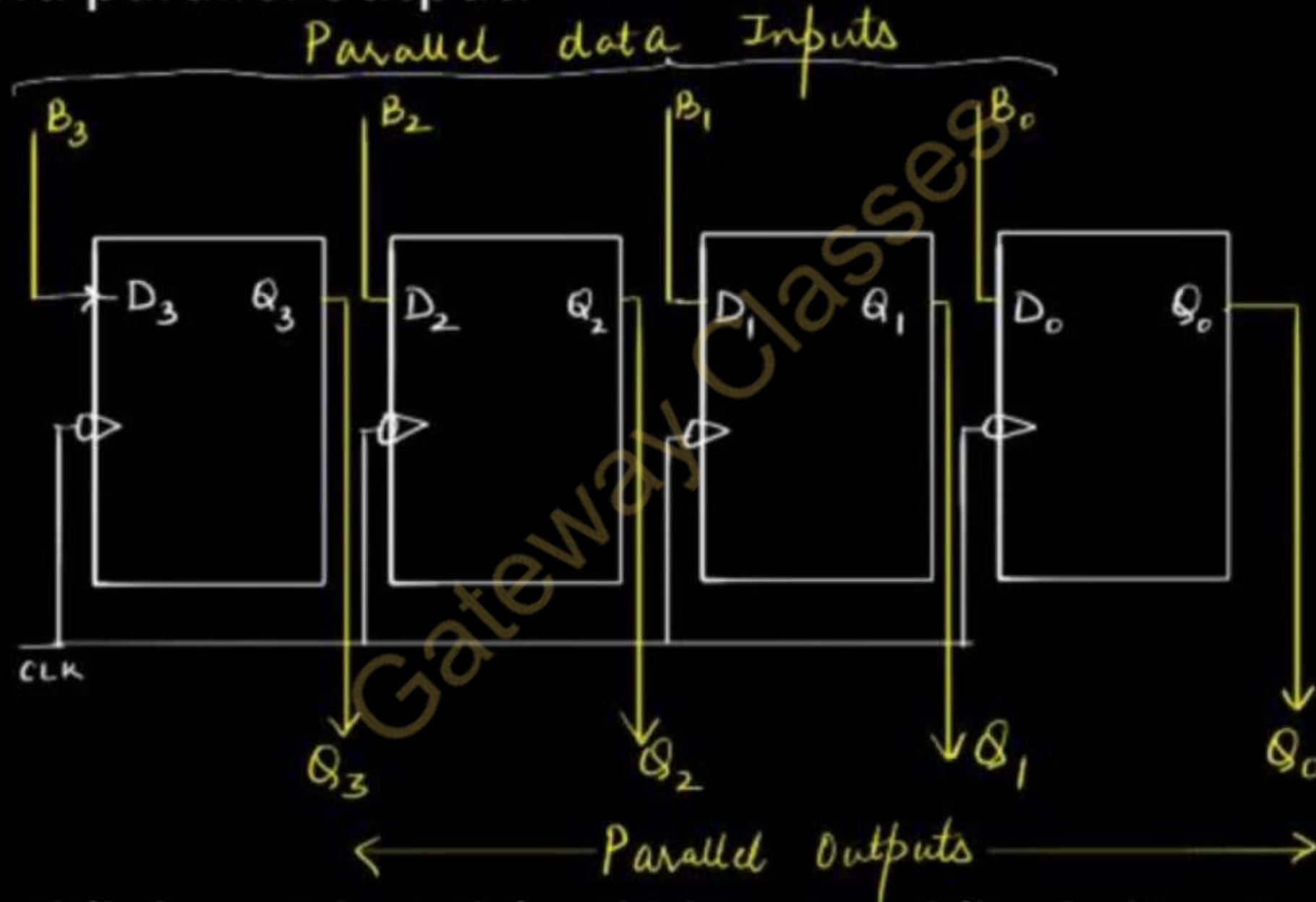
PISO (Parallel-In Serial-Out)

Parallel data loaded simultaneously, then output serially. (Shift mode or Load mode)



PIPO (Parallel-In Parallel-Out)

- Parallel input and parallel output.

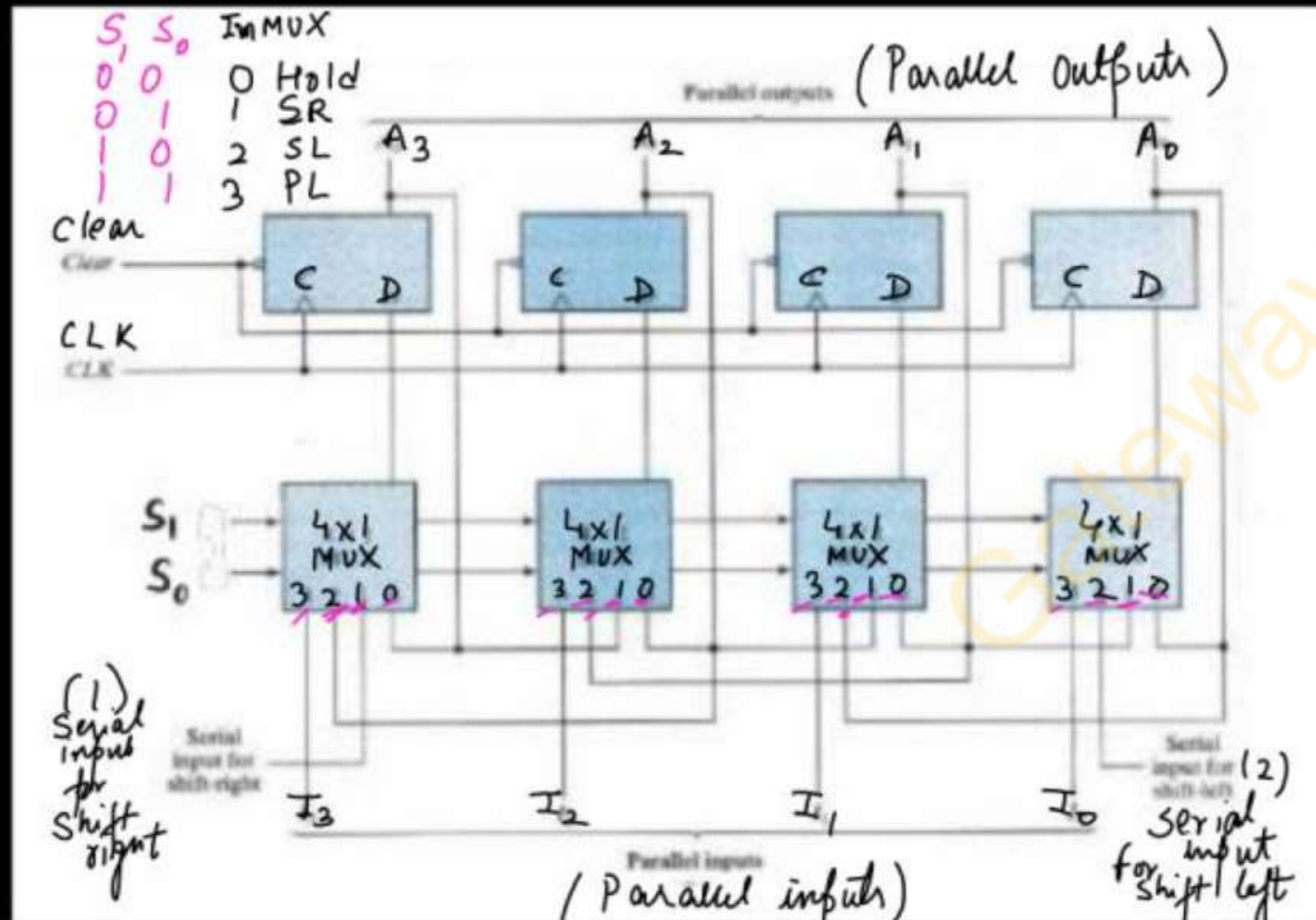


Can be unidirectional (left or right only) or bidirectional (both directions).

✓ **Universal Shift Registers:**(Bidirectional Shift Register)

It operates in three modes—

Shift Left, Shift Right, and Parallel Load—based on control signals, and retains its current data when no control signal is active.



3 **Parallel Load:** Loads data into all flip-flops simultaneously.

$$\begin{matrix} I_0 \rightarrow A_0 \\ I_1 \rightarrow A_1 \\ I_2 \rightarrow A_2 \\ I_3 \rightarrow A_3 \end{matrix}$$

Handwritten diagram showing data flow for Parallel Load:
 $A_3 \rightarrow \text{next}$
 $A_2 \rightarrow A_3$
 $A_1 \rightarrow A_2$
 $A_0 \rightarrow A_1$

1 **Shift Right:** Moves data one position to the right, with serial input entering the MSB side.

2 **Shift Left:** Moves data one position to the left, with serial input entering the LSB side.

Handwritten diagram showing data flow for Shift Left:
 $A_0 \rightarrow \text{next}$
 $A_1 \rightarrow A_0$
 $A_2 \rightarrow A_1$

0 **Hold:** Retains the current data without change.

Handwritten diagram showing data flow for Hold:
 $A_3 \rightarrow A_3$
 $A_2 \rightarrow A_2$
 $A_1 \rightarrow A_1$
 $A_0 \rightarrow A_0$

Operation

Flip-Flops: An n-bit universal register uses n flip-flops (like D flip-flops) to store the bits.

Multiplexers (MUX): Each flip-flop's input comes from a multiplexer (typically 4x1).

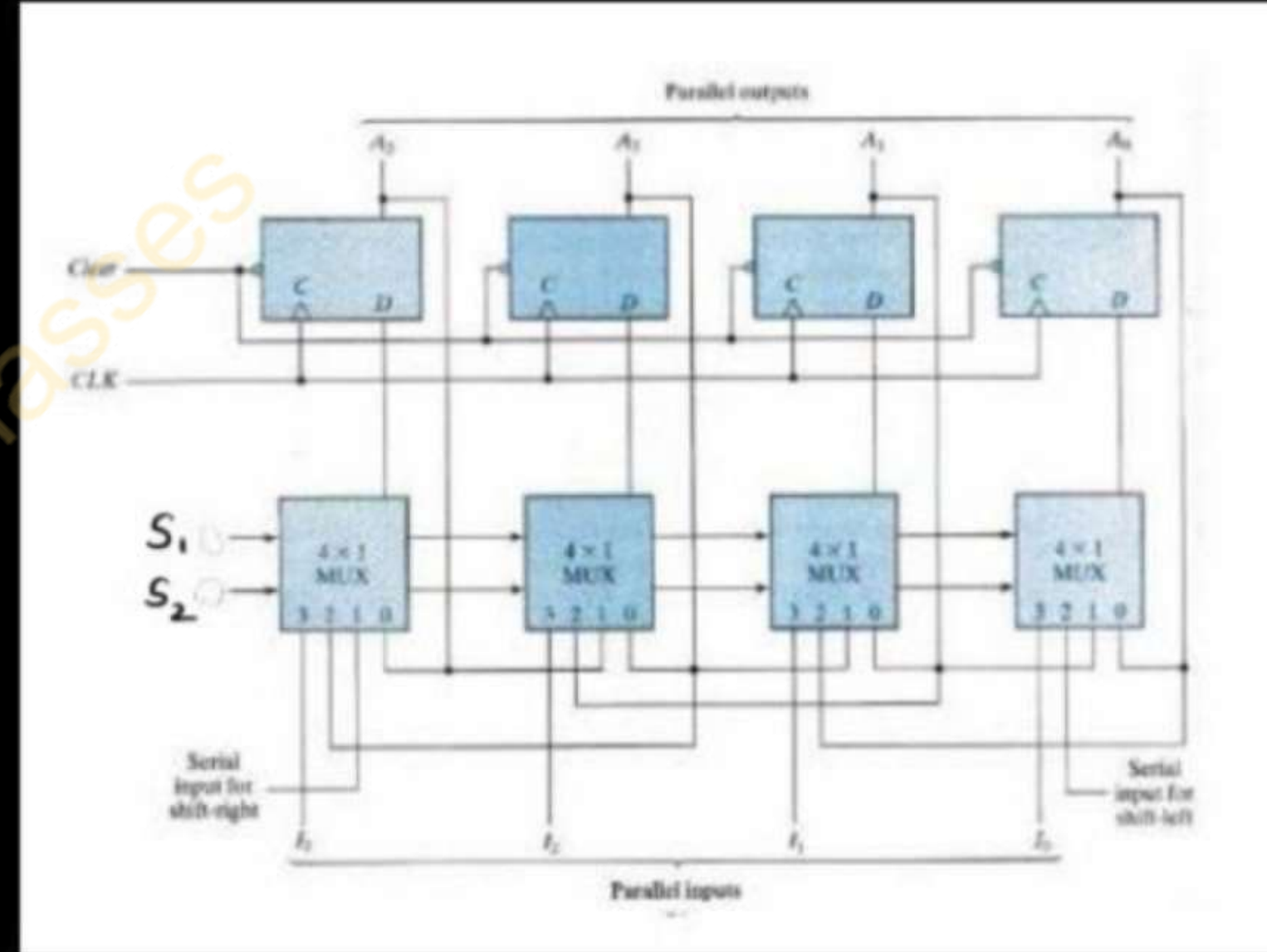
Control Signals: Common select lines (S_1 , S_0) on the MUXes determine the mode:

$S_1, S_0 = 00$: Hold (current Q output fed back).

$S_1, S_0 = 01$: Shift Right (input from previous flip-flop/serial input).

$S_1, S_0 = 10$: Shift Left (input from next flip-flop/serial input).

$S_1, S_0 = 11$: Parallel Load (parallel data input).



COUNTERS

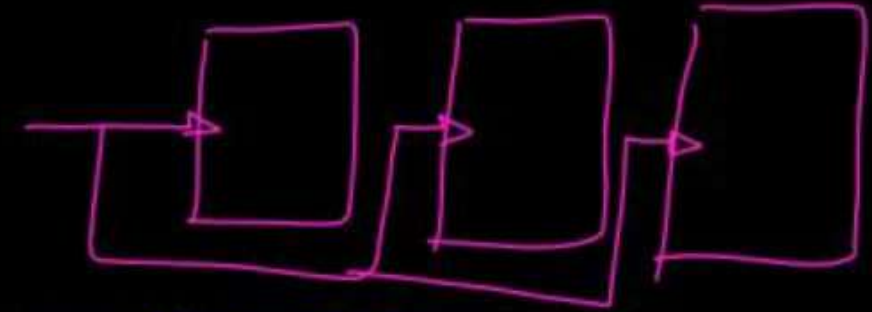
Counters in digital electronics are sequential logic circuits built with flip-flops that **count clock pulses or events**, storing the count in binary form and changing states in a predictable sequence (up, down, or random) for applications like timing, frequency division, and digital systems control.



TYPES OF COUNTERS

Asynchronous(Ripple) Counters

Flip-flops are clocked by the **output of the previous flip-flop**; creates propagation delay (**ripple effect**).



Synchronous Counters

All flip-flops receive the **same clock** signal, offering faster operation and no ripple delay.

ADVANTAGE OF SYNCHRONOUS COUNTERS OVER AN ASYNCHRONOUS COUNTER

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In Synchronous Counters, all flip-flops receive the same clock signal, offering faster operation and no ripple delay. On the other hand, in Asynchronous(Ripple) Counters Flip-flops are clocked by the output of the previous flip-flop; creates propagation delay (ripple effect).

Gateway Classes

Functional Types (By Counting Behavior)

Up Counter: Counts only in the increasing direction. (0, 1, 2, 3...).

Down Counter: Counts only in the decreasing direction. (3, 2, 1, 0...).

Up/Down Counters: Can count in both directions.

Modulus Counter: Counts up to a specific number (modulus) before resetting, rather than its full binary capacity.

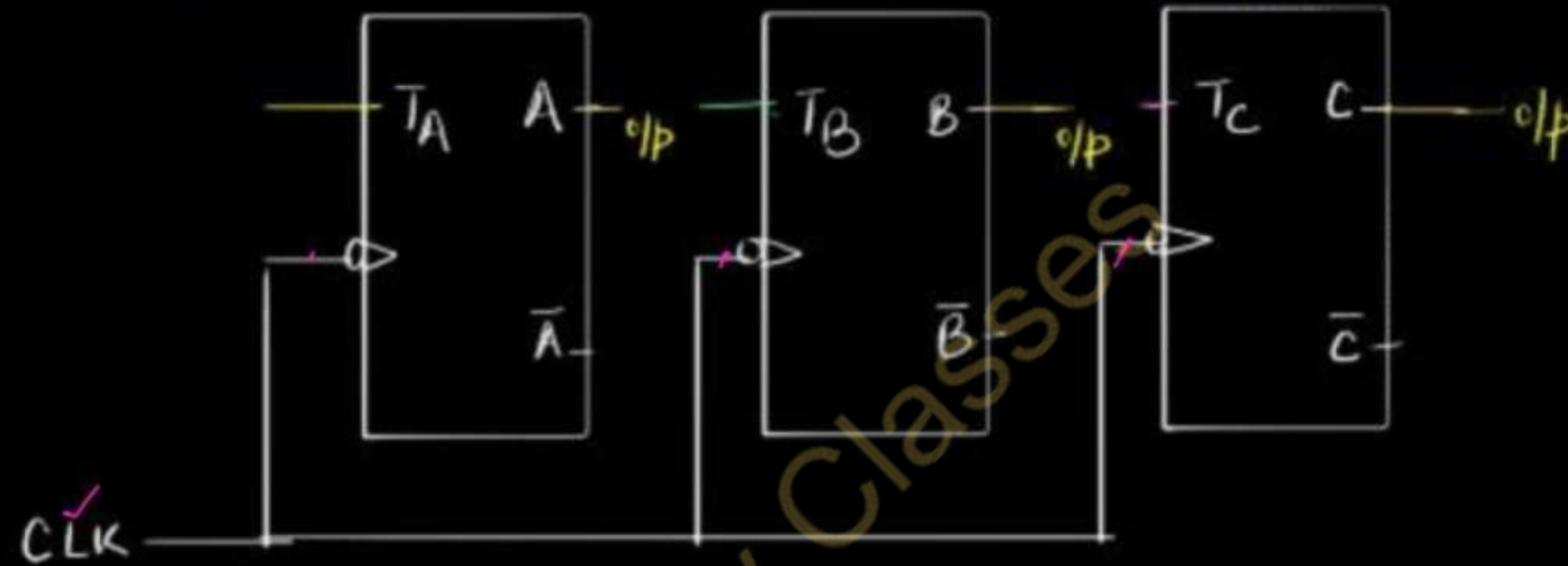
Decade Counter: Counts from 0 to 9 (0000 to 1001 in binary), often used in digital displays.

Ring Counter: A shift register configured to have a single '1' circulating, forming a loop; often used for sequential output.

Johnson Counter: A ring counter with the inverted output of the last FF fed back to the first, creating a longer sequence before repeating.

SYNCHRONOUS COUNTERS

In these Counters , Clock is given simultaneously to all FlipFlops.



Designing Steps

Step 1 :-Decide the number of Flipflops

Step 2 :-Draw State Diagram

Step 3 :-Draw Excitation Table of Flipflop

Step 4 :- Draw Circuit Excitation Table using Step 2 & 3

Step 5 :- K –Map to find expressions

Step 6 :- Draw Logic Diagram

To find number of Flipflops & States.

Mod-N Counter :- N represents number of states.

n - bit Counter :- n represents number of Flipflops used in Counter.

$$2^n \geq N$$

3 bit Counter
 $n = 3 \rightarrow 3 \text{ FFs}$
 $2^3 = 8$
 $N = 8 \rightarrow 8 \text{ states}$

Counter	No. of States	Relation	Flipflops	States	Counts from
Mod-5 ^N	<u>N = 5</u>	$2^n \geq 5$ $2^2 = 4$ $2^3 = 8$ so, $n = 3$	3 FFs	5 States	Since 5 States 0 to 4 i.e (000 to 100) (N-1) 3 bit
Mod-10 ^N	<u>N = 10</u>	$2^n \geq 10$ $2^3 = 8$ $2^4 = 16$ so, $n = 4$	4 FFs	10 States	Since 10 states 0 to 9 i.e (0000 to 1001) (N-1) 4 bit
✓ 3 bit	$N = 2^3 = 8$	$n = 3$	3 FFs	8 states	0 to 7 i.e (000 to 111) 3 bit
✓ 4 bit	$N = 2^4 = 16$	$n = 4$	4 FFs	16 states	0 to 15 i.e (0000 to 1111) 4 bit

Que-1) Design a **2-bit Synchronous Counter**

Step 1:-

2 bit

2 FFs

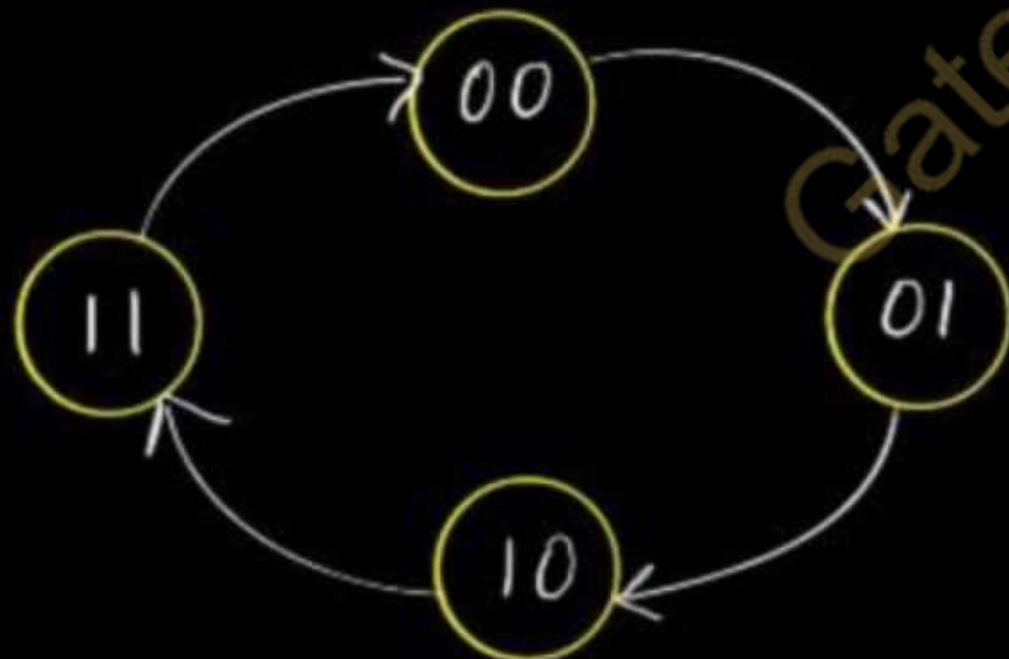
$n=2$, so $N = 2^2 = 4$ states

Counts from 0 to $(2^2 - 1)$ i.e 0 to 3

(00 to 11)

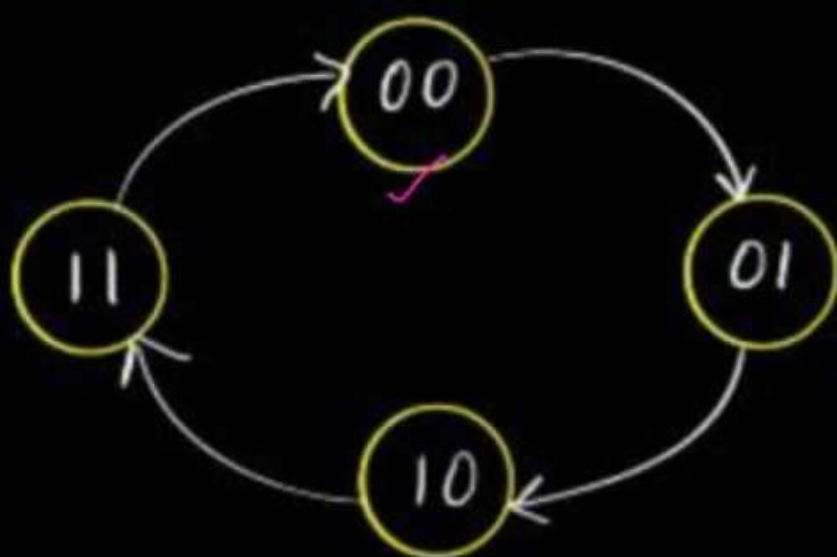
2 bit

Step 2:- State Diagram



Step 3:- Excitation Table for JK-FF (Suppose)

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0



Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Step 4:- Circuit Excitation Table

(made from State Diagram and Excitation Table)

PS NS FF I/p's

PRESENT STATE		NEXT STATE		FLIPFLOP INPUTS			
A	B	A	B	J_A	K_A	J_B	K_B
0	0	0	1	0	X	1	X
0	1	1	0	1	X	X	1
1	0	1	1	X	0	1	X
1	1	0	0	X	1	X	1

Step 5:- K Maps (Present States are taken)

For J_A

B \ A	0	1
0		X
1	1	X

$\rightarrow B$

$$\therefore J_A = B$$

For K_A

B \ A	0	1
0	X	
1	X	1

$\rightarrow B$

$$\therefore K_A = B$$

For J_B

B \ A	0	1
0	1	1
1	X	X

$\rightarrow 1$

$$J_B = 1$$

For K_B

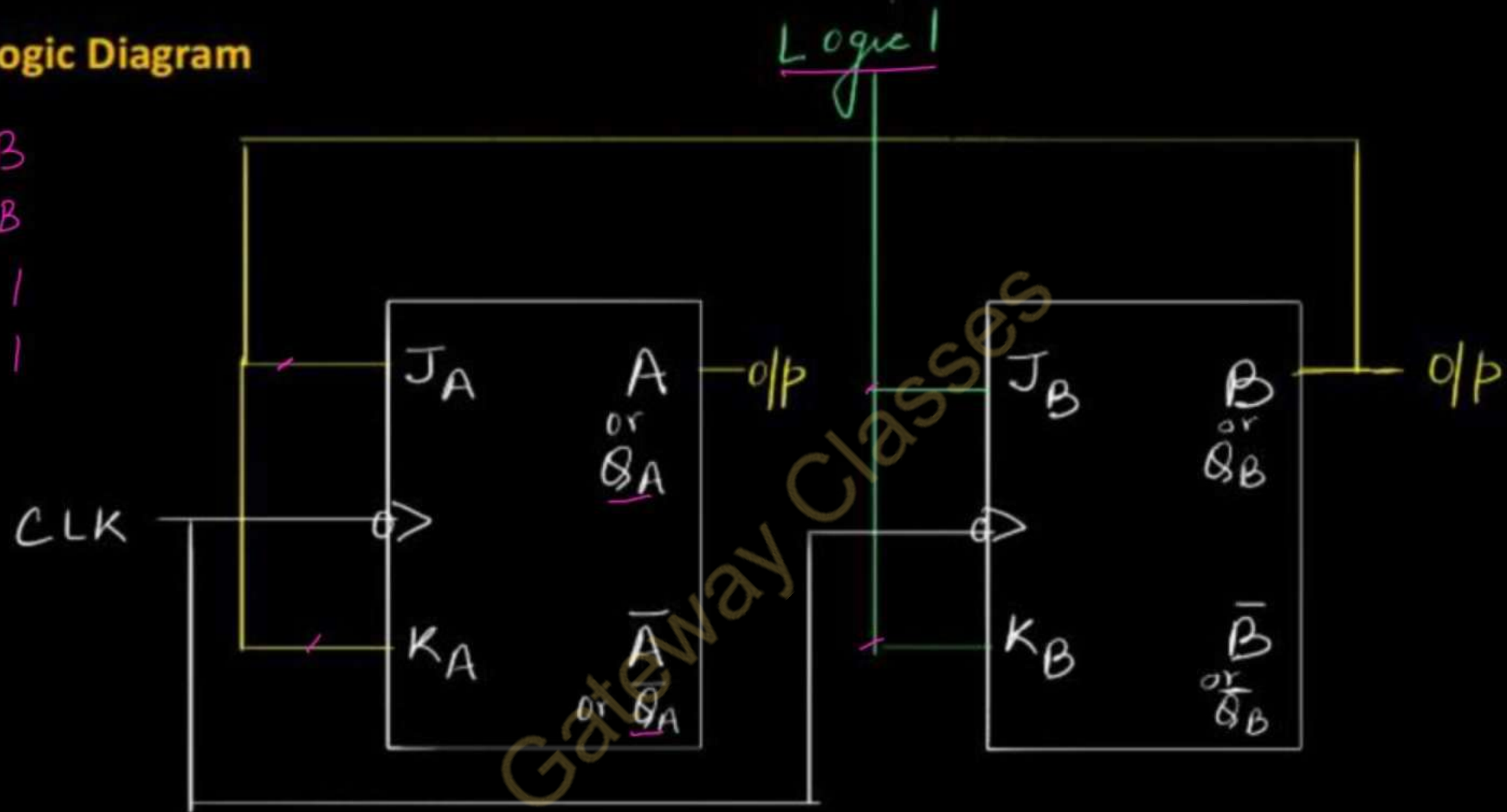
B \ A	0	1
0	X	X
1	1	1

$\rightarrow 1$

$$K_B = 1$$

Step 6:- Logic Diagram

$$\begin{cases} J_A = B \\ K_A = B \\ J_B = 1 \\ K_B = 1 \end{cases}$$



Que-2) Design a 3-bit Synchronous Counter

Step 1:-

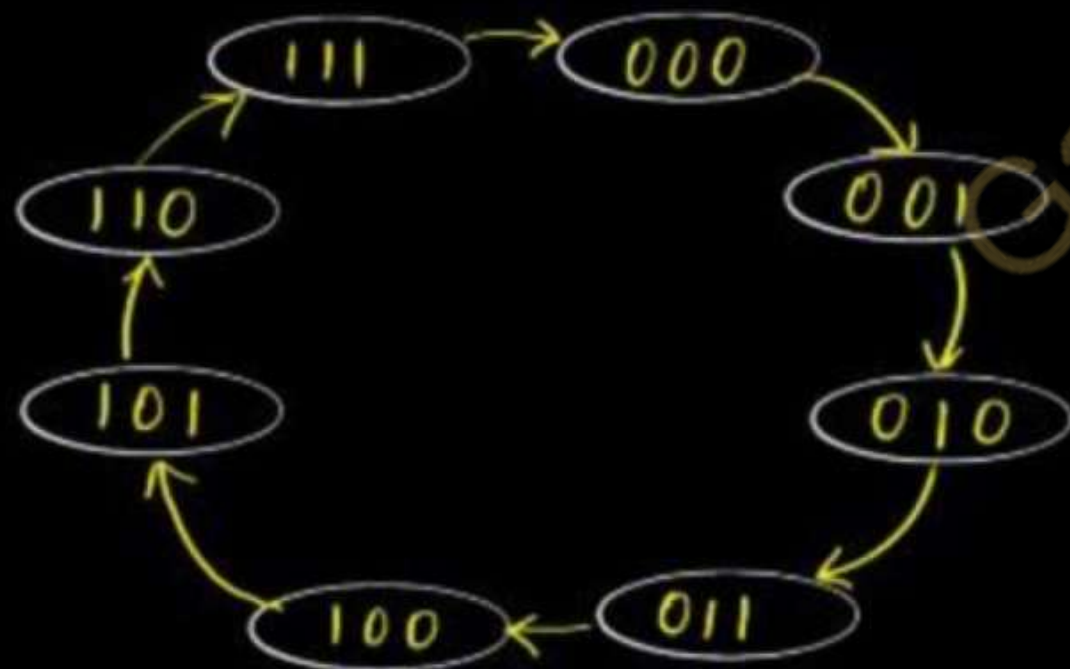
3 bit

3 FFs ✓

$n=3$, so $N = 2^3 = 8$ states

Counts from 0 to $(2^3 - 1)$ i.e 0 to 7

Step 2:- State Diagram

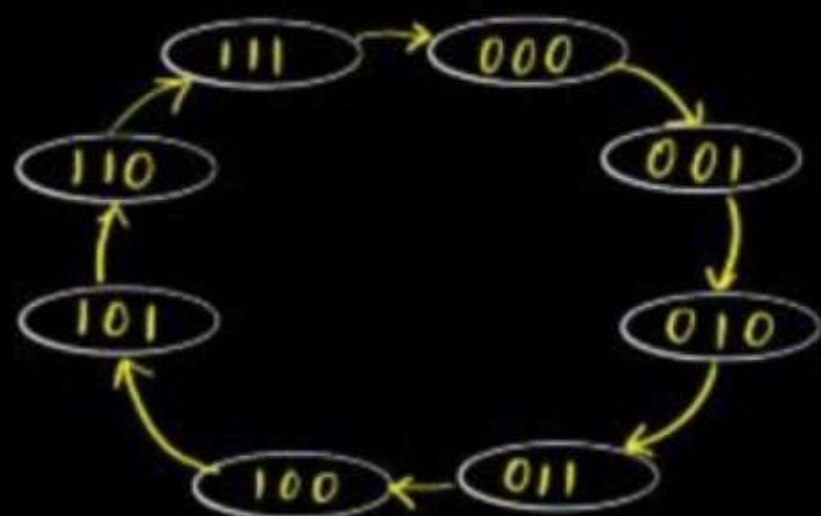


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Step 3:- Excitation Table for T-FF (Suppose)

Similar inputs, $Q/p = 0$
Disimilar inputs, $Q/p = 1$

Q_n	Q_{n+1}	T
<u>0</u>	<u>0</u>	<u>0</u>
<u>0</u>	<u>1</u>	<u>1</u>
<u>1</u>	<u>0</u>	<u>1</u>
<u>1</u>	<u>1</u>	<u>0</u>



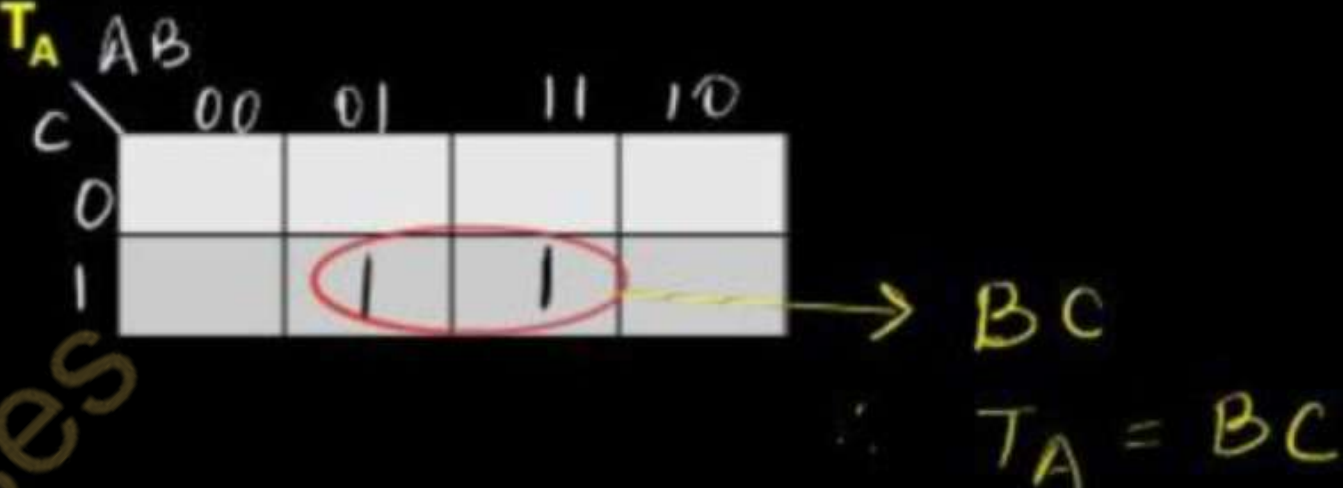
Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Step 4:- Circuit Excitation Table

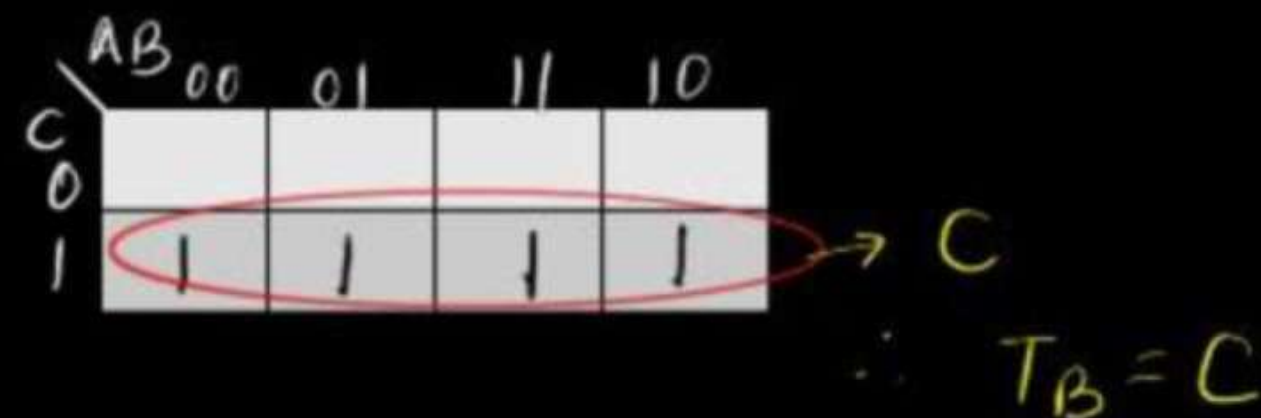
PRESENT STATE			NEXT STATE			FLIPFLOP INPUTS		
A	B	C	A	B	C	T_A	T_B	T_C
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

Step 5:- K Maps

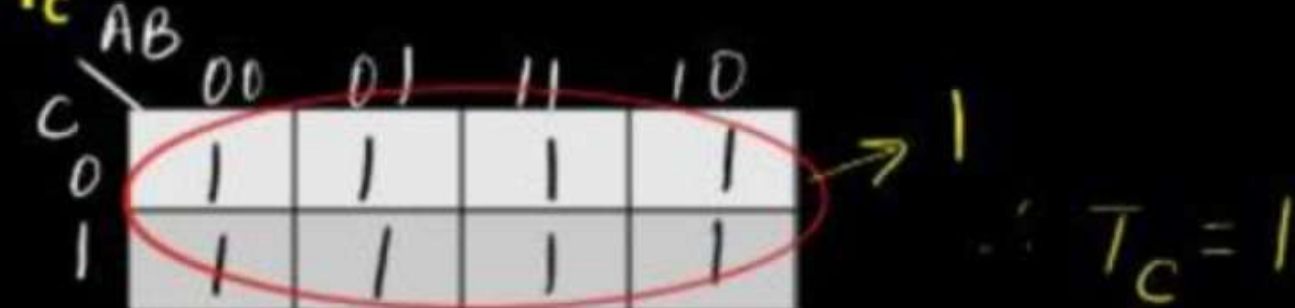
For T_A



For T_B

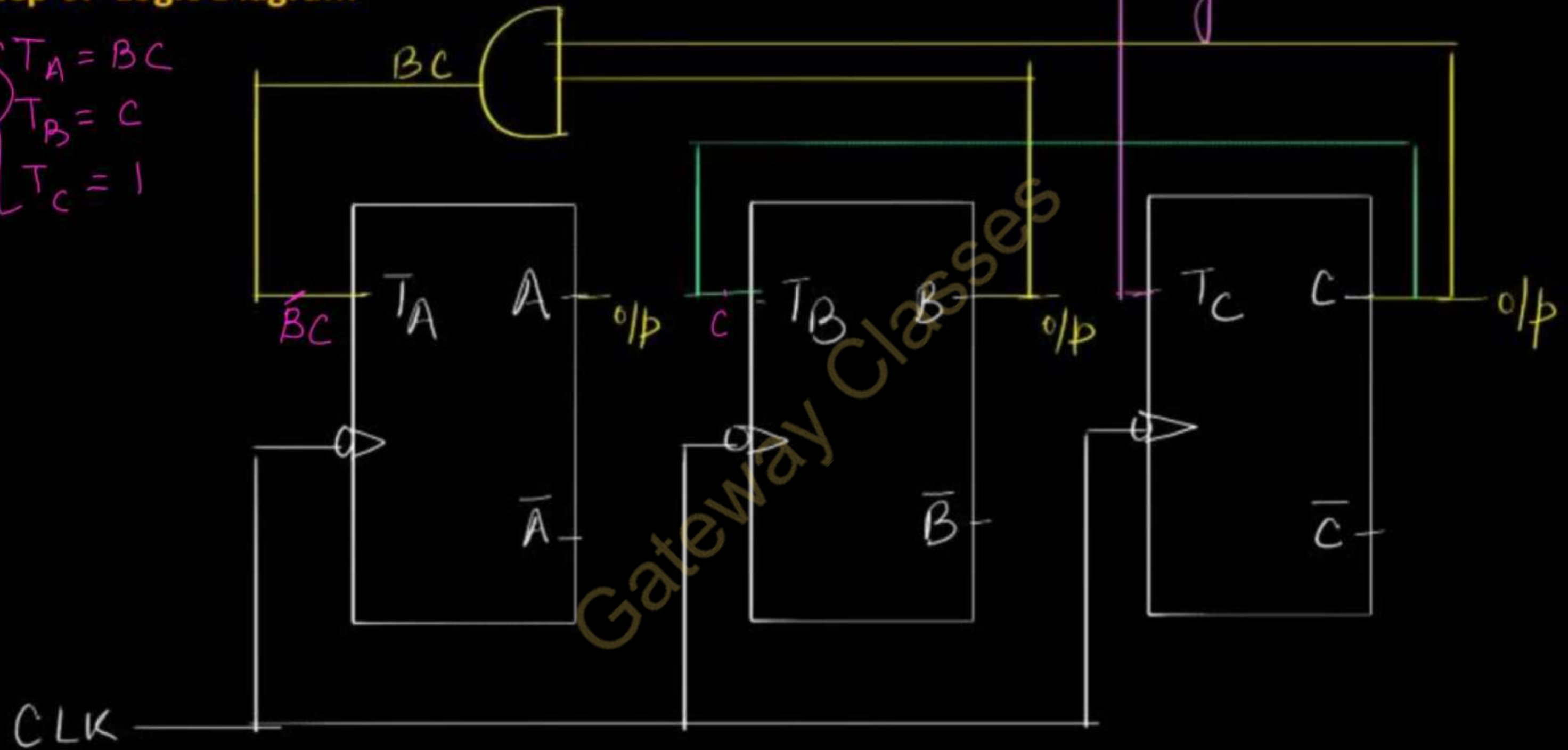


For T_C



Step 6:- Logic Diagram

$$\begin{cases} T_A = BC \\ T_B = C \\ T_C = 1 \end{cases}$$



Que -3) Design a 4-bit Synchronous Counter

Step 1:-

4 bit ,4 FFs

$n=4$, so $N = 2^4 = 16$ states

Counts from 0 to $(2^4 - 1)$ i.e 0 to 15 i.e (0000 to 1111)

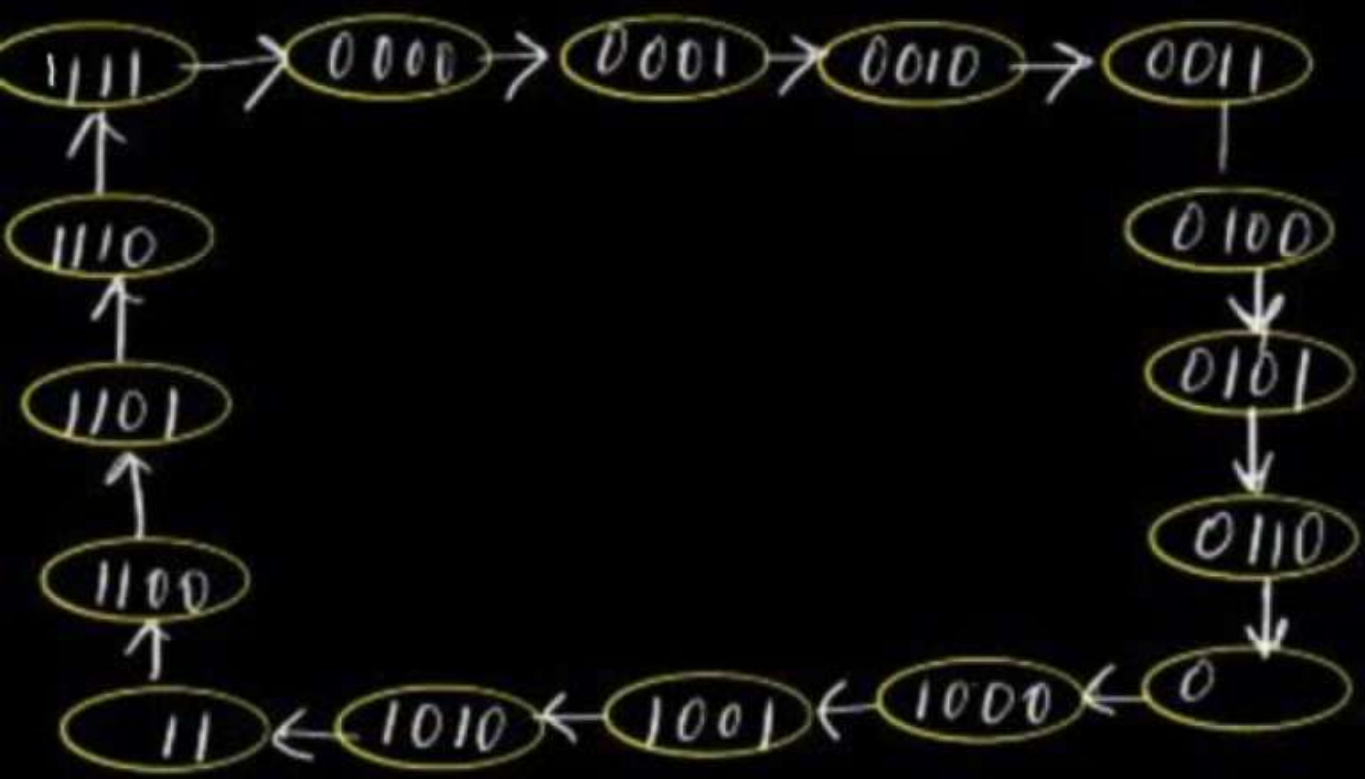
Step 2:- State Diagram



Step 3:- Excitation Table for T-FF (Suppose)

Q_n	Q_{n+1}	T
<u>0</u>	<u>0</u>	<u>0</u>
<u>0</u>	<u>1</u>	<u>1</u>
<u>1</u>	<u>0</u>	<u>1</u>
<u>1</u>	<u>1</u>	<u>0</u>

Step 4:- Circuit Excitation Table



PRESENT STATE				NEXT STATE				FLIPFLOP INPUTS			
A	B	C	D	A	B	C	D	^{A,A} T _A	^{B,B} T _B	^{C,C} T _C	^{D,D} T _D
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	1
0	0	1	1	0	1	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	1
0	1	0	1	0	1	1	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	1
0	1	1	1	1	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	0	1
1	0	0	1	1	0	1	0	0	0	1	1
1	0	1	0	1	0	1	1	0	0	0	1
1	0	1	1	1	1	0	0	0	1	1	1
1	1	0	0	1	1	0	1	0	0	0	1
1	1	0	1	1	1	1	0	0	0	1	1
1	1	1	0	1	1	1	1	0	0	0	1
1	1	1	1	0	0	0	0	1	1	1	1

Step 5:- K Maps

For T_A

CD \ AB	00	01	11	10
00				
01				
11		1	1	
10				

→ BCD

$$\therefore T_A = BCD$$

For T_B

CD \ AB	00	01	11	10
00				
01				
11	1	1	1	1
10				

→ CD

$$\therefore T_B = CD$$

For T_C

CD \ AB	00	01	11	10
00				
01	1	1	1	1
11	1	1	1	1
10				

→ D

$$\therefore T_C = D$$

For T_D

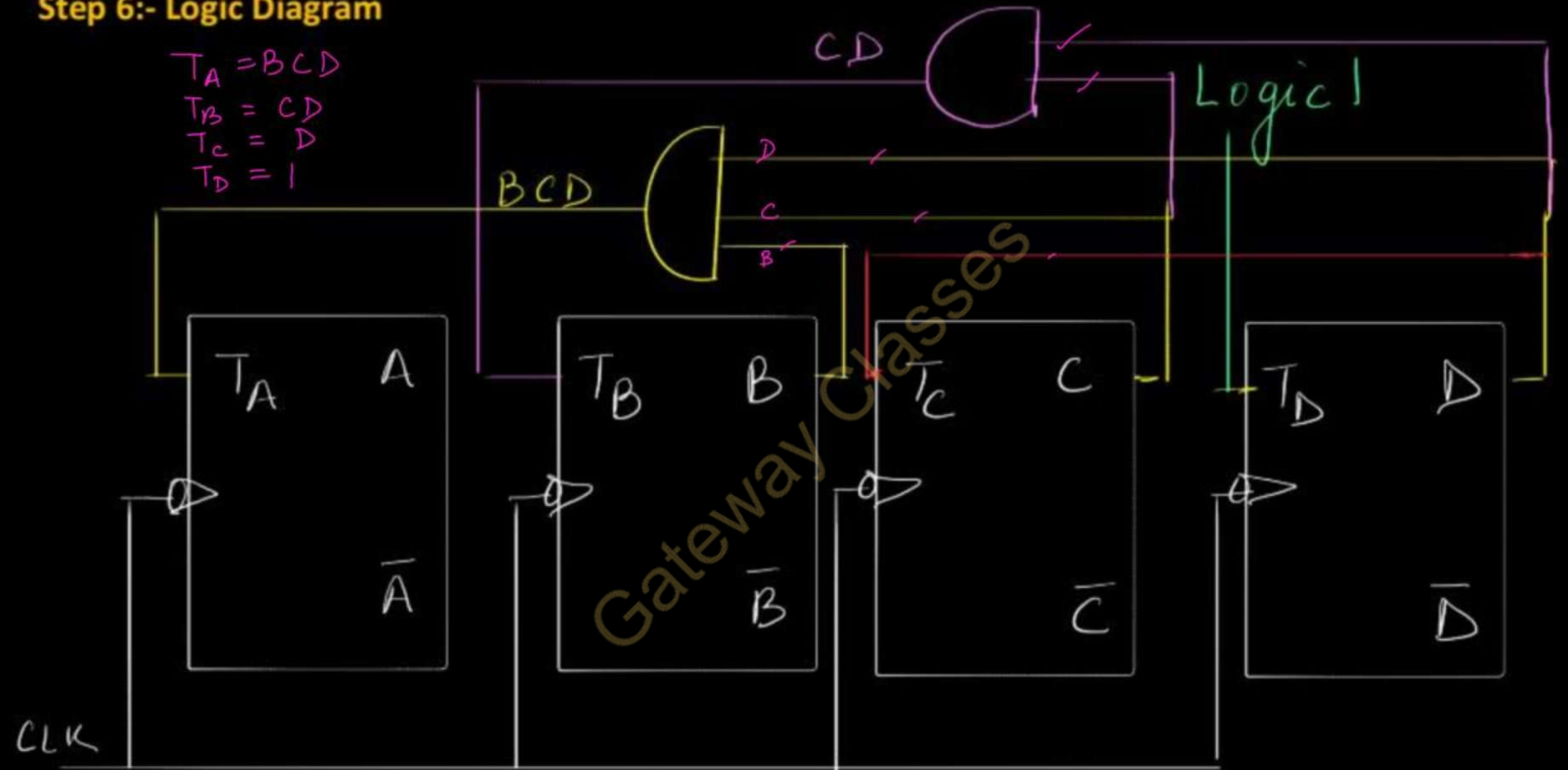
CD \ AB	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

→ 1

$$\therefore T_D = 1$$

Step 6:- Logic Diagram

$$\begin{aligned}T_A &= BCD \\T_B &= CD \\T_C &= D \\T_D &= 1\end{aligned}$$



Que -4) Explain the working and circuit of a **modulo-5 counter using gates.**

Step 1:-

Mod-5, $N=5$, 5 States

$$2^n \geq 5$$

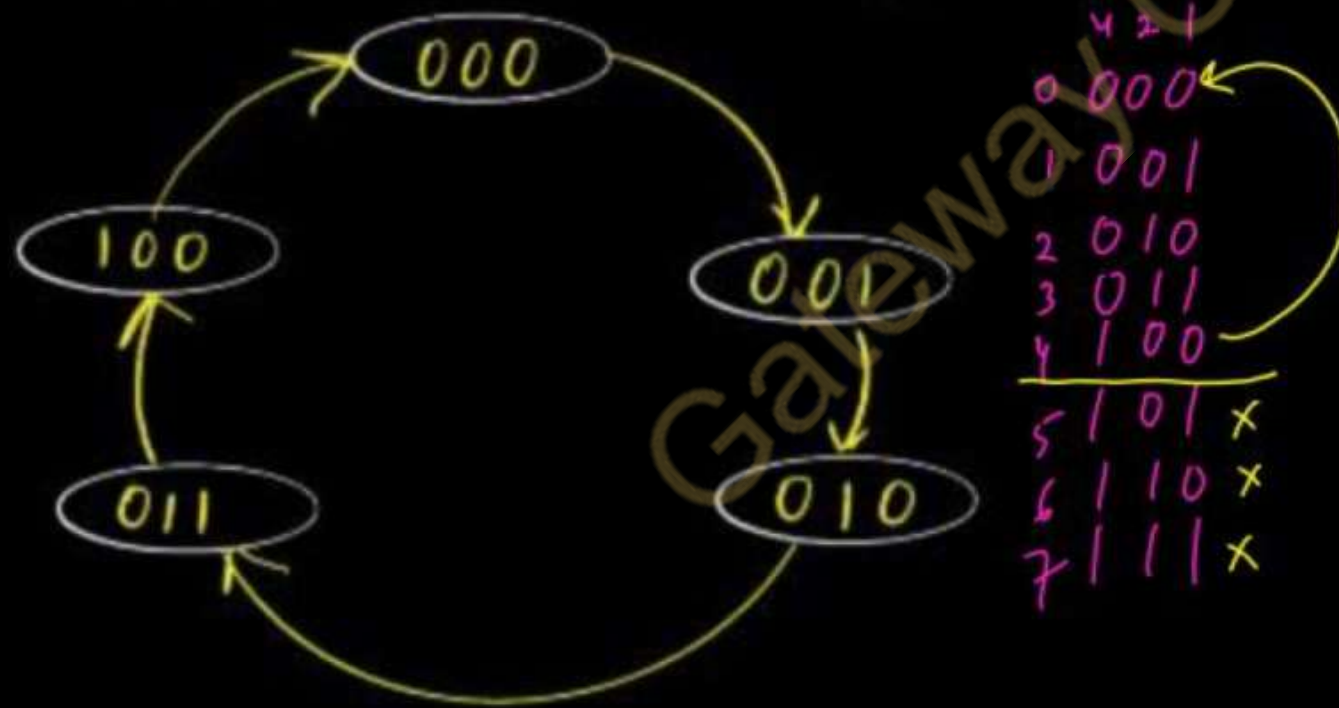
$$2^2=4$$

$$2^3=8$$

so, $n=3$, 3 FFs

Since 5 states, so it will count from 0 to 4 i.e (000 to 100)

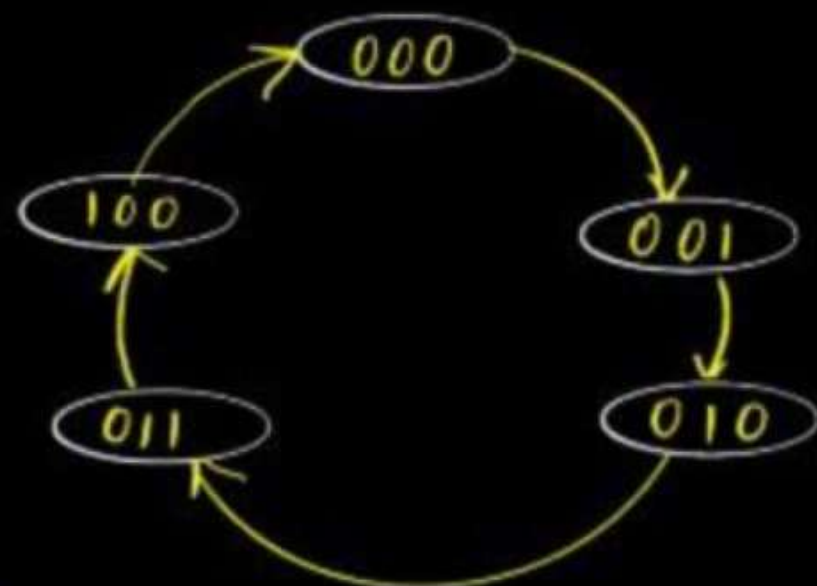
Step 2:- State Diagram



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Step 3:- Excitation Table for T-FF (Suppose)

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0



Step 4:- Circuit Excitation Table

PRESENT STATE			NEXT STATE			FLIPFLOP INPUTS		
A	B	C	A	B	C	T _A	T _B	T _C
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	0	0	0	1	0	0
1	0	1	X	X	X	X	X	X
1	1	0	X	X	X	X	X	X
1	1	1	X	X	X	X	X	X

Step 5:- K Maps

For T_A

AB \ C	00	01	11	10
0			X	1
1		1	X	X

$$T_A = A + BC$$

For T_B

AB \ C	00	01	11	10
0			X	
1	1	1	X	X

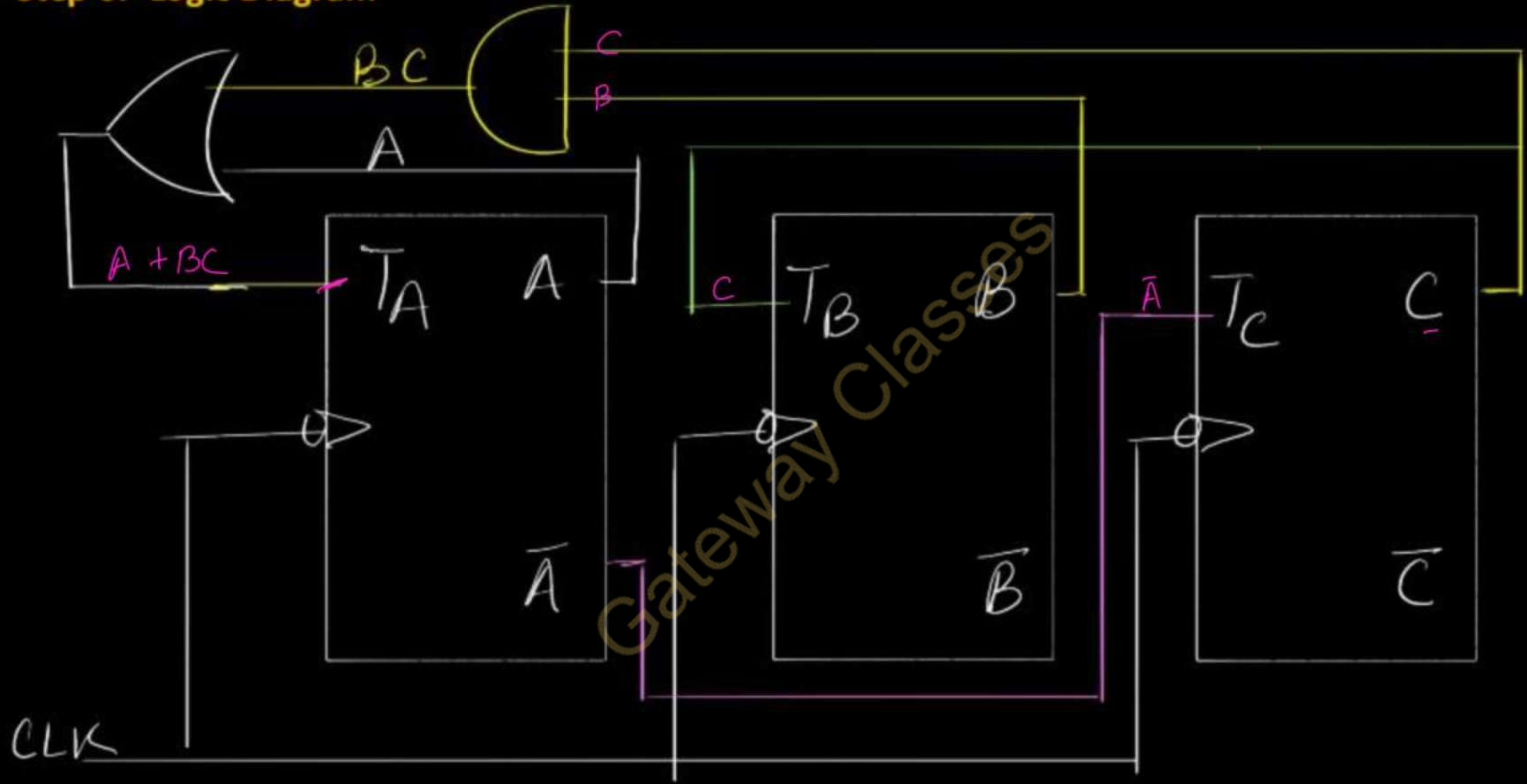
$$T_B = C$$

For T_C

AB \ C	00	01	11	10
0	1	1	X	
1	1	1	X	X

$$T_C = \bar{A}$$

Step 6:- Logic Diagram



Que -4) Explain the working and circuit of a **Mod-10 counter** or **Synchronous Decade Counter**

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Step 1:-

Mod-10, $N=10$, 10 States

$$2^n \geq 10$$

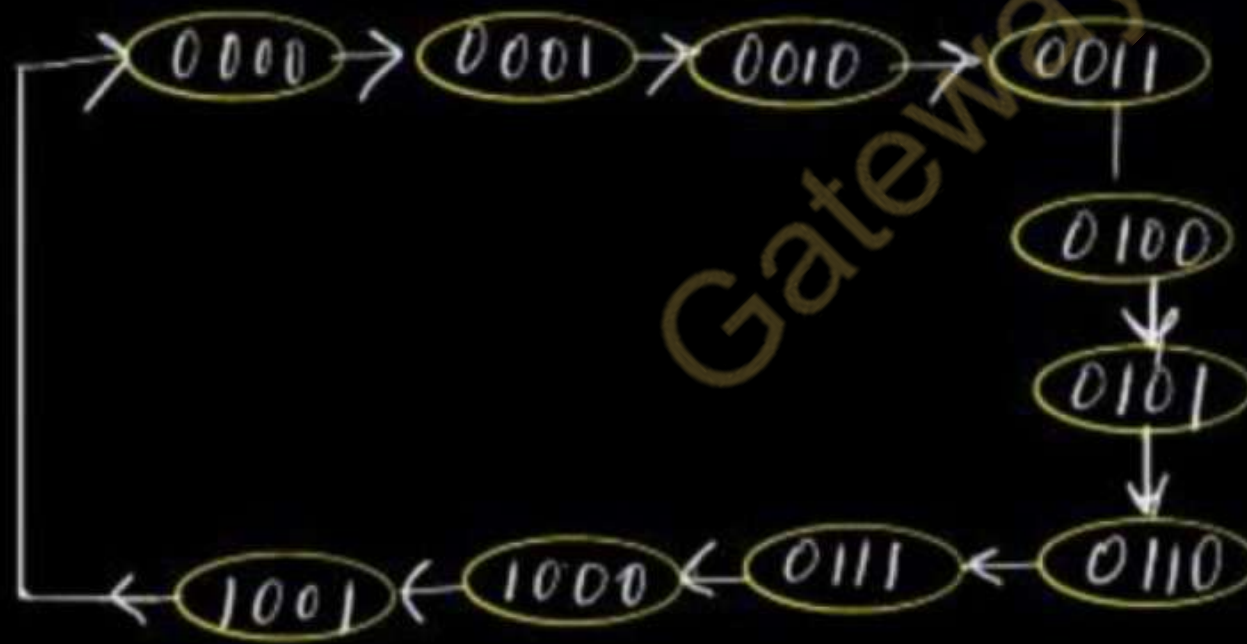
$$2^3=8$$

$$2^4=16$$

so, $n=4$, 4 FFs

Since 10 states, so it will count from 0 to 9 i.e (0000 to 1001)

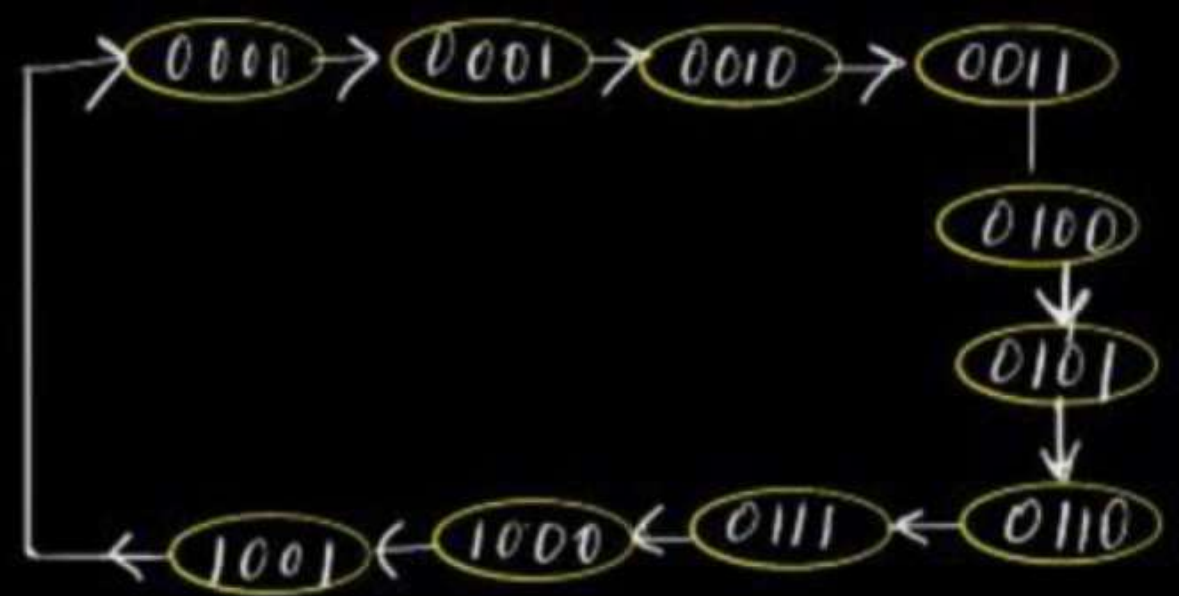
Step 2:- State Diagram



Step 3:- Excitation Table for T-FF (Suppose)

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Step 4:- Circuit Excitation Table



PRESENT STATE				NEXT STATE				FLIPFLOP INPUTS			
A	B	C	D	A	B	C	D	^{A,A} T _A	^{B,B} T _B	^{C,C} T _C	^{D,D} T _D
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	1
0	0	1	1	0	1	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	1
0	1	0	1	0	1	1	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	1
0	1	1	1	1	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	0	1
1	0	0	1	0	0	0	0	1	0	0	1
1	0	1	0	X	X	X	X	X	X	X	X
1	0	1	1	X	X	X	X	X	X	X	X
1	1	0	0	X	X	X	X	X	X	X	X
1	1	0	1	X	X	X	X	X	X	X	X
1	1	1	0	X	X	X	X	X	X	X	X
1	1	1	1	X	X	X	X	X	X	X	X

Step 5:- K Maps

For T_A

CD \ AB	00	01	11	10
00			X	
01			X	1
11		1	X	X
10			X	X

$\rightarrow AD$

$$T_A = AD + BCD$$

$\rightarrow BCD$

For T_B

CD \ AB	00	01	11	10
00			X	
01			X	
11	1	1	X	X
10			X	X

$\rightarrow CD$

$$T_B = CD$$

For T_C

CD \ AB	00	01	11	10
00			X	
01	1	1	X	
11	1	1	X	X
10			X	X

$\rightarrow \bar{A}D$

$$T_C = \bar{A}D$$

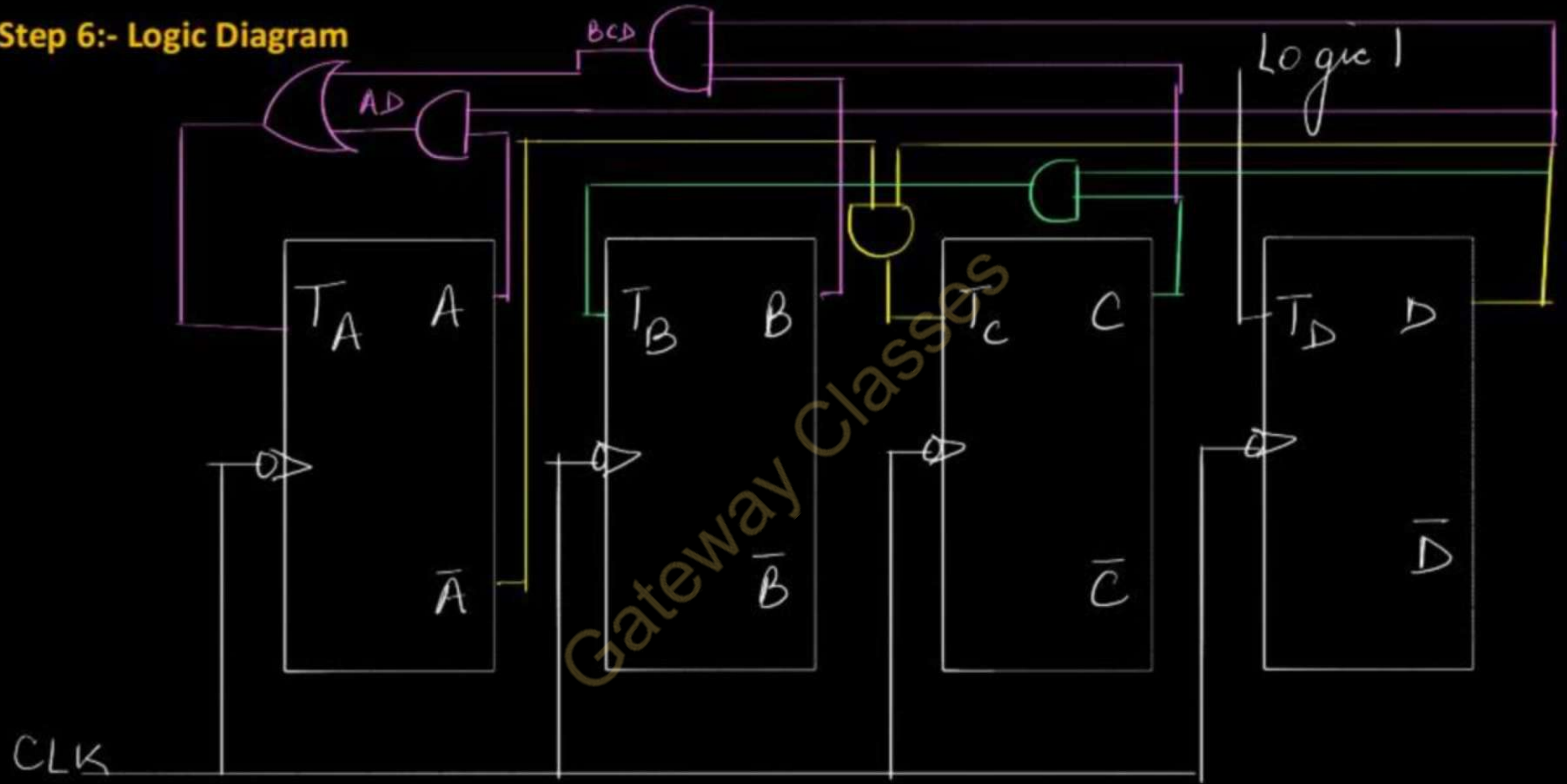
For T_D

CD \ AB	00	01	11	10
00	1	1	X	1
01	1	1	X	1
11	1	1	X	X
10	1	1	X	X

$\rightarrow 1$

$$T_D = 1$$

Step 6:- Logic Diagram



Que -6) Design a 3 bit Up/Down Counter

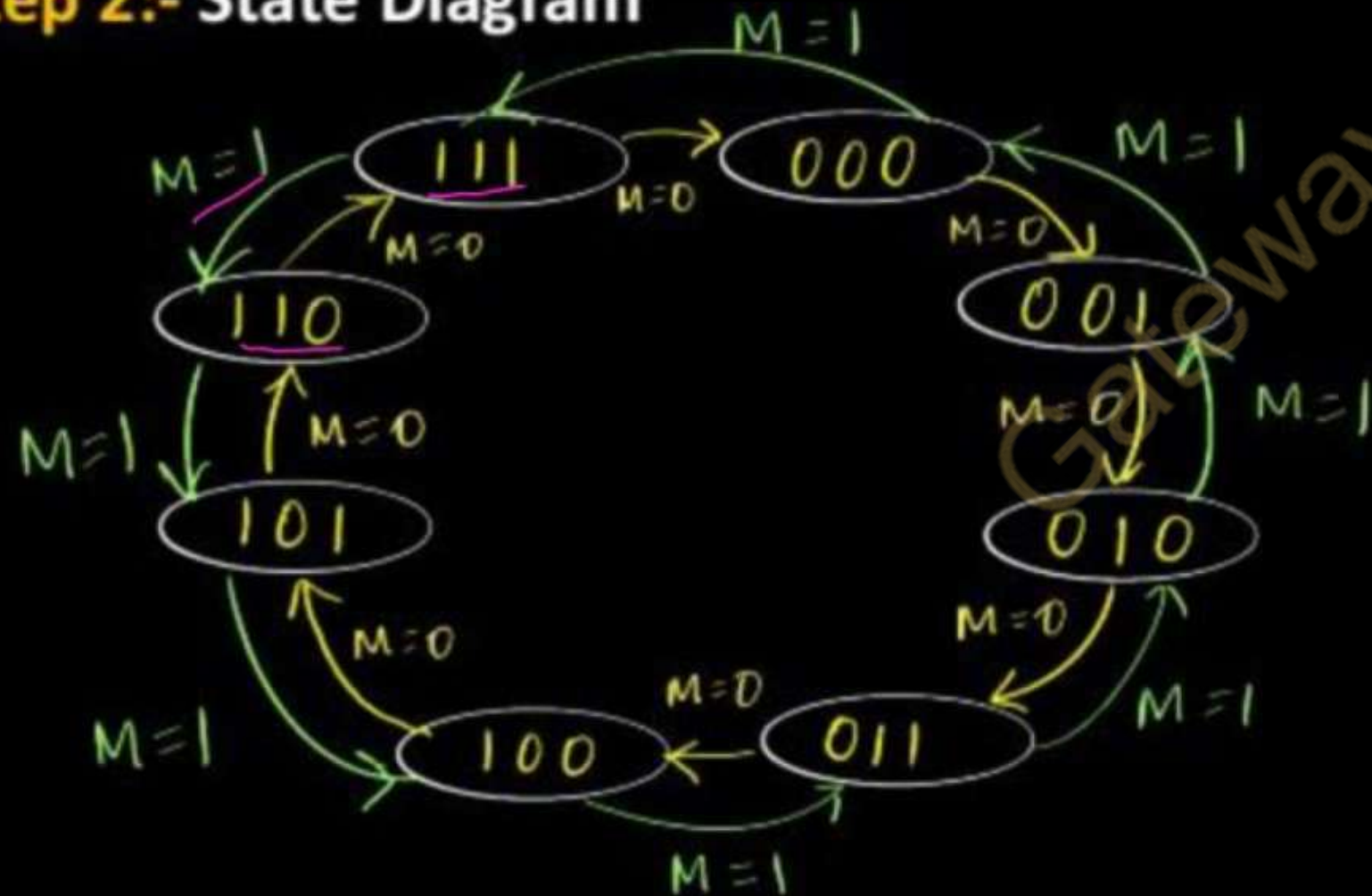
Step 1:- 3 bit , 3 FFs , $n=3$, so $N = 2^3$

= 8 states (Up) + 8 states (down)

Counts from 0 to $(2^3 - 1)$

i.e 0 to 7 (000 to 111 Up) & 7 to 0 (111 to 000 Down) , **Control Bit $M=0$ (Up) & $M=1$ (Down)**

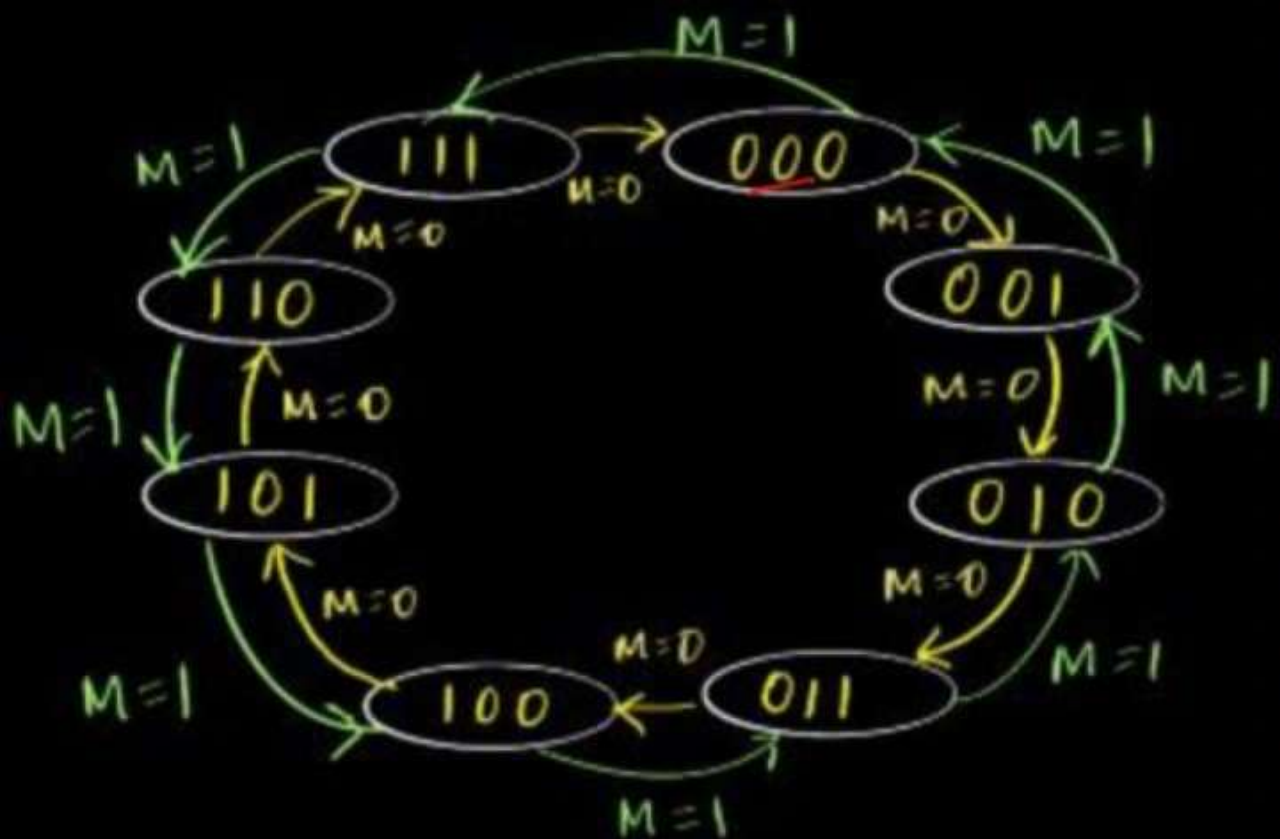
Step 2:- State Diagram



Step 3:- Excitation Table for T-FF (Suppose)

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Step 4:- Circuit Excitation Table



Control Bit	PRESENT STATE			NEXT STATE			FLIPFLOP INPUTS		
	<i>K-Map</i>			<i>x</i>			<i>A, A</i>	<i>B, B</i>	<i>C, C</i>
M	A	B	C	A	B	C	T _A	T _B	T _C
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1
0	0	1	1	1	0	0	1	1	1
0	1	0	0	1	0	1	0	0	1
0	1	0	1	1	1	0	0	1	1
0	1	1	0	1	1	1	0	0	1
0	1	1	1	0	0	0	1	1	1
1	0	0	0	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	0	1	0	1	1
1	0	1	1	0	1	0	0	0	1
1	1	0	0	0	1	1	1	1	1
1	1	0	1	1	0	0	0	0	1
1	1	1	0	1	0	1	0	1	1
1	1	1	1	1	1	0	0	0	1

Step 5:- K Maps

For T_A

MA \ BC	00	01	11	10
00			1	1
01				
11	1	1		
10				

$$\rightarrow M\bar{B}\bar{C}$$

$$\rightarrow \bar{M}BC$$

$$T_A = M\bar{B}\bar{C} + \bar{M}BC$$

For T_B

MA \ BC	00	01	11	10
00			1	1
01	1	1		
11	1	1		
10			1	1

$$\rightarrow M\bar{C}$$

$$\rightarrow \bar{M}C$$

$$T_B = M\bar{C} + \bar{M}C$$

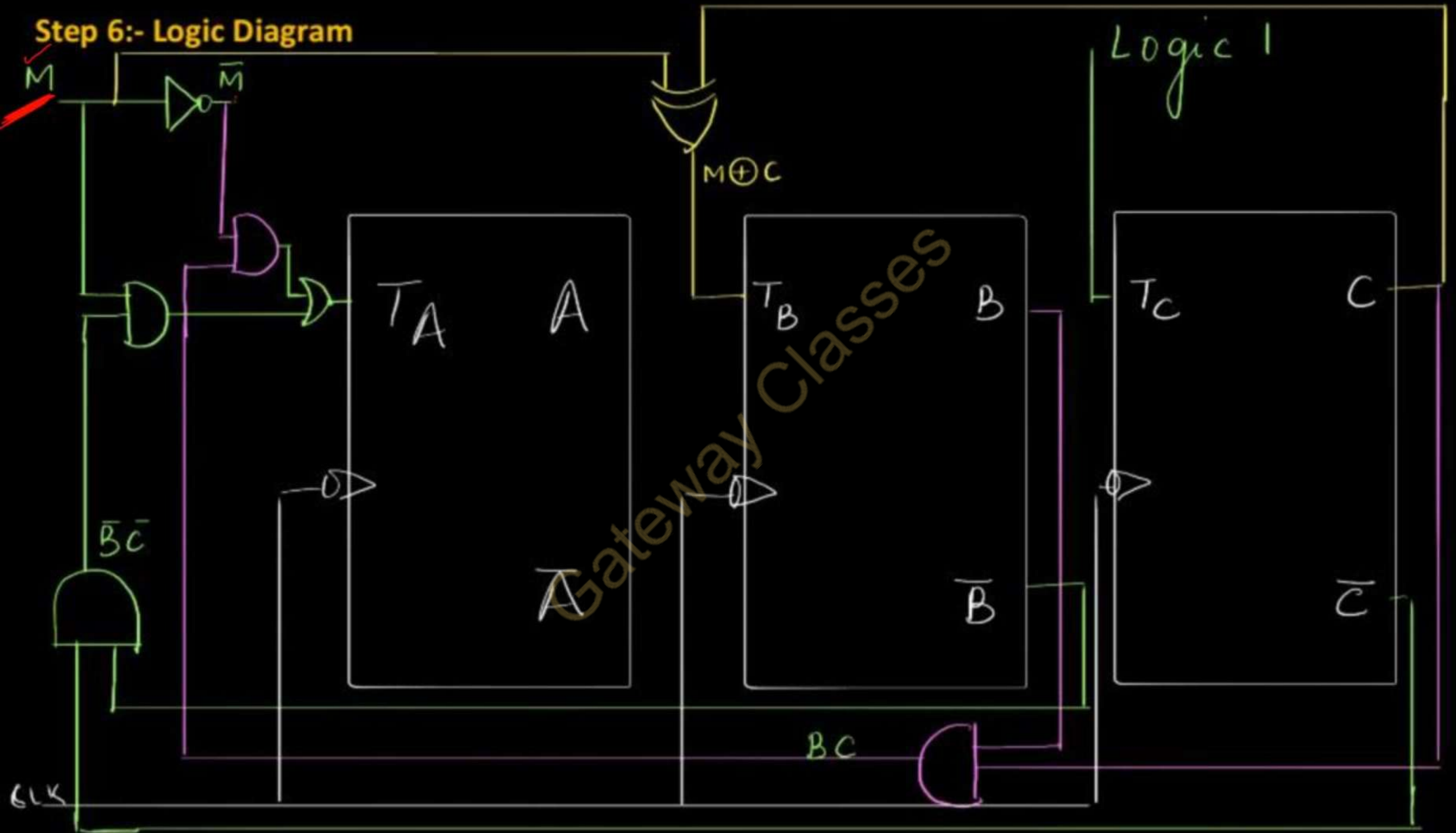
$$= M \oplus C$$

For T_C

MA \ BC	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$T_C = 1$$

Step 6:- Logic Diagram



Que -7) Design a Counter which counts in the following sequence

0 → 2 → 4 → 6 → 7 → 0

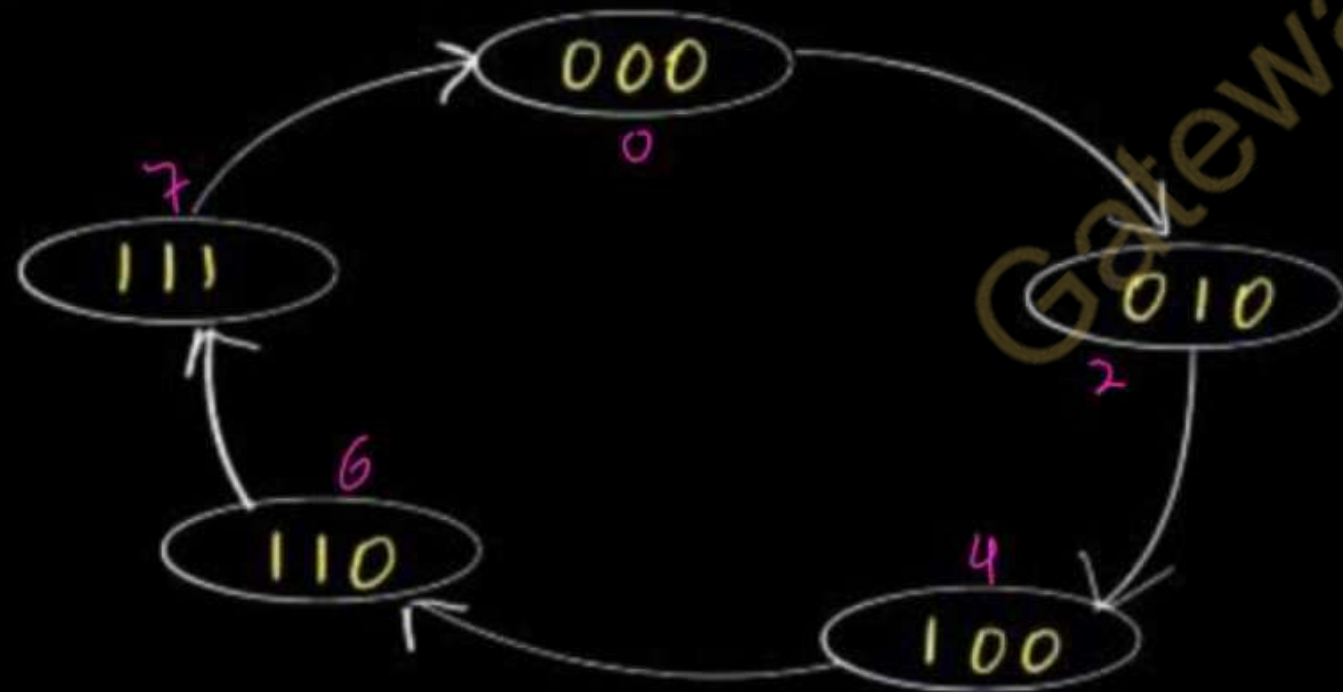
Step 1:- In the sequence given , 7 is the highest number

$$2^2 = 4$$

$$2^3 = 8$$

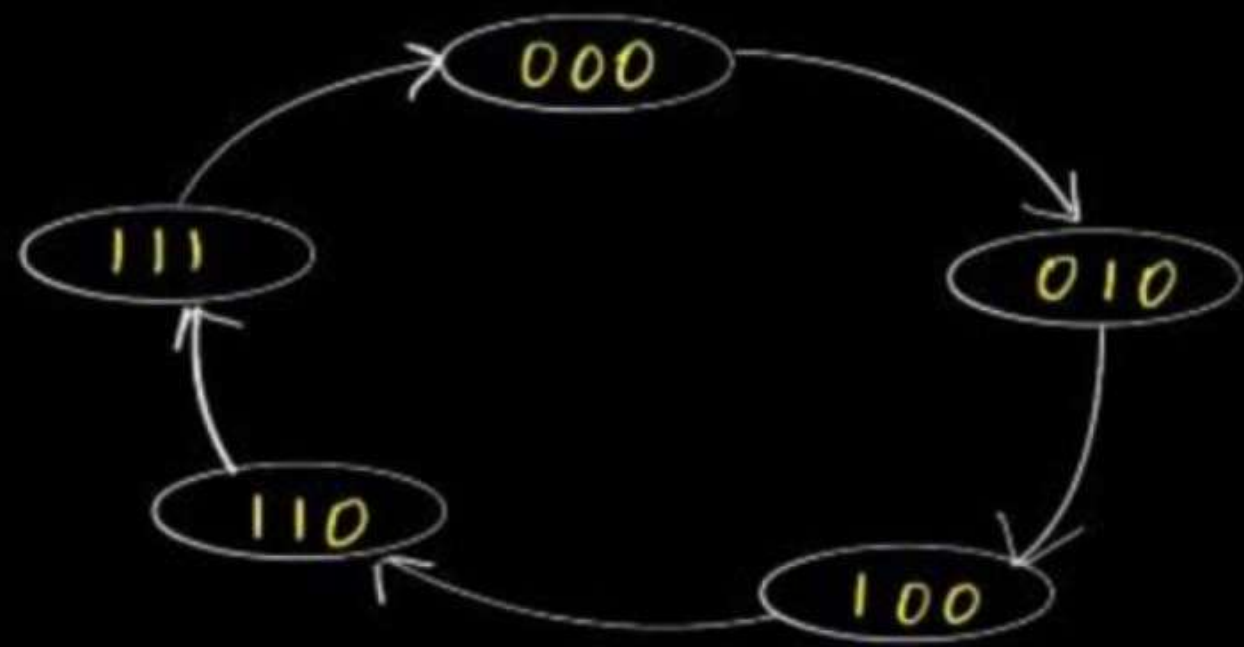
so 3 FFs will be required .(3 bits)

Step 2:- State Diagram



Step 3:- Excitation Table for T-FF (Suppose)

Q _n	Q _{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0



Step 4:- Circuit Excitation Table

PRESENT STATE			NEXT STATE			FLIPFLOP INPUTS		
A	B	C	A	B	C	T_A	T_B	T_C
0	0	0	0	1	0	0	1	0
0	1	0	1	0	0	1	1	0
1	0	0	1	1	0	0	1	0
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

Step 5:- K Maps

For T_A

C \ AB	00	01	11	10
0		1		
1			1	

$\bar{A}B\bar{C}$

ABC

$$T_A = ABC + \bar{A}B\bar{C}$$

For T_B

C \ AB	00	01	11	10
0	1	1		1
1			1	

$\bar{A}\bar{C}$

ABC

$\bar{B}\bar{C}$

$$T_B = \bar{A}\bar{C} + \bar{B}\bar{C} + ABC$$

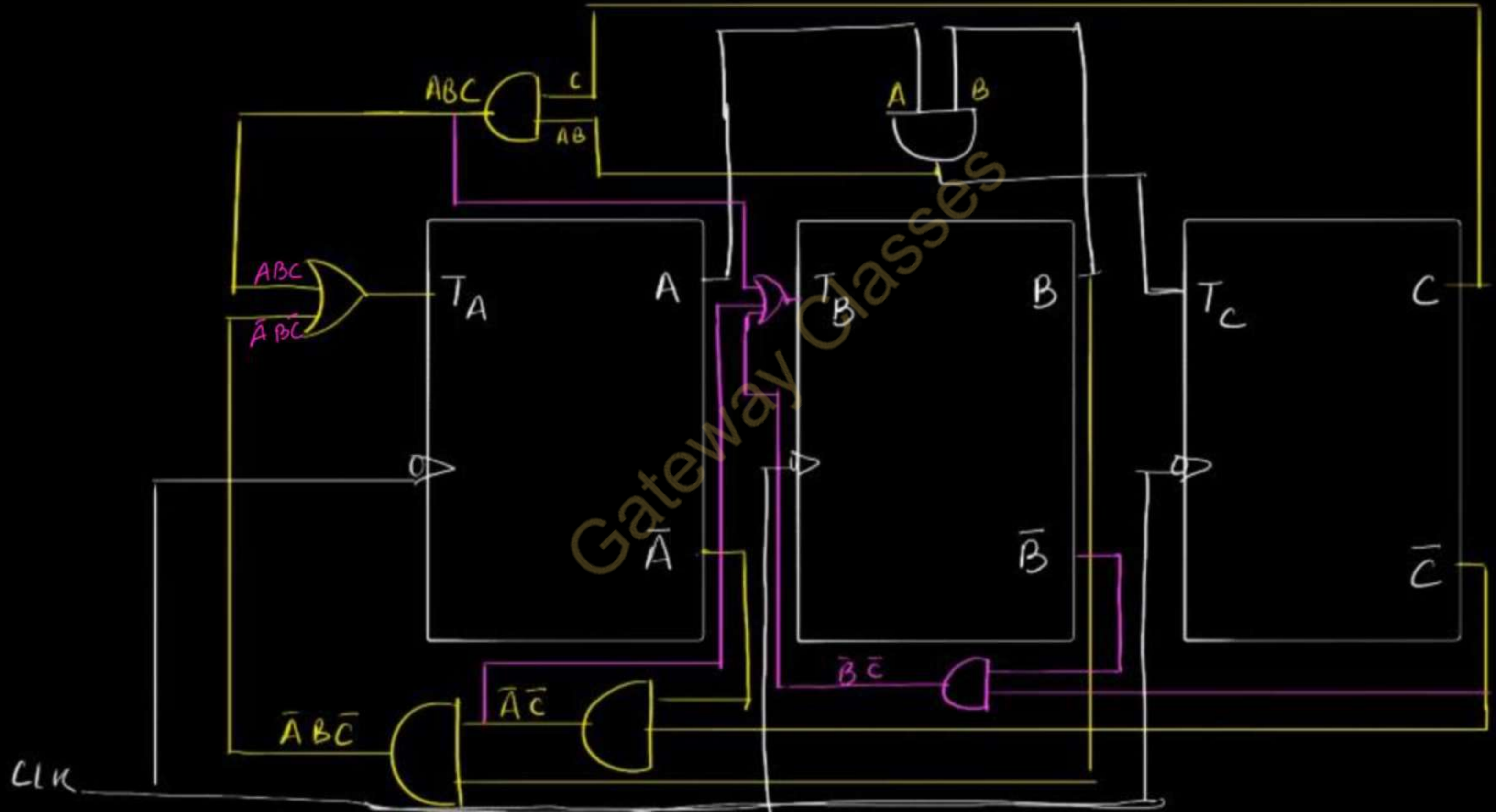
For T_C

C \ AB	00	01	11	10
0			1	
1			1	

AB

$$T_C = AB$$

Step 6:- Logic Diagram



Que -8) Design a Counter which counts in the following sequence

0 → 2 → 4 → 6 → 7 → 0

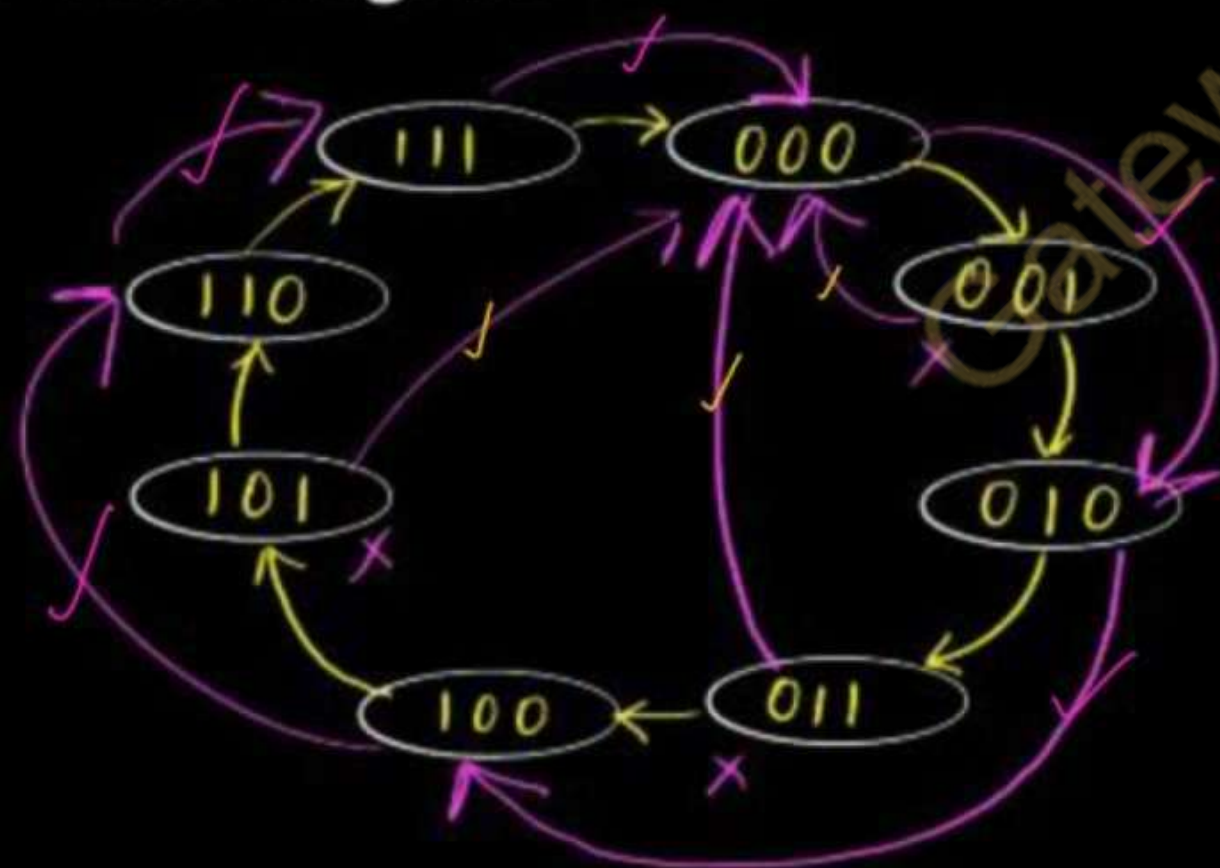
Avoid Lockdown Condition

Step 1:- In the sequence given, 7 is the highest number

$$\begin{aligned} 2^2 &= 4 \\ 2^3 &= 8 \end{aligned}$$

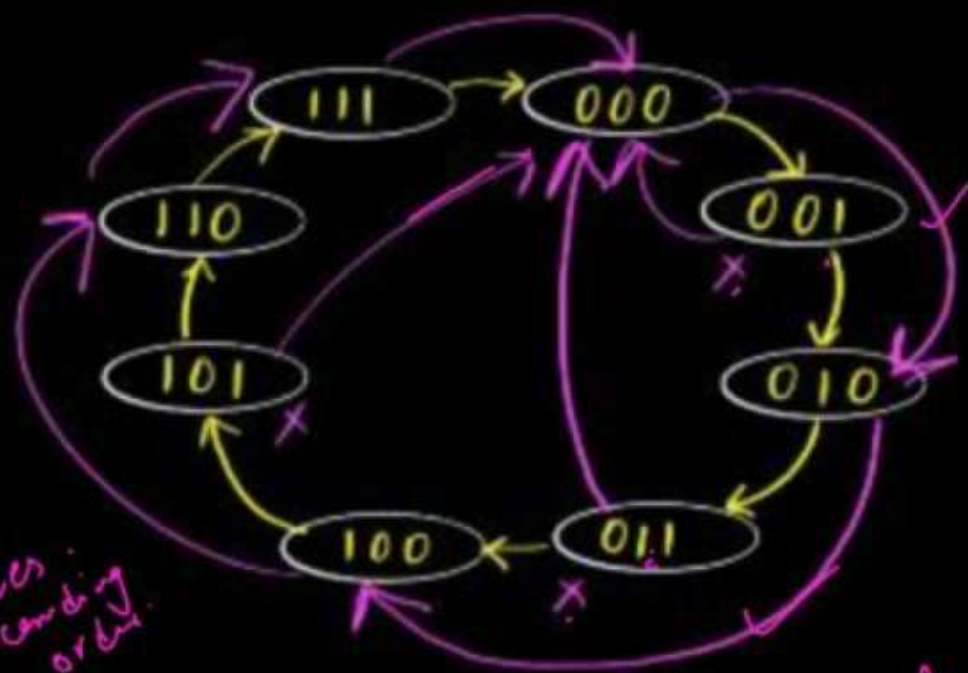
so 3 FFs will be required. (3 bits)

Step 2:- State Diagram



Step 3:- Excitation Table for T-FF (Suppose)

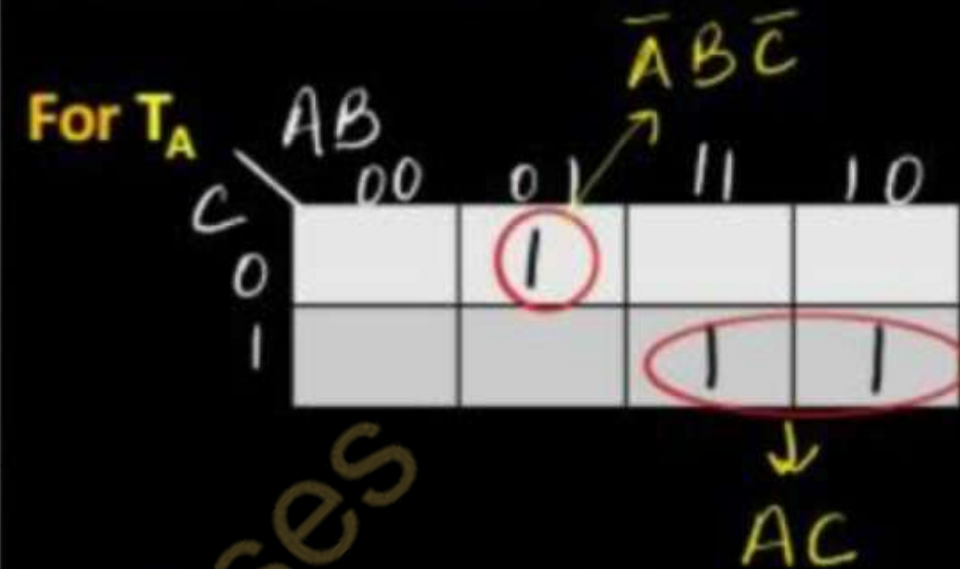
Q _n	Q _{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0



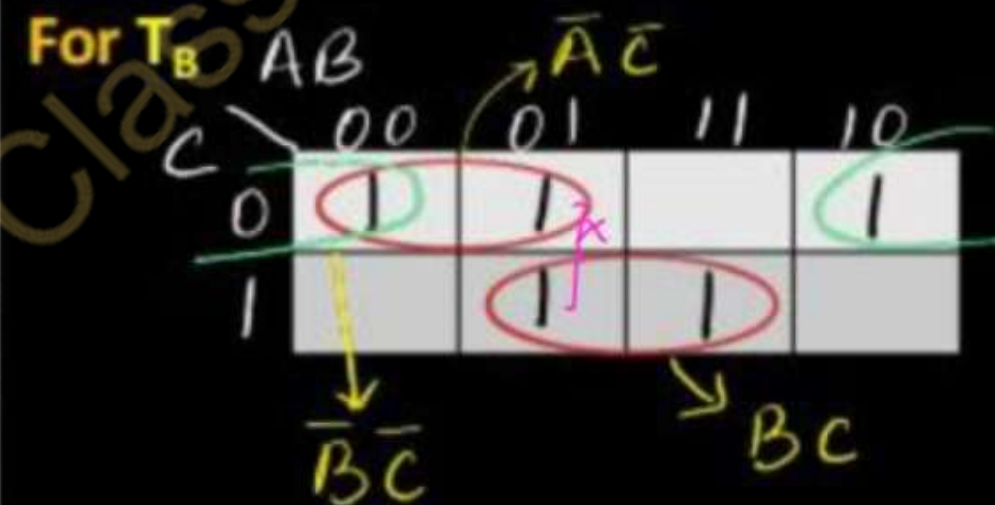
Step 4:- Circuit Excitation Table

PRESENT STATE			NEXT STATE			FLIPFLOP INPUTS		
A	B	C	A	B	C	T _A	T _B	T _C
0	0	0	0	1	0	0	1	0
0	0	1	0	0	0	0	0	1
0	1	0	1	0	0	1	1	0
0	1	1	0	0	0	0	1	1
1	0	0	1	1	0	0	1	0
1	0	1	0	0	0	1	0	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

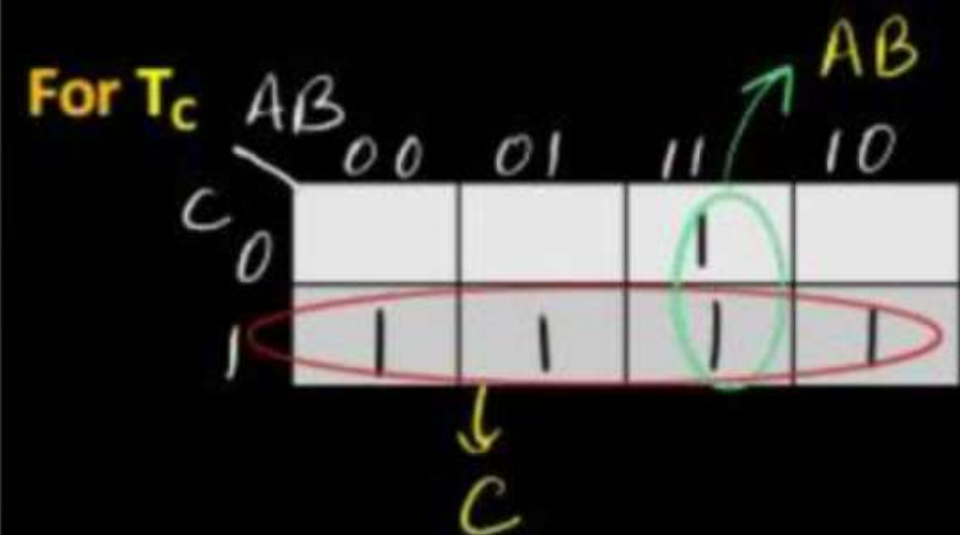
Step 5:- K Maps



$$T_A = AC + \bar{A}B\bar{C}$$

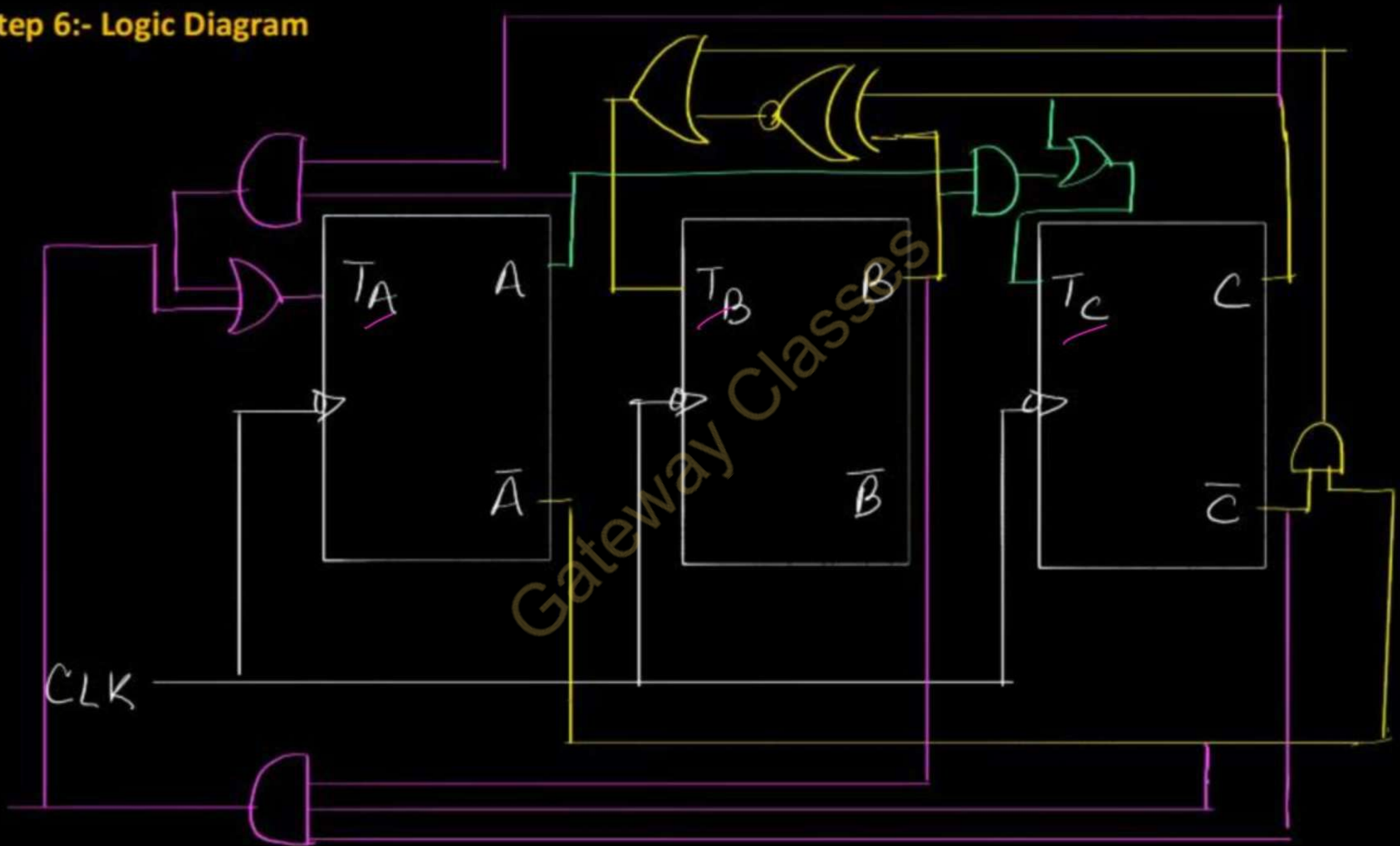


$$T_B = \bar{A}\bar{C} + BC + \bar{B}\bar{C} = \bar{A}\bar{C} + B\odot C$$

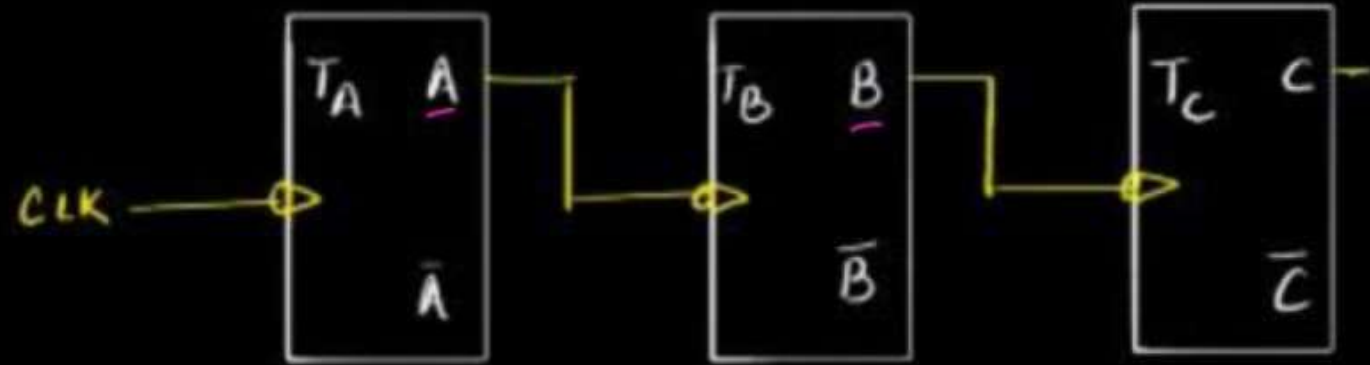


$$T_C = AB + C$$

Step 6:- Logic Diagram



ASYNCHRONOUS COUNTERS (RIPPLE COUNTERS)



Clock is given to 1st FF.

Output of 1st FF is given as clock to 2nd FF and so on.

Designing of Asynchronous Counters

Find number of FFs.

Draw Logic Diagram

Draw Timing Diagram(Waveforms)

Make Table

Table

Up Counter -

Waveforms of Q_A, Q_B, Q_C are considered for Table



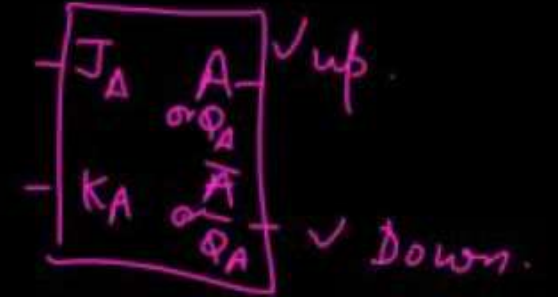
Down Counter -

Waveforms of Q_A', Q_B', Q_C' are considered for Table

Output comes only when **Toggling** is done (Clock is applied) i.e

$J_A K_A, J_B K_B, J_C K_C$ are connected to Logic 1.

Formulae:-



Total States = 2^N

Each State Frequency = $f / 2^N$

Que-9) Design a 3-bit (Asynchronous) Ripple Up Counter

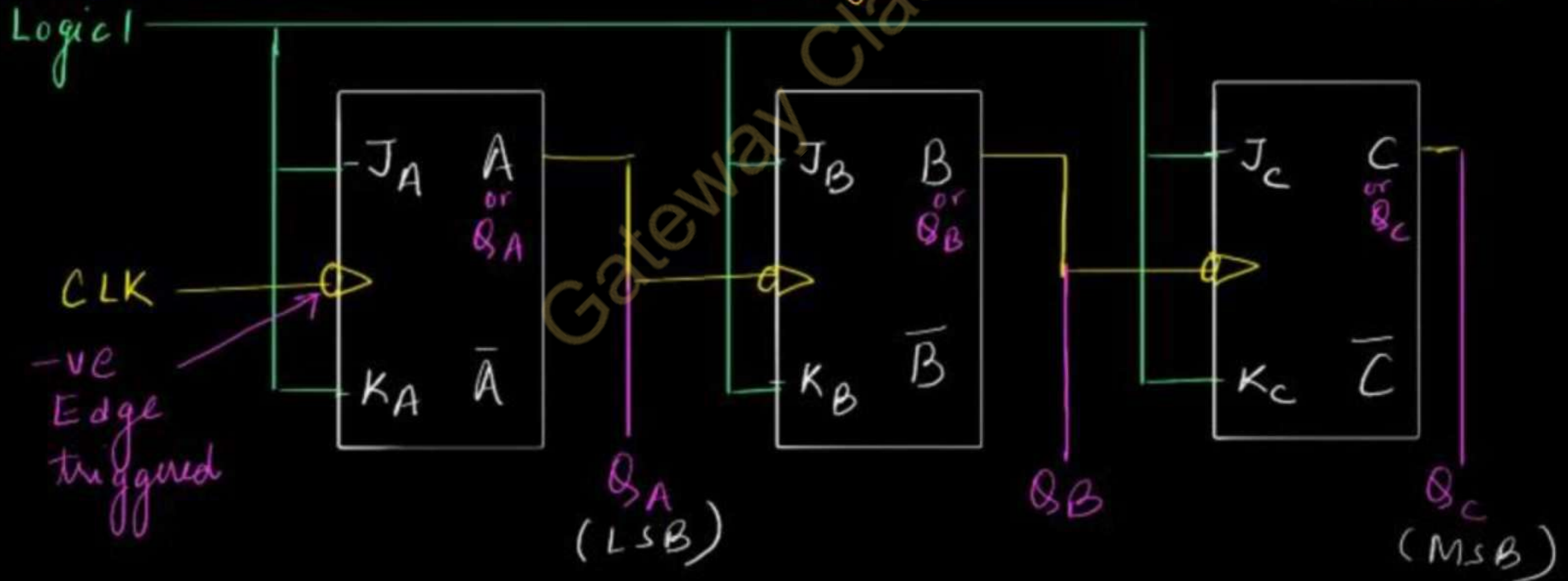
3 bit , 3 FFs

$n=3$, so $N = 2^3 = 8$ states

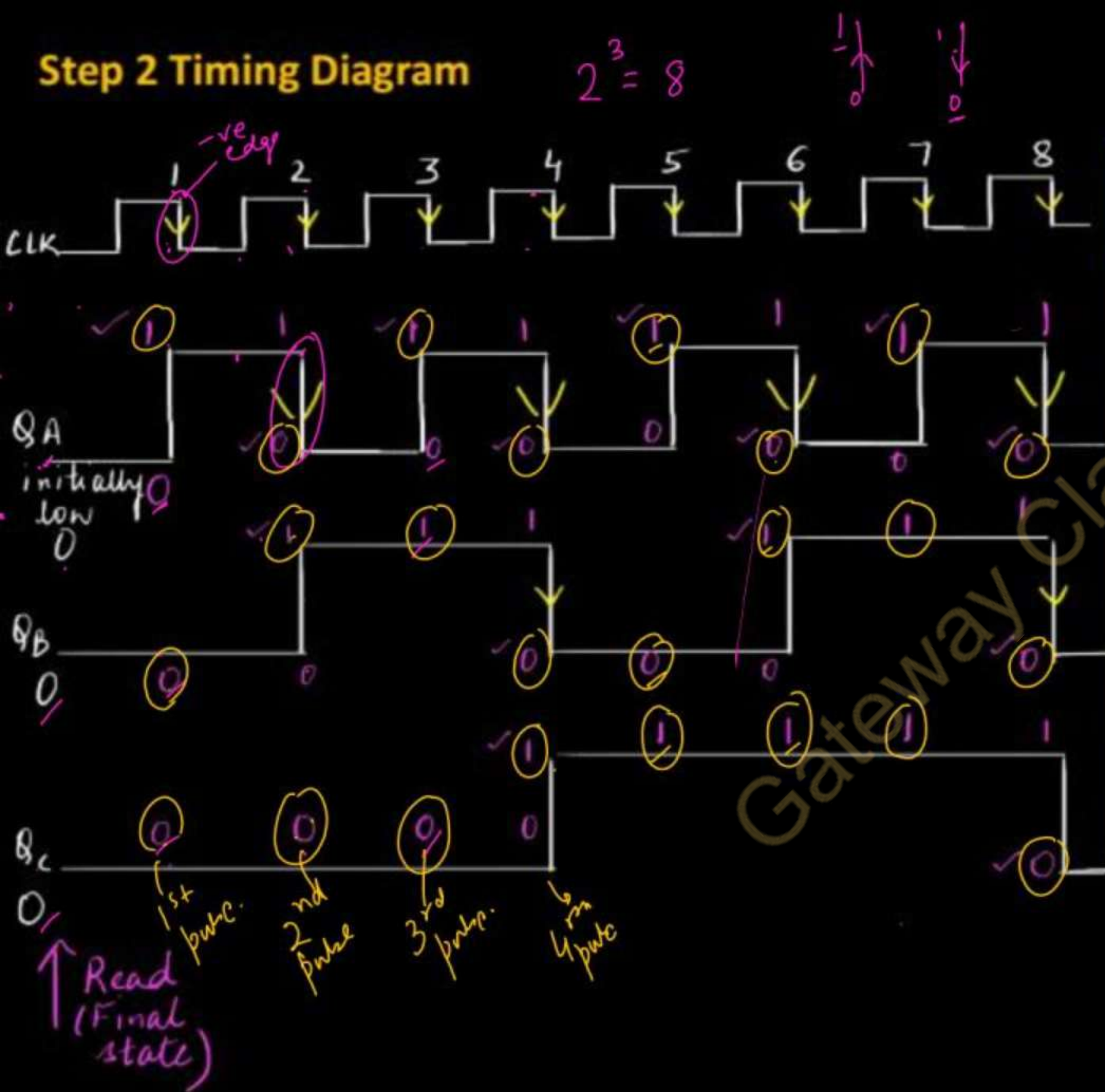
Counts from 0 to $(2^3 - 1)$ i.e 0 to 7 (000 to 111)

Since , it is **Up Counter** , therefore Outputs are taken from Q_A, Q_B, Q_C }

Step 1 Logic Diagram



Step 2 Timing Diagram



Step 3 Table

CLOCK	<u>QC</u>	<u>QB</u>	<u>QA</u>	Decimal Equivalent
Initially	<u>0</u>	<u>0</u>	<u>0</u>	<u>0</u>
1 st	<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>
2 nd	<u>0</u>	<u>1</u>	<u>0</u>	<u>2</u>
3 rd	<u>0</u>	<u>1</u>	<u>1</u>	<u>3</u>
4 th	<u>1</u>	<u>0</u>	<u>0</u>	<u>4</u>
5 th	<u>1</u>	<u>0</u>	<u>1</u>	<u>5</u>
6 th	<u>1</u>	<u>1</u>	<u>0</u>	<u>6</u>
7 th	<u>1</u>	<u>1</u>	<u>1</u>	<u>7</u>
8 th	<u>0</u>	<u>0</u>	<u>0</u>	<u>0</u>

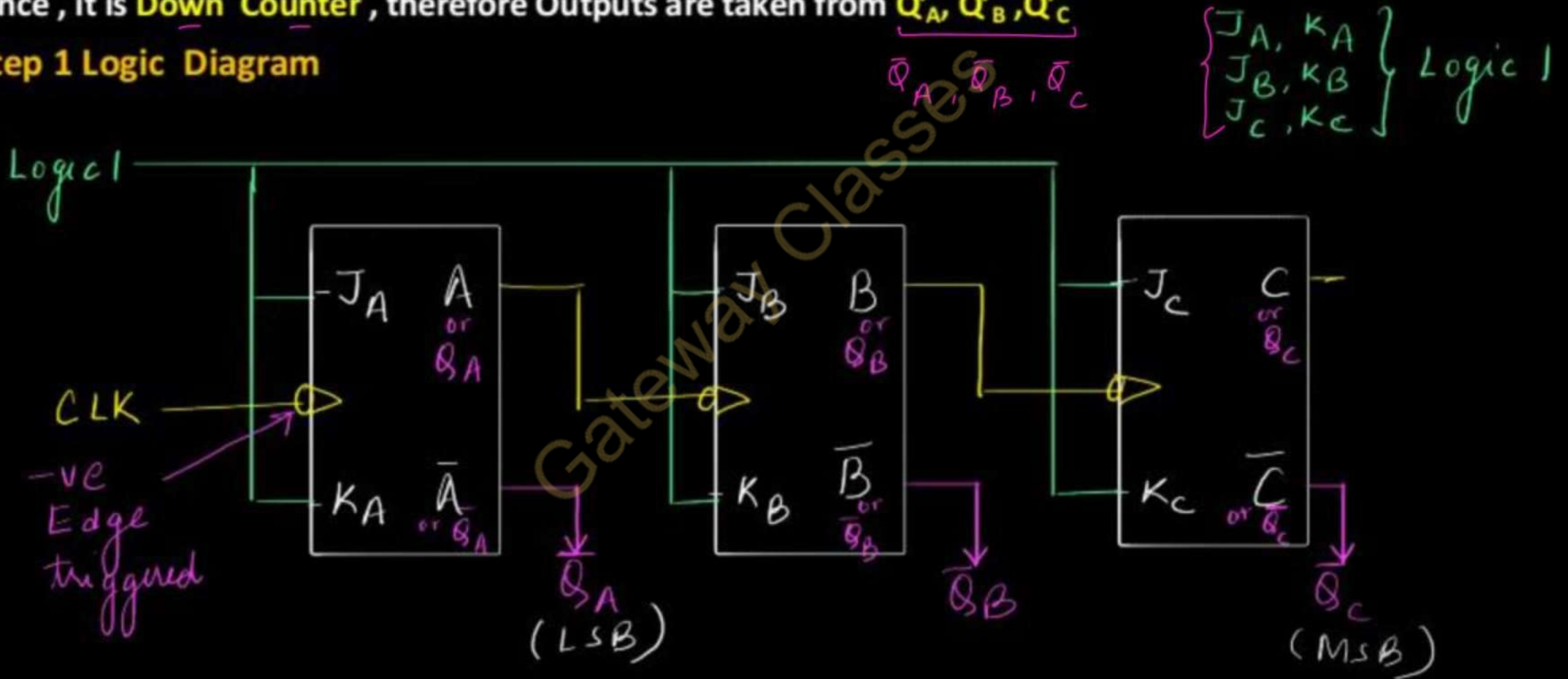
So, it is Counting from 0,1,2,3,4,5,6,7,0 i.e Up

Que -10) Design a 3-bit (Asynchronous) Ripple Down Counter

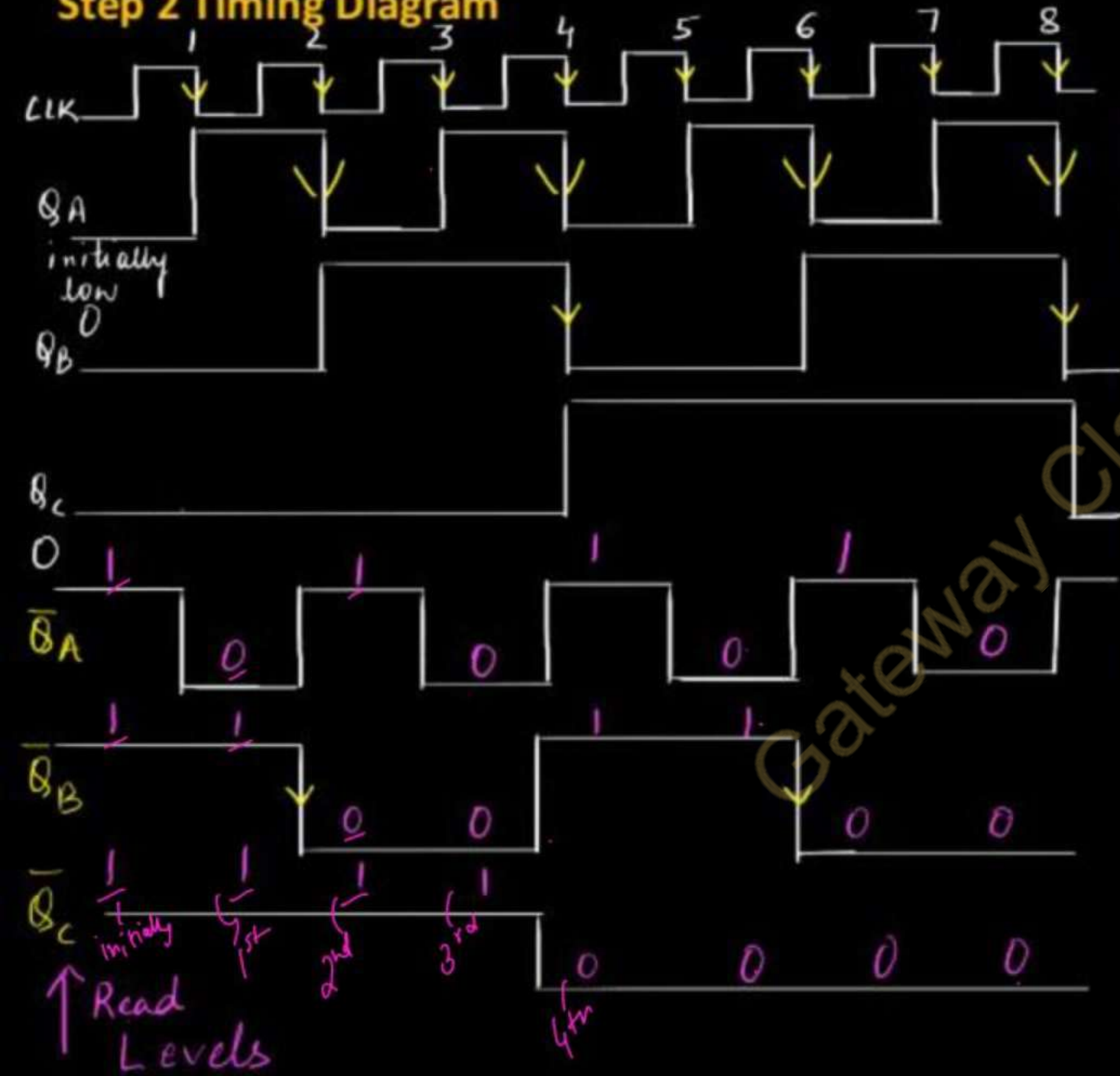
3 bit , 3 FFs , $n=3$, so $N = 2^3 = 8$ states , Counts from $(2^3 - 1)$ to 0 i.e 7 to 0 (111 to 000)

Since , it is Down Counter , therefore Outputs are taken from Q'_A, Q'_B, Q'_C

Step 1 Logic Diagram



Step 2 Timing Diagram



Step 3 Table

CLOCK	Q'_C	Q'_B	Q'_A	Decimal Equivalent
Initially	1	1	1	7
1 st	1	1	0	6
2 nd	1	0	1	5
3 rd	1	0	0	4
4 th	0	1	1	3
5 th	0	1	0	2
6 th	0	0	1	1
7 th	0	0	0	0

So, it is Counting from 7,6,5,4,3,2,1,0 i.e Down

RING COUNTER

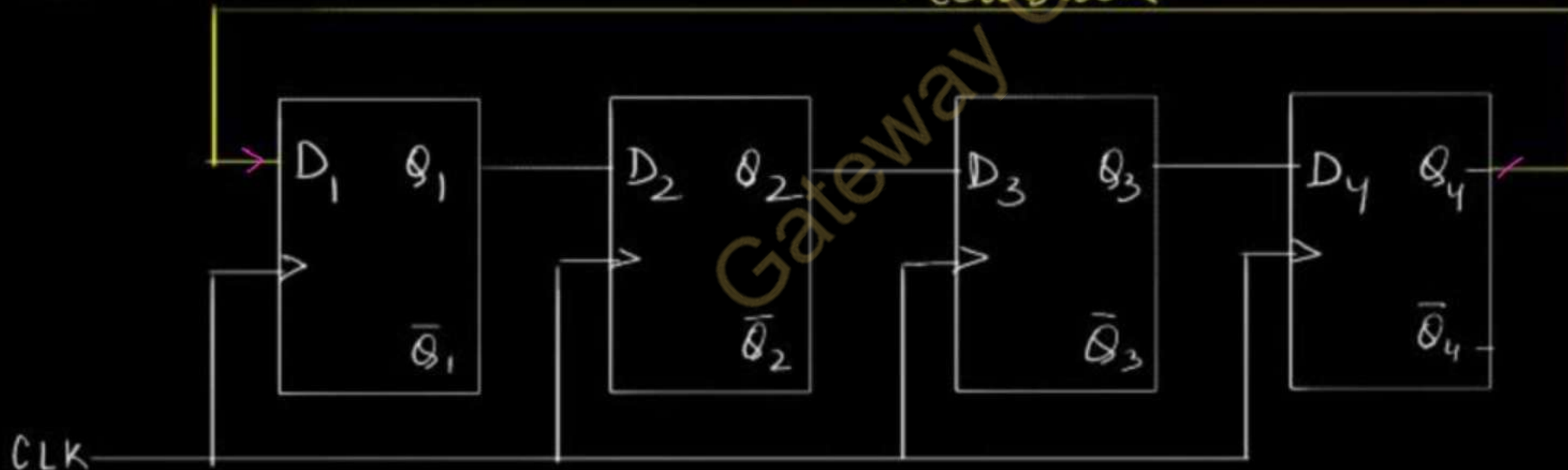
It is also called Shift Register Counter as it is a Shift Register with Feedback from last FF to first FF (**from Q_n**)

Que -11) 4 bit Ring Counter

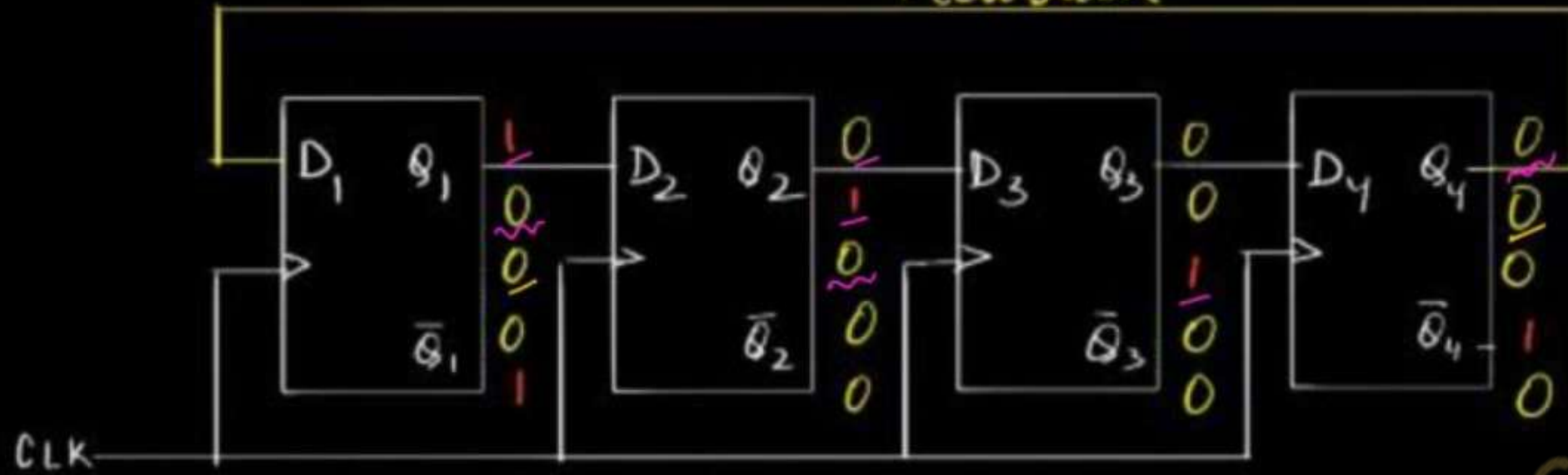
Here $n=4$, $\therefore$ feedback is from Q_4

4 FFs are used.

Feedback



Feedback



N-bit Ring Counter

N FF

N States

N CLK

CLOCK	Q4	Q3	Q2	Q1
0	0	0	0	1 (Let)
1	0	0	1	0
2	0	1	0	0
3	1	0	0	0
4	0	0	0	1 (same)

Say time duration of 1 state = T sec

Therefore ,

Total time for counter = NT Sec

Each state Frequency = f/N

4 CLK are required to get initial data.

same after 4 CLK.

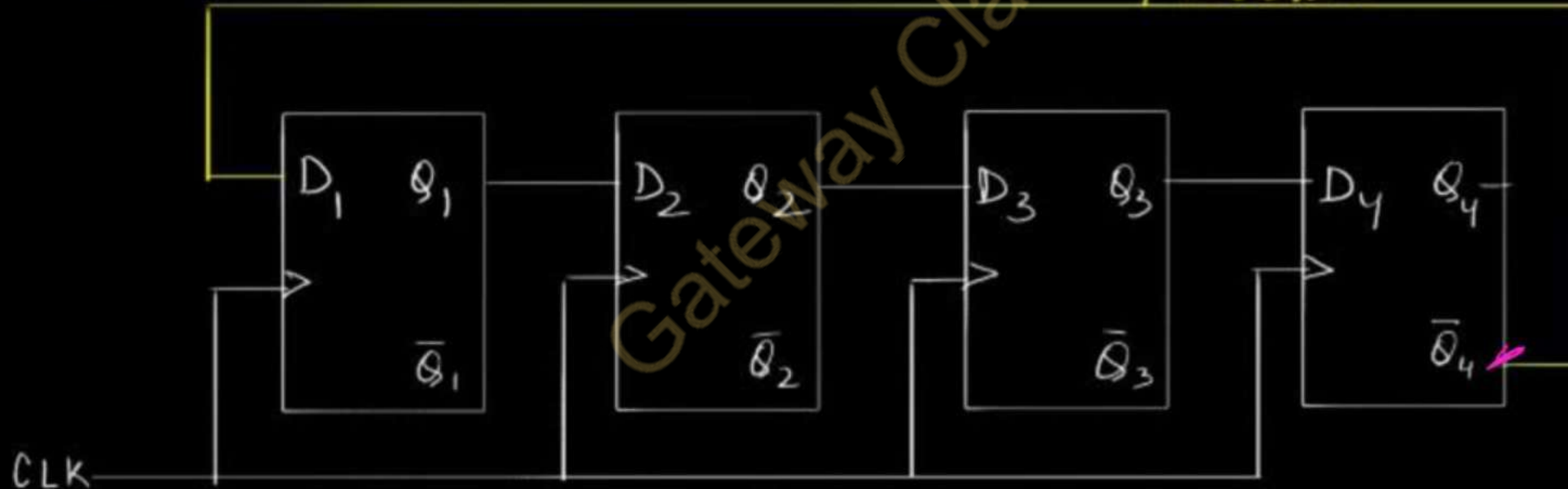
JOHNSON COUNTER (TWISTED RING COUNTER)

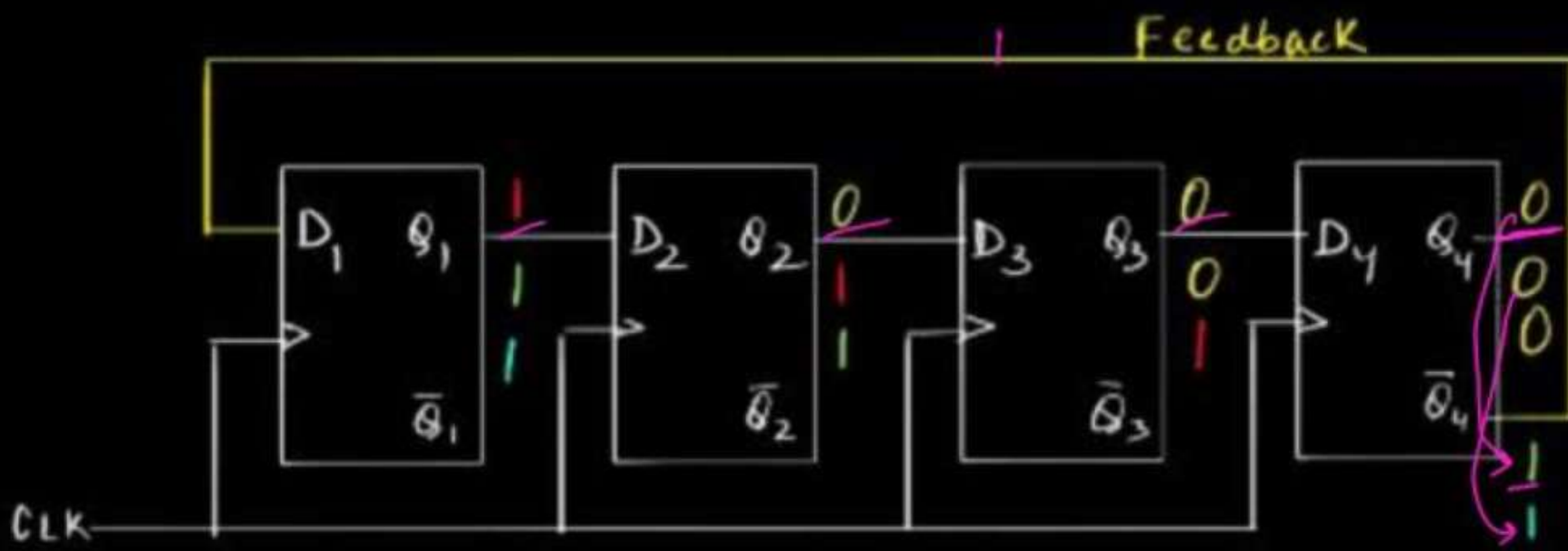
AKTU PYQ-2024-25

It is a Shift Register with Feedback from last FF to first FF (from Q'_n)

Que-12) 4 bit Johnson Counter

$n=4$ *How feedback from $\bar{Q}_4$*
Feedback





N-bit Johnson Counter

N FF ✓

2N States ✓

2N CLK ✓

Say time duration of 1 state = T sec

Therefore ,

Total time for counter = $2NT$ Sec

Each state Frequency = $f/2N$

CLK	Q_4	Q_3	Q_2	Q_1
0	0	0	0	1
1	0	0	1	1
2	0	1	1	1
3	1	1	1	1
4	1	1	1	0
5	1	1	0	0
6	1	0	0	0
7	0	0	0	0
8	0	0	0	1

same after 8 CLK



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